# The Boundaries of the Farm and the Death of Small Banks

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Job Market Paper

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November 11, 2026

**Abstract**

How does firm consolidation affect the structure of banking markets? I show that exogenous changes in subsidy payments nearly double farm concentration without increasing deposits, land values, investment, or production. As farms consolidate, large banks gain a greater share of local deposits and expand their branch networks, while mergers among small banks increase. Local banking markets become more concentrated, and the resulting decrease in competition spills over into consumer credit markets, where households earn less on deposits and face higher borrowing costs. These findings show how consolidation in the real economy can alter banking markets and credit prices.

*JEL Classification*: G21, L10, L11, G50

*Keywords:* Banking, Industrial Organization, Household Finance, Financial Intermediation

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# 1 Introduction

Over 75% of U.S. industries have become more concentrated over the past few decades. Prior research on bank consolidation has focused on drivers within banking markets, including deregulation, technological change, and economies of scale, as well as on the effects of bank consolidation on firms. However, banks do not operate independently of the firms they finance. Large and small banks specialize in lending to borrowers of different sizes, so changes in the size distribution of firms may affect which banks are best positioned to finance them and, in turn, the structure of local banking markets.[1] In this paper, I show how changes in borrower concentration affect the composition of local banking markets and spill over into the pricing of consumer financial products.

In product markets, buyer consolidation leads suppliers to consolidate. Whether borrower consolidation has similar implications for banks, the suppliers of credit, is less clear. As borrowers consolidate and expand in scale, banks with a comparative advantage in serving larger borrowers may gain market share, increasing local banking concentration. However, existing lending relationships may limit this reallocation. Incumbent lenders have informational advantages that make it costly for borrowers to switch lenders and for new banks to enter consolidating markets, so borrowers that expand may remain with the banks that initially financed them.[2]

I use the 1996 Federal Agriculture Improvement and Reform (FAIR) Act to study whether changes in the size distribution of borrowers affect local banking markets. The FAIR Act replaced six decades of price-dependent subsidies with fixed payments that followed a predetermined annual schedule. Payments were based on each farm's average acreage planted in program crops (wheat, rice, cotton, corn, and other feed grains) from 1981 to 1985. This historical acreage, known as a farm's base acres, did not change with current planting decisions. Because subsidy payments were proportional to base acres, larger farms generally received larger payments.

The program provides a setting to study how government policies that allocate support in proportion to firm size affect the distribution of firms. Farms have limited internal liquidity (Thakor, 2023), so predictable payments may help finance the upfront costs of expansion.

[1] For literature studying industry consolidation, see Grullon, Larkin, and Michaely (2019) and Kwon, Ma, and Zimmermann (2024). For the drivers of bank consolidation, see Berger, Demsetz, and Strahan (1999); Jayaratne and Strahan (1996); and Kroszner and Strahan (1999). For its effects on firms, see Strahan and Weston (1998); Cetorelli and Strahan (2006); Huber (2021); and Saidi and Streitz (2021). For lending by bank size, see Petersen and Rajan (1994); Stein (2002); Boot and Thakor (2000); and Berger et al. (2005).

[2] For literature studying buyer and supplier consolidation, see Galbraith (1952); Fee and Thomas (2004); Becker and Thomas (2009); and Bhattacharyya and Nain (2011). For lending relationships, see Diamond (1984); Rajan (1992); Boot (2000); Dell'Ariccia (2001); and Brennecke, Jacewitz, and Pogach (2025).

Larger farms also benefit from economies of scale in program crops, which decrease unit production costs and allow them to retain more of their subsidy payments (Duffy, 2009; Langemeier, 2013). These differences in payments and production costs between large and small farms may allow larger farms to acquire the land of smaller, higher-cost farms. A *Washington Post* article at the time documented that these payments encouraged owners of large farms to expand and outbid their smaller neighbors for land.[3] Given farms' reliance on local lenders, changes in the size distribution of farms may also affect the structure of local banking markets (Akhavein, Goldberg, and White, 2004).

I use a continuous difference-in-differences design based on variation in predicted subsidy payments across counties and over time. I construct predicted subsidy payments by combining each county's historical crop shares with national crop-level subsidy allocations. The historical crop shares reflect planting decisions made before farmers could anticipate the FAIR Act, and federal policy sets the national crop-level subsidy allocations. I find that subsidy payments increase local banking concentration. A within-county one-standard-deviation increase in payments increases the bank deposits Herfindahl-Hirschman Index (HHI) by about 380 points within the first decade of the program, nearly twice the 200-point increase that triggers further review of a bank merger for potentially being anticompetitive (U.S. Department of Justice, Antitrust Division, 1995).

I then determine whether farm consolidation is the primary channel through which subsidy payments increase banking concentration. Subsidy payments may encourage larger farms, which receive greater payments and benefit from lower production costs, to acquire land from smaller farms, increasing farm concentration. In addition to farm consolidation, I consider three alternative channels. First, payments deposited in local banks may increase deposits per capita and the funds available for lending (Gilje, Loutskina, and Strahan, 2016). Second, payments may increase demand for eligible land, increasing land values and the collateral available to borrowers. Third, farmers may use payments to increase production or invest in equipment, increasing credit demand.

Farm concentration is the only one of these outcomes that increases with subsidy payments. I measure county-level farm concentration using the median farm size weighted by harvested acres, or the "midpoint," from the Census of Agriculture (Roberts and Key, 2008). Subsidy payments nearly double farm concentration within the first decade of the program, increasing the midpoint by about 500 acres, or three quarters of a square mile. Payments do not increase county land values or deposits per capita. They also do not increase produc-

[3]See Federal Subsidies Turn Farms Into Big Business (https://www.washingtonpost.com/archive/politics/2006/12/21/federal-subsidies-turn-farms-into-big-business/733c4ae8-6064-4505-aa92-db34f4c11f23/).

tion or investment in equipment. Instead, the payments increase farm concentration as the largest farms acquire additional farmland from smaller farms.

Borrower consolidation may change which banks are best positioned to serve the local market. Large banks tend to lend to larger, more transparent borrowers, while small banks specialize in relationship lending that relies on soft information (DeYoung, Hunter, and Udell, 2004; Berger et al., 2005). Changes in the distribution of borrowers may also affect bank entry. When lenders differ in their costs of acquiring information and extending credit, entrants target the borrowers they are relatively better positioned to serve (Gormley, 2014). At the same time, incumbent lenders retain an informational advantage over borrowers they have previously financed, which may limit entry and preserve their advantage among relationship borrowers (Dell'Ariccia, 2001).

To isolate the borrower consolidation channel, I use variation in terrain. If subsidy payments affect local banking markets through channels other than farm consolidation, they should affect the structure of local banking markets even in counties where they have a limited effect on farm concentration. In rugged counties, where expanding acreage is more difficult, subsidy payments weakly predict farm consolidation and do not affect the structure of local banking markets. In flat counties, subsidy payments strongly predict farm consolidation and increase large banks' presence, with effects on branches and deposits about two to three times as large as those for small banks.[4] The terrain difference remains unchanged after controlling for how the effects of payments differ by population, urbanization, and land values.

Having shown that subsidy payments affect banking concentration primarily through farm consolidation, I use subsidy payments as an instrument for farm concentration. A within-county one-standard-deviation increase in farm concentration of about 460 acres increases local banking concentration by 9%, comparable to one out of seven banks exiting a county. Absent subsidy-induced farm consolidation, local banking concentration would have been about 13% lower by 2014.

Borrower consolidation increases the presence and activity of large banks and leads small banks to consolidate through mergers. A within-county one-standard-deviation increase in farm concentration increases the probability that a large bank is present by about 5 percentage points, more than twice the increase for small banks. On the intensive margin, large banks increase their number of branches by 16% and their deposits per capita by 33%. In contrast, small banks expand their branch networks and deposits per capita by about two thirds as much and increasingly merge with one another. Mergers among small banks increase by about 23%, and no other merger type increases.

To understand how large banks gain deposit market share and why small banks con-

[4]Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise.

solidate, I study their agricultural lending activity. If large banks enter these markets to serve larger borrowers, their lending should increase as farm concentration increases. Using Community Reinvestment Act (CRA) loan originations, which directly report large banks' agricultural lending at the bank-county level, I find that large banks increase lending to larger farms by about 30%, with the increase concentrated in the largest observable loan sizes.[5] To compare agricultural lending at both large and small banks, I also use county-level lending imputed from bank Call Reports (Key, Burns, and Lyons, 2019; Ifft et al., 2024). I find that a within-county one-standard-deviation increase in farm concentration increases large banks' agricultural lending by 46%, with real estate farm lending increasing by 40% and non-real estate farm lending by 46%. At small banks, non-real estate farm lending increases, but by only about one eighth of the corresponding increase at large banks.

The limited increase in lending at small banks may be due to funding constraints. Unlike large banks, which have access to internal capital markets, small banks depend more heavily on local deposits to fund loans (Gilje, Loutskina, and Strahan, 2016; Cortés and Strahan, 2017). Because core deposits provide a relatively stable source of funding, small banks with a larger share of these deposits may be better able to finance larger agricultural loans. I use the pre-program core deposit share as a measure of funding capacity (Berlin and Mester, 1999; Cornett et al., 2011). Using the imputed Call Report data, I find that only small banks with a high core deposit share increase lending through larger agricultural loans, for both real estate and non-real estate farm lending. These results suggest that stable deposit funding allows small banks to compete for larger agricultural loans. In contrast, neither group of small banks increases lending in the smallest loan sizes, consistent with farm consolidation generating greater demand for larger agricultural loans.

My results show that banks reorganize in response to farm consolidation, but information frictions and industry expertise may still play an important role in local banking markets. I therefore study how farm consolidation affects banks that specialize in agricultural lending. A within-county one-standard-deviation increase in farm concentration increases the concentration of agricultural lending by 14%, nearly twice the increase in deposit market concentration. Large agricultural banks are 15 percentage points more likely to be present, more than three times the increase for large banks overall, and their deposits per capita increase by 75%, compared with 24% at small agricultural banks. These results suggest that industry expertise remains important, as large banks with existing agricultural lending experience benefit disproportionately from farm consolidation.

Changes in local banking competition following farm consolidation may also spill over

[5]Beginning in 2005, banks with assets below about $1 billion were exempt from CRA reporting requirements, so the main CRA lending analysis covers large banks (see Appendix B.3 for details).

to the prices banks offer on consumer financial products. Buyer consolidation may affect supplier prices differently depending on supplier market power and the scope for price discrimination (Galbraith, 1952; Snyder, 1996; Chipty and Snyder, 1999; Inderst and Valletti, 2011). In agricultural credit markets, Kropp and Whitaker (2011) documents that farms with a greater share of base acres receive slightly lower interest rates on their operating loans. In consumer credit markets, banks could increase prices by exercising market power in more consolidated local markets or lower them to attract deposits and retail customers.

To test these spillover effects, I study interest rates offered on savings, checking, and time deposits (Drechsler, Savov, and Schnabl, 2017). Savings and checking interest rates decrease by 32 and 22 basis points, or about 26% and 14% relative to their unconditional means. When I compare the same bank in the same county over time, large banks lower interest rates on all three deposit products, but small banks increase interest rates on time deposits by 30 basis points. Time deposits fund commercial lending, so the increase is consistent with small banks competing for local funding to serve larger borrowers (Supera, 2021).

I also study interest rates on auto loans, which are comparable across lenders, unrelated to farm lending, and subject primarily to local competition (Argyle, Nadauld, and Palmer, 2023). At the county level, auto loan interest rates increase by about 103 basis points, while mortgage interest rates do not increase. Because mortgage rates are closely tied to national secondary markets and are less sensitive to local competition (Fuster, Lo, and Willen, 2024), this result is consistent with the increase in auto loan rates reflecting a decrease in local competition instead of a general increase in borrowing costs. Large banks increase auto loan interest rates by about 97 basis points, and small banks lower them by about 37 basis points. Overall, these differences in consumer credit pricing across large and small banks are consistent with small banks competing for local funding and consumer borrowers and with large banks exercising greater pricing power in more concentrated markets.

The same mechanism may operate in other industries experiencing firm consolidation in which bank debt is an important source of financing, such as construction, trucking, and manufacturing. In these settings, consolidation may similarly change which banks are best positioned to finance the remaining firms. More broadly, these results suggest that the consolidation of any industry should be studied together with the consolidation of its upstream and downstream industries, because in equilibrium the market structures of these industries are jointly determined.

The remainder of the paper is structured as follows. Section 2 reviews the related literature, Section 3 provides institutional background on the fixed subsidy payment program, and Section 4 presents a conceptual framework. Section 5 describes the data and sample construction, and Section 6 outlines the identification strategy. Section 7 presents the results,

and Section 8 concludes.

# 2 Literature Contributions

This paper contributes to several strands of literature by showing that subsidy payments that increase with firm size change the distribution of firm size and, in turn, the concentration and composition of local banking markets.

First, this paper contributes to the literature on product and credit markets by showing how consolidation among borrowers affects the banks operating in the same market. Existing research shows that leverage and bank behavior affect output, entry, and price competition among firms (Brander and Lewis, 1986; Chevalier, 1995; Cetorelli and Strahan, 2006; Bertrand, Schoar, and Thesmar, 2007; Huber, 2021; Saidi and Streitz, 2021). The effect of lenders on borrowers is unlikely to be the only connection between lender and borrower concentration. As borrowers increase in scale and complexity, their financing needs evolve, which may change both how much they borrow and from whom. I establish a causal connection in the other direction, showing that changes in borrower structure reorganize the local banking markets that finance these borrowers.

Within this literature, I build on work in industrial organization (IO) and banking showing that buyer consolidation across industries leads to supplier consolidation (Becker and Thomas, 2009). Brennecke, Jacewitz, and Pogach (2025) applies this logic to banking, using a Bartik instrument based on national industry trends to show that decreases in small firm growth decrease deposit growth at small banks. Because the identification in Brennecke, Jacewitz, and Pogach (2025) aggregates variation across many industries and geographies, the resulting estimates reflect the broad relationship between small firm activity and small bank health. However, this approach does not identify how the full distribution of banks changes within local markets following changes in borrower structure. Using a policy-driven shock to one place-based industry, I follow the composition of banks over time as borrowers consolidate. When small businesses expand and consolidate, the additional credit demand does not flow primarily to small banks. Instead, large banks enter affected markets and expand their deposits and branch networks, and households face less favorable terms on both deposits and consumer loans as local banking markets concentrate.

Second, this paper contributes to the literature on the effects of government transfers to firms. Existing work shows that transfers, tax incentives, and collateral shocks can relax financing constraints, increase investment, and change firms' demand for external finance (Fazzari, Hubbard, and Petersen, 1988; Gan, 2007; Chaney, Sraer, and Thesmar, 2012; Zwick and Mahon, 2017; Bahaj, Foulis, and Pinter, 2020; Garmaise and Natividad, 2024).

Where support reaches firms through banks, the banks determine which firms receive it (Granja et al., 2022), and the support changes which firms employ workers (Barrot et al., 2024). I add to this literature by showing that government payments that increase with firm size may also change the structure of the borrower market, with consequences for the banks that allocate credit.

Third, this paper contributes to the literature on the boundaries of the firm (Coase, 1937; Williamson, 1975; Grossman and Hart, 1986; Hart and Moore, 1990). This literature studies which assets and activities are coordinated within a firm rather than through market transactions. In agriculture, the horizontal boundary of the firm is how much land and production a single farmer operates (Allen and Lueck, 1998). I show that subsidy payments concentrate land and production on larger farms, and that as farms become larger, banks that specialize in serving larger borrowers gain market share. Whereas Brickley, Linck, and Smith (2003) studies the boundaries of banking organizations, I show that farm expansion changes which banks lend in a county.

Fourth, this paper contributes to work on the structure of banking markets, which emphasizes supply-side drivers of consolidation such as deregulation, technology, and economies of scale in banking (Jayaratne and Strahan, 1996; Berger, Demsetz, and Strahan, 1999). I highlight a complementary demand-side channel. Large and small banks have different comparative advantages. Small banks rely on soft information and local ties (Petersen and Rajan, 1994; Berger and Udell, 1995; DeYoung, Hunter, and Udell, 2004; Berger, Bouwman, and Kim, 2017), and large banks specialize in standardized and scale-intensive lending (Boot and Thakor, 2000; Petersen and Rajan, 2002; Berger et al., 2005; Alessandrini, Presbitero, and Zazzaro, 2009). Banks also specialize by industry (Paravisini, Rappoport, and Schnabl, 2023; Blickle, Parlatore, and Saunders, 2026). The increase in agricultural lending at large banks suggests that borrowers switch lenders despite established relationships, indicating that growth in borrower scale may alter both the size and type of credit that borrowers demand. Building on Brickley, Linck, and Smith (2003), which documents that large banks concentrate in urban areas, where borrower scale can support centralized operations, I show that borrower consolidation can induce entry by large banks into markets where they previously had little presence.

Fifth, this paper contributes to research on the structure of banking markets and real economic outcomes. Prior studies show that bank consolidation affects firm entry, competition, and community development (Strahan and Weston, 1998; Cetorelli and Strahan, 2006; Garmaise and Moskowitz, 2006; Saidi and Streitz, 2021; Carlson, Correia, and Luck, 2022), and that higher banking concentration is associated with lower deposit rates, higher loan spreads, and higher markups (Berger and Hannan, 1989; Rice and Strahan, 2010; Erel,

2011; Drechsler, Savov, and Schnabl, 2017). Higher concentration also provides banks with more stable funding and longer loan maturities (Li, Loutskina, and Strahan, 2023) and increases the pass-through of cost shocks to auto loan rates (Hankins, Momeni, and Sovich, 2026). My findings add a borrower-driven mechanism to this literature, showing that changes in the structure of borrower industries spill over into the interest rates banks offer households through changes in local banking concentration. Kropp and Whitaker (2011) finds that farms with a greater share of base acres receive slightly lower interest rates on operating loans. I complement this finding by showing that the reorganization of borrowers and lenders increases retail interest margins more broadly, increasing auto loan rates and decreasing deposit rates.

Last, this paper contributes to research using agriculture to study broader financial markets and to the agricultural finance literature. Agriculture is well suited for this purpose because farms borrow from local lenders, produce commodities that are comparable across crops, and receive policy-driven shocks that provide plausibly exogenous variation in credit demand (Butler and Cornaggia, 2011; Rajan and Ramcharan, 2015; Carlin and Mann, 2024). Shocks to farm income have been used to trace how banks move funds across markets and how local cash flows affect land values and bank failures (Bustos, Garber, and Ponticelli, 2020; Bergman, Iyer, and Thakor, 2020). Existing work documents how decoupled subsidy programs influence land allocation, land values, farm consolidation, and credit access (Burfisher and Hopkins, 2003; Roberts and Key, 2008; Bhaskar and Beghin, 2010; Gardner, Hardie, and Parks, 2010; Bekkerman, Belasco, and Watson, 2015; Ifft, Kuethe, and Morehart, 2015). I show that the farm consolidation these programs produce pushes credit demand toward large banks and changes the structure of local banking markets.

# 3 Institutional Background

The 1996 FAIR Act marked a sharp break from 60 years of farm support tied to commodity prices. Since the New Deal, federal policy had stabilized farm income through a combination of price floors, nonrecourse loans, and countercyclical deficiency payments (Effland, 2000). In exchange for this support, farms had to maintain base acreage allocations, meaning that payments were tied to the production of designated crops on enrolled land. The FAIR Act replaced this system with seven years of fixed annual subsidy payments determined solely by a farm's historical acres and yields, regardless of current production or prices (Kirwan, 2009). The program decoupled subsidy payments from planting decisions, current yields, and market prices, allowing farmers to respond to market signals instead of government incentives. The reform was intended to reduce production distortions and reliance on

government support, and to comply with World Trade Organization (WTO) rules.[6]

Payments were determined by the number of base acres, defined as the average number of acres planted in program crops (wheat, corn, sorghum, barley, oats, rice, and upland cotton) between 1981 and 1985. Because larger farms had more base acres, they received larger payments. The FAIR Act also removed planting restrictions, so farms could change production without affecting payment eligibility.[7]

Between 1996 and 2014, the federal government disbursed about $89 billion, or about $4–6 billion per year, in direct payments. During the first three years of the FAIR Act, these fixed payments represented about 79% of subsidy payments distributed under the new farm policy (Williams-Derry and Cook, 2000). The payments were concentrated among the largest recipients. According to the Government Accountability Office (GAO), the top 25% of recipients collected 73% of all direct payments in 2011 (United States Government Accountability Office, 2012). Congress renewed the program in 2002 under the Farm Security and Rural Investment (FSRI) Act, expanding eligibility to oilseeds and peanuts, and allowing base acres to be updated using average plantings between 1998 and 2001. The program was renewed again in 2008 without additional crops or base acre updates and ended with the 2014 Agricultural Act. Figure 1 shows a timeline outlining the policy changes.

[Insert Figure 1 Here]

There was substantial consolidation in farming during the period I study. Although the total number of farms nationwide was similar between 1992 and 2012, their size distribution changed. From 1992 to 2012, the number of large farms and farms with very low sales increased, and the number of small commercial and midsize farms declined (Burns and Kuhns, 2016). In my sample, the average number of farms per county remains in a narrow range of about 620 to 760 from 1992 to 2012, but the average midpoint farm size nearly doubles. The main change over this period is a reallocation of agricultural activity toward larger farms rather than a large change in the total number of farms.

# 4 Conceptual Framework

This section outlines the mechanisms through which borrower consolidation influences the structure of local banking markets. The argument proceeds in three steps. First, fixed subsidy payments relax farms' financial constraints and enable expansion, particularly among

[6]See Statement on Signing the Federal Agriculture Improvement and Reform Act of 1996.

[7]Although the federal government limited fixed payments to $40,000 annually per person, many large farms bypassed this limit by subdividing operations into multiple entities or assigning ownership shares to passive investors (United States General Accounting Office, 2004).

larger farms, increasing farm concentration. Second, as farms consolidate, credit demand becomes concentrated in larger loans. Third, this change in borrowers alters which banks are best positioned in the local market. Banks equipped to supply larger loans enter and expand in these markets, and other banks exit, merge, or contract. These changes in the composition of borrowers and lenders may ultimately affect the pricing of consumer financial products in rural communities.

Commodity crop producers have strong incentives to expand. Advances in mechanization, chemical inputs, and crop management practices decrease the marginal cost of managing additional acres (Paul and Nehring, 2005; Duffy, 2009). Because of high fixed costs and economies of scale, larger farms also have lower average production costs than smaller farms (Hoppe and Banker, 2006). In competitive commodity markets where farms are price takers (Stiers, 2022), expanding acreage allows farms to spread fixed costs over more output and improve cost efficiency (Langemeier, 2013; Hoppe, 2014).

However, expansion requires substantial upfront capital for land acquisition, machinery, and seasonal operating inputs. With limited internal liquidity or collateral, many farms rely on external finance to cover these costs (Hubbard and Kashyap, 1992; Barry, Bierlen, and Sotomayor, 2000). Interviews with agricultural lenders also indicate that most farmers are cash constrained (Thakor, 2023).

Subsidy payments may help relax these constraints. The 1996 FAIR Act provided farms with predictable annual subsidy payments tied to historical base acres. Because base acres were correlated with farm size, larger farms systematically received larger total payments (Roberts and Key, 2008). These predictable transfers provided larger farms with more resources to finance expansion. Anecdotally, reports from the time suggest that large farms used these payments to go out "trolling for land."[8] Larger farms may have used these payments to pursue land acquisitions, accelerating the pace of consolidation.

Debt facilitates farm expansion. Between 1994 and 2017, total farm sector debt increased by 84% in real terms, reaching $390 billion in 2017 (U.S. Department of Agriculture, Economic Research Service, 2026). Credit use also differs by farm size. Between 1996 and 2014, about 28% of farms with under $100,000 in sales had debt, compared with 67% of farms with over $1 million in sales (U.S. Department of Agriculture, Economic Research Service and National Agricultural Statistics Service, 2026). These facts suggest that debt markets play an important role in financing agricultural investments, particularly among large farms.

Subsidy payments may also affect credit directly through borrower balance sheets. The-

[8]See Federal Subsidies Turn Farms Into Big Business (https://www.washingtonpost.com/archive/politics/2006/12/21/federal-subsidies-turn-farms-into-big-business/733c4ae8-6064-4505-aa92-db34f4c11f23/).

oretically, higher net worth mitigates moral hazard by improving borrower incentives and decreasing monitoring costs (Holmstrom and Tirole, 1997). Empirically, subsidy payments have small effects on farms' liquidity and repayment capacity (Kropp and Katchova, 2011). Farms whose net worth declines invest less (Hubbard and Kashyap, 1992). Holding farm size fixed, payments increase the net worth of every farm that receives them, so lenders can extend more credit to each farm without a change in loan size. The balance sheet channel then predicts that credit expands across all banks and loan sizes. Sections 7.1 and 7.2 test these predictions directly.

To guide the empirical tests, I develop a model in the Internet Appendix that relates farm scale, collateral, deposits, and production to local banking. For collateral, production, and deposits, the model predicts the same effect of subsidy payments regardless of farm size. Increases in collateral or production decrease borrowing costs and expand credit across loan sizes, particularly at small banks with less funding capacity. Deposit inflows provide small banks with more funds to lend, allowing them to retain larger borrowers that might otherwise move to large banks.

Unlike the other channels, farm consolidation increases average loan size, so the model yields four empirical implications based on how large and small banks finance loans differently. First, as farms consolidate, lending growth is concentrated in larger loans, and large banks provide a greater share of lending. Second, within small banks, lending growth is concentrated at banks with greater core deposit shares, which are less constrained by funding capacity and better able to retain these larger borrowers. Third, farm consolidation increases small banks' incentives to merge, because pooling their deposit bases reduces the funding costs of larger loans. Fourth, where farming is more capital intensive, each acre requires more credit, so the same increase in farm size requires a larger loan. As a result, small banks become capacity constrained at smaller farm sizes and large banks gain market share. Section 7 tests each implication by studying the concentration of lending in larger loans and at large banks, the dependence on small banks' funding capacity, the role of farming costs, and whether the effects operate through collateral, deposits, or production.

# 5 Data Sources and Sample Construction

I construct a panel dataset linking county-level farm structure with banking outcomes and consumer financial product rates from 1992 to 2014. This section describes the banking outcomes, farm concentration measure, controls, and sample. Appendix B provides additional construction details.

## 5.1 Dependent Variables

The main banking outcomes measure local banking concentration, bank composition, and mergers. I measure local concentration using the bank deposits HHI from the Federal Deposit Insurance Corporation's (FDIC) Summary of Deposits (SOD), calculated as the sum of squared bank deposit shares within each county-year. The index ranges from 0 to 10,000, with 10,000 indicating a monopoly and lower values indicating greater competition.

I report many of the banking outcomes separately for small and large banks. I merge the SOD with the Federal Financial Institutions Examination Council's (FFIEC) Consolidated Reports of Condition and Income (Call Reports) and classify banks using a $1 billion threshold in inflation-adjusted assets. This threshold follows the standard definition of community banks and is close to the CRA asset threshold.[9] For each county, I calculate the number of branches and deposits per capita held by small and large banks using population estimates from the Bureau of Economic Analysis (BEA). Bank assets, deposits, agricultural loans, land values, and income per capita are expressed in 2020 dollars using the Consumer Price Index (CPI) from the Bureau of Labor Statistics (BLS).

To track consolidation activity, I use the Chicago Federal Reserve's Bank Merger database to construct three county-level outcomes aggregated over five-year intervals. The first, *Large Acquires Small*, measures the total number of mergers or acquisitions in which a large bank acquires or merges with a small bank. The second and third, *Small Acquires Small* and *Large Acquires Large*, count the total number of mergers or acquisitions between two small banks and between two large banks, respectively. Because the FDIC SOD is recorded each June 30, I measure each bank's size and the target's branch counties at the last SOD before the transaction date. I count each transaction in every county where the target operated branches.

To measure how the composition of agricultural lending changes as farms consolidate, I use Call Reports, which report outstanding farm loans by origination size. I test whether small banks' ability to supply larger loans depends on funding capacity, measured using the pre-program core deposit share and the loans-to-deposits ratio, both averaged over 1990 to 1995. The core deposit share is deposits excluding time deposits of $100,000 or more and brokered deposits, divided by total deposits. The loans-to-deposits ratio is total loans divided by total deposits. A higher core deposit share indicates more stable funding, and a lower loans-to-deposits ratio indicates more funding available for additional lending (Acharya and Mora, 2015). Both measures are fixed at their pre-program values.

I also measure the importance of agricultural lending to each bank using real estate and

[9]For the definition of community banks, see Berger et al. (1998); DeYoung, Hunter, and Udell (2004); Brennecke, Jacewitz, and Pogach (2025); and Federal Deposit Insurance Corporation (2012).

non-real estate farm loans from Call Reports. Real estate farm loans are secured by farmland, and non-real estate farm loans finance farm production and operating expenses, including seed, fertilizer, livestock, and equipment. I classify a bank as an agricultural bank if farm loans exceed 5% of total loans in any of the preceding six years. To construct county-level farm loan volumes, I allocate each bank's agricultural lending across the counties in which it operated a branch at the start of each five-year period. I allocate lending in proportion to each county's share of farm interest expenses in the most recent Census of Agriculture available at the start of the period. Real estate farm lending is weighted by real estate interest expenses, non-real estate farm lending by non-real estate interest expenses, and total agricultural lending by their sum. I aggregate the resulting lending separately for small and large banks, following Key, Burns, and Lyons (2019) and Ifft et al. (2024). For each group, I measure growth in total, real estate, and non-real estate farm lending as the change in the group's lending divided by the county's total volume of that loan type in the prior period.

Although this method provides a proxy for county-level farm loan volumes, it has several limitations. First, the baseline allocation assumes that a bank lends only in counties where it operated a branch at the start of the period. As a robustness test, I also construct an alternative allocation that uses the bank's current branch counties and adds counties in which it originated farm loans reported under the CRA. For counties without a branch, the allocation weight declines with distance to the bank's nearest branch (see Appendix B.1 for details). Second, the allocation assumes that banks lend in proportion to historical local credit demand, proxied by farm interest expenses, which may differ from current lending decisions. Nonetheless, the method provides evidence on the geographic distribution of agricultural lending. I also use CRA loan originations, which report each bank's farm loans of $500,000 or less by the county of the borrowing farm, allowing me to observe county-level lending directly (see Appendix B.3 for details).

To study how borrower consolidation and changes in bank composition affect the interest rates banks offer on consumer deposits and loans, I use interest rate data from RateWatch. I focus on time deposits (12-month $10,000 certificates of deposit), savings deposits ($10,000 money market accounts), interest-bearing checking accounts, auto loans, and residential mortgages. For mortgages, I average the offered rates across the fixed rate, adjustable rate, and balloon products reported by each branch. I convert interest rates to real terms using the CPI. Because the FDIC measures SOD deposits as of June 30, I use each branch's average weekly quoted rate in June so that the RateWatch and SOD data refer to the same month.

To test whether farm consolidation changes access to federal credit for small farms, I also use data on direct loans issued by the U.S. Department of Agriculture's (USDA) Farm

Service Agency (FSA). These loans target small, new, or socially disadvantaged farmers who cannot obtain commercial credit. I aggregate the number and volume of direct farm ownership loans, which finance farmland and farm property purchases or expansion, and direct operating loans, which finance production expenses and equipment, at the county-year level from 1994 to 2014.

## 5.2 Variables to Test Different Channels

The public Census of Agriculture reports only the number of farms within acreage categories (e.g., 1–9, 10–49, 50–69, 70–99, …, 1,000–1,999, and 2,000+ acres), so individual farm sizes are not observed. To compute traditional concentration measures such as the HHI, each farm in a bin would be assigned the same average acreage. For example, a 2,500-acre farm and a 25,000-acre farm in the "2,000+ acres" bin would contribute equally to the HHI calculation. Because the upper bins span wide acreage ranges, this method effectively treats all large farms as the same size. Doing so would tend to introduce bias by Jensen's inequality, because the sum of squared farm sizes exceeds the sum of squared average farm sizes. This bias would be largest in the larger farm size bins, making the HHI an unreliable measure of farm concentration.

Given the structure of the data, I follow standard practice in agricultural economics and measure borrower consolidation using the harvested acre-weighted median farm size (the "midpoint") from the USDA's Census of Agriculture (MacDonald, Korb, and Hoppe, 2013). The midpoint is defined as the acreage at which half of a county's harvested cropland lies on larger farms and half on smaller farms. For example, if a county has 2,800 total harvested acres and the eight smallest farms harvest 1,000 acres in total, then the ninth-smallest farm is the one whose acreage spans the 1,400-acre halfway point and marks the midpoint. If that farm operates 700 acres, then the midpoint farm size is 700 acres. This approach is standard in the literature for two reasons. First, individual farm sizes are not publicly reported. Second, the production distribution is highly skewed, with a relatively small number of large farms operating much of the harvested acreage. The midpoint more accurately reflects changes in the scale of production than the simple median.

I also construct measures of the alternative channels through which subsidy payments may affect local banking markets. I measure land values using the inflation-adjusted market value of agricultural land and buildings per acre from the Census of Agriculture. I measure local deposits as county deposits per capita, using total deposits from the SOD and county population. For investment and production, I use four outcomes: the real value of county machinery, the number of recently manufactured tractors, a Törnqvist crop-yield index that aggregates per acre yield growth across the nine census crops with quantity data using

revenue-share weights, and the share of county crop acres planted in the seven program crops covered by the 1996 FAIR Act.

## 5.3 Treatment Variable

I measure county-level predicted subsidy payments using historical base acres and national crop-level subsidy allocations. I obtain county-level base acres for each eligible crop from the USDA and combine them with the annual national crop-level subsidy allocation for that crop. For each crop, I multiply the county's share of national base acres by the national crop-level subsidy allocation and sum across crops to construct predicted county-level subsidy payments.

## 5.4 Controls

County-level farmland controls from the Census of Agriculture include idle cropland, pastureland, and unharvested cropland (Weber and Key, 2012). I measure local economic conditions using per capita income and the employment rate from the BEA. In additional analyses, I control for the lagged log number of bank branches and the log of total interest expenses on farm loans. To study the role of crop insurance, I use the number of policies covering the relevant commodity crops from the USDA's Risk Management Agency (RMA). I also include the number of Farm Credit System (FCS) branches in each county, provided by the Farm Credit Administration. To study the effect of countercyclical payments on my results, I include predicted countercyclical payments. For each crop, I multiply the county's share of national base acres by the national countercyclical payments on that crop in the same payment year and sum across crops. National countercyclical payments by crop are from the FSA.

I use the 2010 Terrain Ruggedness Index (TRI) from the USDA Economic Research Service (ERS) to test whether the effects differ in areas less conducive to large-scale farming. I classify counties in the top quartile of the national TRI distribution as rugged and all others as flat. Flat counties are generally in the Great Plains and the Mississippi Delta (Figure A.1).

For each major commodity crop (barley, corn, oats, peanuts, rice, soybeans, sorghum, upland cotton, and wheat), I calculate average per acre economic costs from 1990 to 1995 using the Farm Costs and Returns Survey (FCRS) from the ERS. I use this pre-program period to measure production costs before the subsidy payments began. I construct a time-invariant county-level cost measure by weighting each crop's average cost by its share of county harvested acres in the 1992 Census of Agriculture. Counties with weighted average per acre economic costs above the nationwide median are classified as high-cost, and those below

the median are classified as low-cost. High- and low-cost counties are spread throughout the U.S. (Figure A.2).

### 5.5 Sample Construction and Summary Statistics

My primary analysis uses a county-level panel observed at five-year intervals from 1994 to 2014. The USDA Census of Agriculture provides farm structure data every five years, which I merge with banking data from the FFIEC Call Reports and FDIC SOD for the corresponding periods. I lag farm variables by two years to allow changes in agriculture to precede banking outcomes. I exclude 57 large central metropolitan counties from the main sample to focus on counties where agriculture may be a more important part of the local economy. To avoid overstating economic magnitudes, I standardize farm concentration using its within-county variation over time (Liu and Winegar, 2025).

For the analysis of interest rate outcomes, I construct a separate annual panel from 2002 to 2014, merging branch-level data from RateWatch with bank- and county-level information from the SOD and Call Reports. I estimate fitted farm concentration from the Census of Agriculture, available every five years, and use the most recent fitted value in interim years. This method preserves the structural variation identified at the census years while allowing annual fluctuations in local economic conditions to enter through time-varying controls in the second-stage regressions. The approach reflects the gradual nature of farm consolidation and assumes slow-moving structural change between census years. The data support this assumption. Farm concentration increases by only about 2.5% per year on average. To estimate the specification in first differences, I interpolate fitted farm concentration linearly between census years.

Table 1 reports summary statistics. Subsidy payments are widespread and highly skewed. On average, counties receive $1.37 million in subsidy payments per year, with a standard deviation of $1.88 million between counties and $1.15 million within counties over time. These summary statistics show that there is substantial cross-sectional and time-series variation in subsidy payments.

[Insert Table 1 Here]

## 6 Identification Strategy

To study how borrower consolidation affects local banking markets, I use a continuous difference-in-differences design that combines predetermined county crop shares with changes in national crop-level subsidy allocations. First, I estimate the reduced-form effect of subsidy payments on local banking markets. Second, I evaluate the channels through which payments

may affect banks. Last, I instrument farm concentration with subsidy payments to estimate how borrower consolidation affects banking outcomes.

The difference-in-differences design uses county-level predicted subsidy payments under the fixed payment program introduced by the 1996 FAIR Act and extended in 2002 and 2008 (see Section 3 for details). Payments were based on historical planting instead of current production or market conditions. Base acres and yields initially reflected 1981–1985 averages, and farms could update them in 2002 using 1998–2001 plantings.[10] I construct predicted subsidy payments using base acres alone. Excluding historical yields produces a more conservative intent-to-treat measure because yields may be correlated with later banking outcomes through persistent differences in local productivity. The results are similar when yields are included.

Predicted subsidy payments are defined as the inner product of (a) each county's fixed crop-level base acre shares and (b) the national crop-level subsidy allocations in a given year from 1996 to 2014, when the program ended. For example, if a county has 1% of the nation's corn base acres and the national crop-level subsidy allocation for corn is $2 billion in a given year, that county would be assigned predicted subsidy payments of 1% $\times$ $2 billion. The national crop-level subsidy allocations vary over time and apply uniformly across all counties. Identification therefore comes from changes over time in national crop-level subsidy allocations, interacted with county-level crop shares determined by historical base acres. Because both components are determined outside the local banking environment, one by historical planting decisions and the other by national policy, the resulting variation in predicted subsidy payments is plausibly exogenous with respect to local credit conditions.

The timing of the policy and the historical planting decisions underlying predicted payments also help address concerns of reverse causality. For reverse causality to threaten the difference-in-differences design, banking consolidation would need to be systematically correlated with predicted subsidy payments for reasons unrelated to the policy-induced changes in farming. Reverse causality would be a concern if, for instance, banks consolidated in anticipation of the policy or if farmers chose the crops that determine base acres in anticipation of local banking conditions. This is unlikely. The 1996 FAIR Act marked a sharp and largely unexpected break with six decades of price-dependent support, and faced considerable political uncertainty and multiple filibusters prior to passage (Orden, Paarlberg, and Roe, 1999). As a result, banks could not easily anticipate the policy, and planting decisions made more than a decade earlier are unlikely to reflect expectations about future credit conditions.

[10]The results are similar when I construct predicted subsidy payments using only pre-program data (Table A.1). I hold the 1996 base acre shares fixed for the original program crops and use 1992 Census of Agriculture acreage for crops added in 2002.

Using predicted subsidy payments, I estimate the following difference-in-differences specification:

$$Y_{i,t} = \alpha \sum_{c} \left( \frac{\textit{Base Acres}_{c,i,t}}{\textit{Base Acres}_{c,t}} \times \textit{Natl Allocation}_{c,t} \right) + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}, \qquad (1)$$

where $c$ represents the crop type, $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The interpretation of $\alpha$ depends on the outcome. When $Y_{i,t}$ is a local banking outcome, $\alpha$ measures the effect of subsidy payments on the structure of local banking markets. When $Y_{i,t}$ is a possible channel outcome, $\alpha$ measures the effect of subsidy payments on that channel. For banking outcomes, the subsidy term and the controls enter with a two-year lag. For channel outcomes from the Census of Agriculture, they enter in the same census year. $\textit{Base Acres}_{c,i,t}$ is the number of historical base acres of crop $c$ in county $i$ and year $t$, $\textit{Base Acres}_{c,t}$ is the total number of historical base acres in the U.S. of crop $c$ in year $t$, and $\textit{Natl Allocation}_{c,t}$ is the national crop-level subsidy allocation for crop $c$ in year $t$. $\mathbf{X}_{i,t}$ is a vector of control variables, including county-level economic characteristics and farm characteristics that control for land availability. These controls are the county-level employment rate, log income per capita, and lagged measures of idle cropland, pastureland, and cropland left fallow, lost to crop failure, or otherwise unharvested. I include a fixed effect for state-level branching deregulation using the index from Célérier and Matray (2019), which reflects the staggered implementation of interstate branching reforms between 1994 and 2005. County and year fixed effects absorb time-invariant county characteristics and national shocks.

To understand how subsidy payments affect local banking markets, I study the three main channels through which payments may affect banks: farm concentration, land values, and deposits. Figure 2 shows the geographic distribution of subsidy payments and these three channels. Panel A maps the change in cumulative county-level subsidy payments, and Panels B–D map changes in farm concentration, land values per acre, and deposits per capita.

[Insert Figure 2 Here]

The unconditional correlations with subsidy payments are 51% for farm concentration, −6% for land values per acre, and 12% for deposits per capita. The geographic correlation is strongest for farm concentration, motivating a closer examination of borrower consolidation relative to the alternative channels. Section 7.1 studies each channel formally, testing whether payments change land values, deposits, investment, or production, and shows that farm concentration is the only outcome that increases with subsidy payments. To separate farm consolidation from the other channels, Section 7.2 tests whether subsidy payments affect local banking markets in counties where they have a limited effect on farm concentration.

There are two main challenges with studying the effect of borrower consolidation on the structure of local banking markets. First, borrower consolidation and bank consolidation may be jointly driven by local characteristics. For example, rural areas with limited economic activity may have fewer banks and fewer, but larger, farms because of low population density or substantial land availability. Second, bank consolidation may drive borrower consolidation. In areas with more concentrated banking markets, limited competition may restrict capital access for smaller farms. These farms may be forced to exit or sell to larger operators, accelerating consolidation. Alternatively, a more consolidated banking sector may prefer lending to larger farms because it is less suited to lending to informationally opaque borrowers (Berger and Udell, 2002). In both scenarios, the structure of local banking markets could cause borrower consolidation.

To address these concerns, I use predicted subsidy payments as a shift-share-like instrument for borrower consolidation. Because predicted payments depend on national subsidy allocations and historically determined base acres, they are less likely to reflect local factors that jointly drive farm and bank consolidation. Moreover, changes in local banking structure cannot determine the historical plantings or national subsidy allocations underlying predicted payments, mitigating concerns about reverse causality. Last, the exclusion restriction requires that subsidy payments affect local banking markets through farm consolidation and not through the alternative channels discussed above. As mentioned above, Sections 7.1 and 7.2 test the alternative channels and whether subsidy payments affect local banking markets in the absence of substantial farm consolidation.

The first stage is given by the difference-in-differences specification above with $Y_{i,t}$ equal to farm concentration, measured as the midpoint farm size described in Section 5. As shown in the following sections, the first-stage $F$-statistics exceed the Stock and Yogo (2005) thresholds for weak instruments, supporting the relevance condition. The strength of the first stage supports the interpretation that the policy alters the structure of farms. In the baseline specification, I lag farm concentration by two years to allow changes in farm structure to precede changes in local banking structure. The results are similar with three- and four-year lags (Table A.2).

The second-stage estimates measure how exogenous changes in borrower structure affect the organization of local banking markets. The baseline second-stage specification is:

$$Y_{i,t} = \beta_2 \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}, \tag{2}$$

where subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The outcome variable $Y_{i,t}$ represents various measures of the structure of local banking markets at the county level, including the bank deposits $\text{HHI}_{i,t}$. All variables

are measured in five-year intervals from 1994 to 2014. The vector of control variables $\mathbf{X}_{i,t-2}$ includes the lagged economic variables, log income per capita$_{i,t-2}$ and employment rate$_{i,t-2}$, aligned with the timing of the farm data in the first stage. The land availability controls from the first stage also enter every specification that instruments farm concentration or has an outcome from the Census of Agriculture.

Because farm consolidation should affect small and large banks differently, I construct separate county-level measures of local banking outcomes for each group. For these outcomes, I use a Poisson control function approach (Cohn, Liu, and Wardlaw, 2022). About 22% of county-year observations in my sample do not contain a large bank, so this method addresses the high incidence of zeros in the data. Following Wooldridge (2015), I estimate the first stage via OLS, obtain the residual, and include it as a correction term in the Poisson model:

$$E\left[Y_{i,t} \,\middle|\, \textit{Farm Concentration}_{i,t-2},\ \hat{\varepsilon}_{i,t-2},\ \mathbf{X}'_{i,t-2},\ \gamma_i,\ \delta_t,\ \phi_{d(i,t)}\right] = \exp\left(\beta_2\ \textit{Farm Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\,\hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\right), \quad (3)$$

where subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The vector of control variables $\mathbf{X}_{i,t-2}$ includes the same control variables as in Equation 2, with the addition of the log number of branches$_{i,t-2}$ to compare counties with similar pre-existing banking infrastructure. For deposits per capita, the first stage also includes log population$_{i,t-2}$ to adjust for the per capita scaling of the outcome. Because the model is estimated in two stages, I bootstrap the full procedure 500 times, clustering standard errors at the county level to preserve the panel structure. All analyses that separate large and small banks follow this estimation strategy, with the exception of linear probability models. For ease of interpretation, coefficients in these tables are reported as decimal percentages.[11]

# 7 Empirical Results

In this section, I show that subsidy payments increase local banking concentration and consider several channels that could produce this result. Subsidy payments do not increase local land values, total deposits, or farm production but substantially increase farm concentration. Using subsidy payments as an instrument for farm concentration, I show that increased farm concentration leads large banks to enter and increases large banks' share of branches and deposits, while small banks merge with one another in response to farm consolidation. These results reflect large banks' increase in local agricultural lending and

[11] All Poisson (PPML) coefficients are reported as implied percentage changes. The relevant tables report $e^{\beta} - 1$ in decimal percentage form instead of the raw Poisson estimate $\beta$. For example, a raw estimate of $\beta = 0.53$ is reported as 0.70, corresponding to a 70% increase in the outcome ($e^{0.53} - 1 = 0.70$).

small banks' limited funding capacity, and the changes in presence and deposits are largest among banks that specialize in agricultural lending. The resulting change in the structure of local banking markets spills over to consumer credit markets, where households face higher borrowing costs and earn lower returns on deposits.

## 7.1 Subsidy Amount, Banking Concentration, and the Candidate Channels

Subsidy payments increase banking concentration. Figure 3 reports dynamic estimates during the first decade of the program using the estimator developed by de Chaisemartin and D'Haultfœuille (2026) to accommodate continuous treatment. Panel A shows that the bank deposits HHI increases by about 380 points. At the sample average level of banking concentration, an increase of this magnitude is nearly twice the 200-point increase that triggers further review of a bank merger for potentially being anticompetitive (U.S. Department of Justice, Antitrust Division, 1995).

[Insert Figure 3 Here]

Subsidy payments may affect banking concentration through several channels, including changes in the composition of borrowers, agricultural land values, and local deposits. First, subsidy payments may help larger farms acquire land from smaller farms, increasing farm concentration. Large banks, with a comparative advantage in serving larger borrowers, may then gain market share. Second, because the payments were predictable and tied to the land, some of their value may be capitalized into agricultural land values, providing farmers with additional collateral to borrow against. Higher collateral values may particularly benefit small banks, which specialize in lending to smaller, bank-dependent borrowers who rely more heavily on real estate collateral (Gupta, Sapriza, and Yankov, 2022). Third, subsidy payments may increase local deposits, providing banks with additional funding to expand lending. Because small banks rely more heavily on local deposits, these inflows may help them retain borrowers and compete with large banks. Panels B through D provide evidence on these three potential channels.

Of these candidate channels, only farm concentration increases with subsidy payments. Panel B shows that farm concentration increases by about 500 acres, or 67% relative to its unconditional mean. Larger farms receive greater payments because they have more base acres (Roberts and Key, 2008). These fixed, predictable payments may support farm expansion by providing a stable cash flow to service additional debt. Larger farms also benefit from economies of scale in commodity crop production (Duffy, 2009; Langemeier, 2013). It

follows that larger farms, with lower production costs and more predictable cash flows, have an advantage in acquiring land that would be relatively less profitable for smaller farms to operate. In 1998, for example, large farms received an average annual subsidy payment of about $30,000, which was enough to cover the annual interest cost of financing about 500 acres in Texas, an area equivalent to about 380 football fields.[12]

The other two candidate channels, land values and deposits per capita, do not change with subsidy payments (Panels C and D). For additional deposits or higher collateral values to affect the structure of local banking markets, the increases generated by subsidy payments would need to be substantial relative to banks' existing funding and farmers' existing collateral.

Even for small banks, subsidy payments are small relative to their deposits. A within-county one-standard-deviation increase in annual payments is $1.15 million. If this entire increase were deposited at a single small bank, it would equal only about 1% of the median small bank's $104 million in total deposits.[13] The implied gains in land values are similarly small. Kirwan (2009) estimates that about a quarter of each subsidy dollar accrues to landowners as higher land rents. If landowners' share of each payment were fully priced into farmland, this estimate implies that payments increase county land values by at most about 1%, an increase that is within the confidence intervals reported in Panel C.[14] However, for a representative farm to finance an expansion from 1,000 to 1,500 acres, the land it already owns would have to appreciate by about 24%, even after a mortgage on the newly owned

[12]Based on Texas average cropland values in 1998 of $701 per acre and the average interest rate on farm loans in Texas of 8%. See Agricultural Land Values (https://downloads.usda.library.cornell.edu/usda-esmis/files/5425k968s/5138jh927/zc77st01m/aglndvalcashrent_Agricultural_Land_Values_-_Final_Estimates__1994-98.pdf).

[13]Median total deposits across banks with less than $1 billion in inflation-adjusted assets and at least one branch in a sample county, computed from the FDIC SOD over the sample years.

[14]A within-county one-standard-deviation increase in payments is about $1.15 million per year. Applying the 25% incidence estimate in Kirwan (2009), landowners collect about $290,000 per year of this flow in higher rents. Capitalized as a perpetuity at 6%, the long-run average 10-year Treasury rate used by the Economic Research Service (Nickerson et al., 2012), the additional rent adds about $4.8 million to county land values, a 0.5–1.0% increase against county farmland worth $0.5–0.9 billion. On a recipient farm's own acres, the fixed payment at the 2002 rate is about $19 per acre at the sample's average corn program yield, and the same share and rate add about $78 per acre, a 3% increase in land value. This calculation likely overstates the increase in land values for the farms driving the expansion because Kirwan and Roberts (2016) finds that rental incidence decreases by 5–15 cents when total operated acreage doubles.

acres covers 85% of their price.[15]

Farm consolidation may increase agricultural credit demand without substantially increasing land values if farms expand by renting land instead of purchasing it. I find that subsidy payments reallocate harvested acreage from midsize farms toward the largest farms, and farms expand more through rented acreage than through owned acreage (Table A.3). This result suggests that larger farms can expand their operations and associated financing needs without a proportional increase in land purchases, which limits the extent to which subsidy payments are capitalized into land values.

I also test farm concentration, land values, and deposits per capita jointly, allowing residuals to be correlated across outcomes within a county. The results show that farm concentration increases with subsidy payments, but land values and deposits per capita do not increase (Table A.4). Last, subsidy payments do not lead to greater agricultural investment or productivity. Crop yields and the share of acreage planted in program crops do not change, and machinery and new tractors decrease (Table A.5). These results indicate that payments primarily increase farm concentration.

## 7.2 Falsification Test: Terrain

If payments affect local banking through channels independent of farm consolidation, such as land values or deposits, local banking markets should also change in counties where subsidy payments do not have a large effect on farm concentration. To isolate the borrower consolidation channel, I use variation in terrain ruggedness. Large agricultural machinery operates less efficiently on rugged land, which reduces the benefits of farm expansion in rugged counties (MacDonald, Korb, and Hoppe, 2013). Table 2 shows that in rugged counties, subsidy payments weakly predict farm consolidation and do not affect the structure of local banking markets. In flat counties, subsidy payments strongly predict farm consolidation and increase large banks' number of branches and deposits per capita, with effects about two to three times as large as those for small banks.

[Insert Table 2 Here]

[15]The representative farm owns 600 of its 1,000 acres (U.S. Department of Agriculture, National Agricultural Statistics Service, 2004), values land at 2002 Corn Belt cropland values of $2,340 per acre (U.S. Department of Agriculture, National Agricultural Statistics Service, 2003), borrows against land up to the 85% statutory maximum under the Farm Credit Act, and uses $350 per acre in operating credit (U.S. Department of Agriculture, Economic Research Service, 2006). Expanding to 1,500 acres under the same tenure share means buying 300 acres and operating 500 more. A mortgage on the newly owned acres covers 85% of their price. The down payment and the operating credit, about $280,000 in total, must be borrowed against the 600 acres the farm already owns, which are worth $1.4 million. Borrowing $280,000 at the 85% cap requires those acres to gain about $330,000 in value, a 24% increase, about seven times the increase implied by applying the 25% incidence estimate in Kirwan (2009).

The difference between flat and rugged counties remains after controlling for population, urbanization, and land values (Table A.6). These results suggest that farm consolidation is the primary channel through which subsidy payments affect local banking markets.

## 7.3 Farm Concentration and Banking Concentration

Large banks tend to serve larger firms, so they may gain market share when larger farms acquire land from smaller farms. Small banks, which specialize more in relationship lending to smaller borrowers, may instead exit, merge, or be acquired (Stein, 2002; Berger et al., 2005). This prediction is consistent with the broader IO idea that changes in the distribution of buyers may affect the organization of suppliers (Snyder, 1996). I use subsidy payments as an instrument for farm concentration to measure how much borrower consolidation increases local banking concentration.

Table 3 reports the effect of subsidy payments on farm concentration and on banking concentration, followed by IV and OLS estimates of the effect of farm concentration on banking concentration. In Panel A, a within-county one-standard-deviation increase in subsidy payments increases farm concentration by about 105–107 acres. For the average corn farmer in my sample, an additional 107 base acres would generate $2,012 in extra fixed cash payments.[16] The same increase in subsidy payments increases the bank deposits HHI by about 64 points.

[Insert Table 3 Here]

Instrumenting for farm concentration with subsidy payments, I find that a within-county one-standard-deviation increase in farm concentration increases the bank deposits HHI by about 275 points, or 9% relative to its unconditional mean (Panel B). An increase of this size is comparable to one out of seven equally sized banks exiting a county.[17] I use these estimates to construct a counterfactual that removes the subsidy-induced component of farm consolidation and measures its contribution to banking concentration. Absent subsidy-induced farm consolidation, local banking concentration would have been about 13% lower

[16] The average historical program yield for corn in my sample is 79 bushels per acre. At the 2002 payment rate of $0.28 per bushel and the program's adjustment factor of 0.85, an additional 107 base acres would generate $79 \times 0.28 \times 107 \times 0.85 = \$2{,}012$.

[17] Assuming deposits are initially evenly distributed across the seven banks and that when one bank exits, its deposits are distributed equally across the six remaining banks. Results are similar when I use the share of harvested acres on farms of at least 1,000 acres or the log number of farms as alternative measures of farm concentration (Table A.7), and in a first-difference specification that addresses concerns about autocorrelated residuals (Table A.8). Price-based measures also indicate greater bank market power. Banks' total interest expense beta (Drechsler, Savov, and Schnabl, 2021) and deposit interest expense beta (Narayanan, Ratnadiwakara, and Strahan, 2025) both decline with farm concentration, indicating that deposits are held at banks that pass less of a change in policy rates through to depositors (Table A.9).

by 2014, which suggests that borrower consolidation is an important determinant of local banking concentration.

Panel B also reports OLS estimates that do not correct for the endogeneity of farm concentration. These coefficients are smaller than the IV estimates, which may be due to downward bias from measurement error in the midpoint. When I instrument the twenty-year change in the midpoint with changes in alternative measures of the farm size distribution, the coefficient is similar to the IV estimate (Table A.10).[18] Following Ishimaru (2024), I decompose the gap into differences in covariate weights, treatment-level weights, and marginal effects to further verify that the gap between the IV and OLS coefficients does not come from differences in how the two estimators weight observations. The marginal effect component, which collects measurement error and unobserved shocks that jointly determine farm structure and local banking markets, constitutes between 95% and 99% of the gap across specifications, including one that controls for land values and deposits (Table A.11).[19]

Last, to test whether banking concentration increases progressively as farms become more concentrated, I follow Saidi and Streitz (2021) and measure changes in the bank deposits HHI across deciles of fitted lagged farm concentration. Figure 4 shows that the bank deposits HHI increases steadily across the deciles, consistent with greater borrower consolidation increasing banking concentration.

[Insert Figure 4 Here]

## 7.4 Farm Concentration and the Structure of Local Banking Markets

Borrower consolidation also alters the competitive dynamics of local banking. Unlike many buyer-supplier relationships, bank-firm relationships involve information asymmetries and monitoring costs that give incumbent lenders relationship-specific advantages. Small community banks have historically relied on relationship lending to serve smaller or more opaque borrowers (Haynes, Ou, and Berney, 1999; Berger et al., 2005). However, as farms consolidate and expand, large banks may become better positioned to serve the local market because they have greater funding capacity, access to wholesale funding and internal capital

[18]The alternative measures come from census years inside the twenty-year window, so they do not share the midpoint's survey error.

[19]The covariate weight component reflects differences in the weights the two estimators place on counties with different observable characteristics, and the treatment-level weight component reflects differences in the weights placed on ranges of farm concentration.

markets, and higher regulatory lending limits.[20]

Table 4 shows that large banks are more likely than small banks to enter the consolidating markets. Following a within-county one-standard-deviation increase in farm concentration, the probability of large bank presence increases by about 5 percentage points, compared with about 2 percentage points for small banks. These results suggest that large banks enter as the composition and scale of local credit demand change.

[Insert Table 4 Here]

Large banks also expand their branch networks and deposits per capita more than small banks. Branch networks show whether banks commit physical resources to a market (Cohen and Mazzeo, 2010), and deposits per capita reflect the extent to which customers move deposits across institutions. Large banks expand their local branch presence by about 16%, compared with 10% for small banks, and increase their deposits per capita by about 33%, compared with 22% for small banks. These results remain similar after excluding the 2008 Financial Crisis and after controlling for land values, total county deposits, and production measures (Tables A.12 and A.13).

In contrast, small banks consolidate among themselves. Table 5 shows that mergers and acquisitions among small banks increase by about 23%. Mergers between two large banks decline, and mergers in which a large bank acquires a small bank do not change. These results suggest that small banks respond to competitive pressure from larger banks and changes in credit demand by merging. The remaining small banks become fewer but larger.

[Insert Table 5 Here]

These findings align with theoretical predictions that structural changes in downstream markets, such as an increase in borrower size, can trigger consolidation among upstream suppliers (Galbraith, 1952; Snyder, 1996).

## 7.5 Mechanisms: Agricultural Lending and Small Bank Constraints

This section studies how farm consolidation affects agricultural lending and why large banks gain market share. First, I study which banks provide additional agricultural credit and whether the increase in lending is concentrated among larger farms and larger loans.

[20]For literature studying large banks' funding capacity and access to wholesale funding, see LaDue and Duncan (1996); DeYoung, Hunter, and Udell (2004); and Park and Pennacchi (2009). For literature studying internal capital markets, see Stein (1997) and Cortés and Strahan (2017). For regulatory lending limits, see Office of the Comptroller of the Currency (2004).

Second, I study the funding and lending constraints that limit small banks' ability to increase agricultural lending. Last, I test whether farm consolidation has larger effects on banks that specialize in agricultural lending.

### 7.5.1 Farm Concentration and Agricultural Lending

If farm consolidation drives changes in local banking, the effects should be strongest in agricultural credit markets, where borrower consolidation is most directly relevant. As farms consolidate, their financing needs increase, allowing large banks to increase agricultural lending and gain local deposit market share.

I first test whether large banks increase lending to larger farms and, if so, in which loan size categories using CRA loan originations. CRA loan data directly measure county-level bank lending to farms and report borrower revenue and loan size.[21] Table 6 shows that a within-county one-standard-deviation increase in farm concentration increases large banks' lending to farms with more than $1 million in annual revenue by about 30%. The increase is concentrated in loans between $250,000 and $500,000, the largest loan size category reported in the CRA data. In contrast, large banks do not increase lending to farms with $1 million or less in annual revenue or in the smallest loan size category of $100,000 or less. These differences suggest that large banks' additional agricultural lending is concentrated among larger borrowers and larger loans, consistent with their specialization in serving larger firms.

[Insert Table 6 Here]

Among the subset of small banks reporting farm loan originations under the CRA through 2004, lending does not increase in any borrower revenue or loan size category (Table A.14).

To study how total agricultural lending changes at both large and small banks, I complement the CRA loan analysis with agricultural lending imputed from bank Call Reports. Because Call Reports provide each bank's agricultural lending only at the bank level, I allocate each bank's lending across counties where it operates branches in proportion to farm interest expenses, which proxy for the geographic distribution of agricultural credit demand for each loan type. Table 7 shows that the increase in agricultural lending is concentrated at large banks. A within-county one-standard-deviation increase in farm concentration increases large banks' total agricultural lending by about 46%. Within total agricultural lending, real estate farm lending increases by about 40% and non-real estate farm lending by about 46%, which suggests that the increase in large banks' agricultural lending is not due to an advantage in valuing real estate collateral. Small banks' total agricultural lending

[21] CRA reporting requirements apply only to banks above an asset threshold. Through 2004, the threshold was $250 million in bank assets or $1 billion in holding company assets. Beginning in 2005, it increased to about $1 billion, and most small banks stopped reporting, so I restrict the main analysis to large banks.

and real estate farm lending do not increase. Their non-real estate farm lending increases by about 6%, about one eighth of the increase at large banks.

[Insert Table 7 Here]

The results are robust to including counties where banks originate CRA farm loans outside their branch networks, with the allocation adjusted for distance (Table A.15). These results suggest that large banks gain local deposit market share in part by providing more agricultural credit as farms consolidate.

### 7.5.2 Small Bank Funding Constraints

Why does agricultural lending increase at large banks but not at small banks? Small banks fund their lending mainly from local deposits, whereas large banks can finance larger loans by moving funds across branches through their internal capital markets (Gilje, Loutskina, and Strahan, 2016; Cortés and Strahan, 2017).[22] For small banks that rely primarily on local deposits, financing consolidating farms may require a large share of the funds available for new lending. Using the same farm example as in Section 7.1, an expansion from 1,000 to 1,500 acres requires about $880,000 in additional credit, about 40% of the median small bank's annual change in core deposits.[23] Financing several such expansions could require a substantial share of a small bank's annual deposit growth. When deposit growth is limited, the composition of a bank's existing funding may determine how much it can lend. Core deposits are a stable source of funding, so banks with a larger share of core deposits may be better able to increase lending when farms expand (Berlin and Mester, 1999; Cornett et al., 2011).

To test whether small banks with more stable funding are better able to increase lending as farms consolidate, I split small banks according to their core deposit share before the FAIR Act. Because only a subset of small banks reports under the CRA, I use the agricultural lending imputed from bank Call Reports, as in Table 7. Table 8 shows that farm concentration increases agricultural loans above $500,000 at small banks with a larger share of core deposits, but not at small banks with a smaller share. Loans of $100,000 or less do not increase in either group. The additional lending at small banks with a larger share of core deposits is concentrated in larger loans, which suggests that it does not reflect a general increase in agricultural lending. Farm concentration also increases both real estate and

[22] Only 278, or about 0.2%, of 116,057 syndicated loans from 1994 to 2014 are to U.S. crop farms (DealScan, SIC 01).

[23] The median annual change in core deposits across all small-bank-years from 1997 to 2007, including years in which core deposits declined, is $2.3 million, and core deposits increased by less than $880,000 or declined in 37% of bank-years.

non-real estate farm lending at these banks, indicating that their greater lending activity is not due to an advantage in lending against real estate collateral. These results suggest that stable deposit funding may help small banks compete for larger agricultural loans, even though large banks gain market share overall.

[Insert Table 8 Here]

Small banks' ability to increase agricultural lending may also depend on how much of that funding is already committed to existing loans. Small banks with both a larger share of core deposits and unused lending capacity, a pre-program ratio of loans to deposits below the median, show the largest increases in agricultural loans above $500,000 (Table A.16).

Lending capacity constraints may drive banks' differential responses to farm consolidation. In counties where production requires more fixed credit per acre, small banks may face binding funding constraints at smaller farm sizes, allowing large banks to gain more market share in counties with higher production costs. I classify counties as high- or low-cost by the average economic cost per acre of their crops before the program, weighted by harvested acres (Figure A.2). Large banks increase deposits per capita in high-cost counties but not in low-cost counties, whereas small banks increase deposits per capita about three times more in low-cost counties than in high-cost counties (Table A.17). Land values appreciate by similar amounts in the two groups (Figure A.3). Large banks gain more market share in counties where production requires more credit per acre, consistent with small banks' funding constraints limiting their ability to finance larger farms.

A related constraint comes from regulatory limits on lending to a single borrower. The Office of the Comptroller of the Currency (OCC) restricts these loans to 15% of a bank's capital and surplus. In 2004, the OCC noted that farm consolidation had "resulted in fewer, but larger, farms with expanded credit needs" and that existing limits were too restrictive for community banks (Office of the Comptroller of the Currency, 2004). Over time, higher land values and production costs further increased the credit that expansion requires. For example, a loan that financed about 290 acres of central Illinois farmland in 2001 financed about 110 acres in 2012 (Ellinger, 2012).

### 7.5.3 The Structure of Agricultural Banking

The prior results are consistent with IO theory and suggest that relationship lending does not limit banks' ability to reorganize, but this finding does not necessarily imply that informational frictions are irrelevant. Agricultural lending becomes more concentrated as farms consolidate. Table 9 shows that a within-county one-standard-deviation increase in farm concentration increases the concentration of agricultural lending by 14%, nearly twice the increase in deposit market concentration.

[Insert Table 9 Here]

For the remaining outcomes, I restrict the sample to agricultural banks, defined as banks whose farm loans exceed 5% of total loans in any of the preceding six years.[24] Large agricultural banks are 15 percentage points more likely to be present, more than three times the increase for large banks overall. Their deposits per capita increase by about 75%, compared with about 24% at small agricultural banks. Large agricultural banks also increase their number of branches by about 55%, but small agricultural banks do not. These results suggest that both bank size and agricultural expertise affect banks' adjustment to borrower consolidation.

To quantify the reallocation of market share, I compare the observed split of subsidy-induced deposit growth between large and small agricultural banks with a counterfactual split in proportion to each group's deposit share before the FAIR Act. In counties where large agricultural banks operate, subsidy-induced deposit growth is about 49% at large agricultural banks and about 7% at small agricultural banks. Had the same total growth been divided in proportion to the two groups' initial shares, small agricultural banks' deposit growth would have been about 66 percentage points higher, and small agricultural banks' deposits about $105 million higher per county.

## 7.6 Farm Concentration and Interest Rates on Broader Consumer Products

IO theory suggests that buyer consolidation can influence prices through two distinct channels. First, consolidated buyers may gain bargaining power over suppliers, negotiating more favorable prices and pushing suppliers to compete on price for all buyers in the market (Snyder, 1996; Chipty and Snyder, 1999). Second, consolidation may trigger restructuring on the supplier side, leading to concentration among the most efficient suppliers, which can result in unchanged or even higher prices (Galbraith, 1952; Tirole, 1988; Farrell and Shapiro, 1990). Kropp and Whitaker (2011) finds that in agricultural credit markets, farms with a greater share of base acres, and thus larger subsidy payments relative to similarly sized farms, receive slightly lower interest rates on their operating loans.

However, the broader effects of borrower consolidation and the structure of local banking markets on consumer financial products are less clear. New banks entering consolidated markets may lower credit prices to attract customers. Banks may instead increase credit prices if they can segment consumer markets from consolidated borrowers (Inderst and Valletti,

[24] The results are monotonic with respect to the 5% threshold. As the share of agricultural lending decreases below 5%, the estimated effects of farm concentration on large agricultural banks' presence and deposits are smaller (Table A.18).

2011). Any pricing effects may also differ between large and small banks. Large banks tend to offer lower deposit rates because they have access to wholesale funding, whereas small banks may offer higher deposit rates to attract local retail deposits and fund their lending activities (Park and Pennacchi, 2009).

To assess these spillover effects, I study how borrower consolidation affects the pricing of consumer deposit and loan products. Among deposit products, I focus on three commonly offered types: time, savings, and checking deposits. These deposit types differ in liquidity, in interest elasticity, and in their role in bank funding strategies. Liquid deposits, such as savings and checking deposits, provide a stable and relatively inelastic source of funds. Because depositors rarely withdraw these funds in response to rate changes, banks can maintain low-cost funding even as they reduce offered interest rates. In contrast, time deposits are more sensitive to interest rates and are often used to fund commercial lending, including agricultural loans (Drechsler, Savov, and Schnabl, 2017; Supera, 2021). I use real, inflation-adjusted interest rates in all analyses. In addition to the baseline farmland and local economic condition controls, I control for changes in local household and farm credit demand by including county-level measures of total deposits and total farm interest expenses.

Panel A of Table 10 shows that a within-county one-standard-deviation increase in farm concentration decreases the county-level average savings deposit interest rate by 32 basis points and the checking deposit interest rate by about 22 basis points, or about 26% and 14% relative to their unconditional means. The highest savings deposit interest rate offered by any bank branch in the county also decreases, by more than the lowest rate offered (Table A.19). County-level time deposit interest rates increase by about 13 basis points, consistent with banks offering higher rates on deposits that are more sensitive to interest rates to preserve their funding base.

[Insert Table 10 Here]

I also analyze offered interest rates on auto loans, which are standardized, broadly held products unrelated to farm lending, and subject primarily to local competition (Argyle, Nadauld, and Palmer, 2023). The county-level average auto loan interest rate increases by about 103 basis points.[25] The lowest and highest auto loan interest rates offered by any bank branch in the county also increase, so households that compare rates across the county's branches still face higher auto loan interest rates (Table A.19). The result is similar in first differences (Table A.20). Mortgage interest rates should be less sensitive to local

[25]This magnitude is within the range of price differences that search frictions sustain in auto lending, where borrowers on opposite sides of a credit score threshold at the same lender face rate differences of about 130 basis points (Argyle, Nadauld, and Palmer, 2023) and dealer-arranged loans carry discretionary markups averaging about 113 basis points (Grunewald et al., 2023).

banking conditions because mortgages are commonly sold into national secondary markets and priced using national benchmarks (Fuster, Lo, and Willen, 2024). Consistent with this prediction, the county-level average mortgage interest rate does not increase.

Small and large banks price consumer products differently as farms consolidate. Panels B and C report the effects separately for small and large banks, comparing interest rates offered by the same bank in the same county over time. Auto loan interest rates increase by about 97 basis points at large banks but decrease by about 37 basis points at small banks. Small banks also increase time deposit interest rates by about 30 basis points, compared with a decrease of about 18 basis points at large banks. Interest rates on savings and checking deposits decrease at both large and small banks. Mortgage interest rates increase by only about 11 basis points at large banks, about 4% relative to the unconditional mean, with no corresponding increase at small banks. The small magnitude is consistent with mortgage rates being less sensitive to local banking conditions. These results suggest that small banks compete by increasing time deposit interest rates to secure local funding and by lowering auto loan interest rates to attract consumer borrowers, while offering lower rates on deposits that are less sensitive to interest rate changes. Large banks, on the other hand, reduce deposit interest rates and increase auto loan interest rates, consistent with greater pricing power in more concentrated local banking markets.

## 7.7 Robustness

This section tests whether the subsidy program changed government lending and controls for greater crop insurance take-up, other government payments, and the number of Farm Credit System branches, which provide agricultural credit outside the commercial banking system and compete with banks for farm borrowers.

### 7.7.1 Entry and Credit Demand

Although my main results show that fixed subsidy payments increase farm consolidation, one alternative mechanism is that these payments decrease income risk and encourage new entry into farming. Subsidy payments may also simply increase credit demand more broadly instead of specifically among large, consolidating operations. If either of these mechanisms were at work, federal direct loans, which are targeted to small, new, or socially disadvantaged farmers, should increase. However, I find no evidence that subsidy-induced consolidation affects the number or volume of direct loans at the county-year level (Table A.21). These results support the interpretation that subsidy payments finance the expansion of existing farms and not entry by smaller farms, which would decrease the midpoint farm size. The absence of an increase in direct loans suggests that the consolidation does not reflect broader

access to credit.

### 7.7.2 Crop Insurance

Crop insurance decreases downside risk, improves credit access, and supports farm expansion (Cornaggia, 2013; Azzam, Walters, and Kaus, 2021; Ifft et al., 2024), so greater take-up may affect these results. Controlling for the log number of county-level crop insurance policies leaves the effects of payments on farm concentration and of farm concentration on banks largely unchanged (Table A.22).

### 7.7.3 Countercyclical Payments

The 2002 FSRI Act included countercyclical payments, which were paid on base acres when crop prices were below thresholds set by the FSRI Act (Monke, 2006). To address the possibility that these payments also affected local banking markets, I control for predicted countercyclical payments, constructed using the same approach as predicted subsidy payments. The main results remain similar (Table A.23).

### 7.7.4 The Farm Credit System (FCS)

Nonbank lenders may also affect the structure of local commercial banking. In agricultural credit markets, the FCS is the main alternative to commercial banks. Established in 1916 as a government-sponsored enterprise, the FCS is a borrower-owned cooperative system designed to provide credit to farmers and ranchers who historically faced limited access to commercial bank financing. FCS institutions may lend only to farmers, and they are prohibited from offering nonfarm loans.

Because the FCS competes with commercial banks for agricultural borrowers, counties with more FCS branches may experience different levels of commercial banking concentration, deposit holdings, or branching activity. To address this possibility, I control for the log number of FCS branches in each county. The effects of farm consolidation on commercial bank outcomes remain similar (Table A.24).

# 8 Conclusion

In this paper, I introduce a new driver of bank consolidation: the consolidation of borrowers. Using U.S. agriculture as a setting, I show that farm consolidation, driven by fixed federal subsidies, increases local banking concentration. As farms consolidate, large banks increase their presence, branch networks, deposits, and agricultural lending, while small banks consolidate through mergers. Large banks gain deposit market share particularly where farms require more capital, and small banks increase their agricultural lending above

$500,000 only when they have a larger share of core deposits. Changes in bank composition also affect the local community. Following an increase in borrower concentration, large banks reduce deposit interest rates and increase auto loan interest rates. In contrast, small banks increase time deposit rates and lower auto loan rates, potentially to attract retail customers and deposits to fund lending.

The mechanism should reverse when borrowers' incomes decrease. Large banks enter these markets to finance larger loans and fund them through internal capital markets, so their lending in a county does not depend on deposits raised there. When farm incomes decreased in the decade after the program ended, several of the largest banks decreased their farm lending.[26] While borrower consolidation may attract large banks in the short run, it may also decrease long-run financial stability.

More broadly, the effect on banking should be strongest where firms depend heavily on bank credit, so changes in firm size alter the distribution of credit demand across banks. Where firms rely less on bank credit or credit is supplied through national or securitized markets, the same mechanism should be weaker, consistent with the absence of an effect of borrower consolidation on county-level average mortgage interest rates. Firms in construction, trucking, and manufacturing rely on bank debt to finance investment, and private equity has consolidated physician, dental, and veterinary practices. As borrowers in these industries become larger, large banks should gain market share. These findings show that consolidation in the real economy alters local banking markets and the pricing of consumer financial products.

[26]See Rabobank takes on US agriculture's lending gap as large banks exit (https://www.bankingdive.com/news/rabobank-us-agriculture-lending-gap/559142/), Big banks exiting farm loan business (https://www.proag.com/news/big-banks-exiting-farm-loan-business/), and Wall Street banks bailing on troubled U.S. farm sector (https://www.reuters.com/article/business/wall-street-banks-bailing-on-troubled-us-farm-sector-idUSKCN1U618E/).

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**Figure 1.** Timeline of U.S. Farm Policy Changes from 1996 to 2014

This figure illustrates the timeline of base acre reference periods and key policy events under successive U.S. farm bills. The 1996 Federal Agriculture Improvement and Reform (FAIR) Act introduced fixed direct payments based on average planted acres from 1981 to 1985. The 2002 Farm Security and Rural Investment (FSRI) Act expanded the program to include oilseeds and peanuts and allowed base acres to be updated to reflect average plantings from 1998 to 2001. The program was renewed in the 2008 Food, Conservation, and Energy Act and ended in the 2014 Agricultural Act.

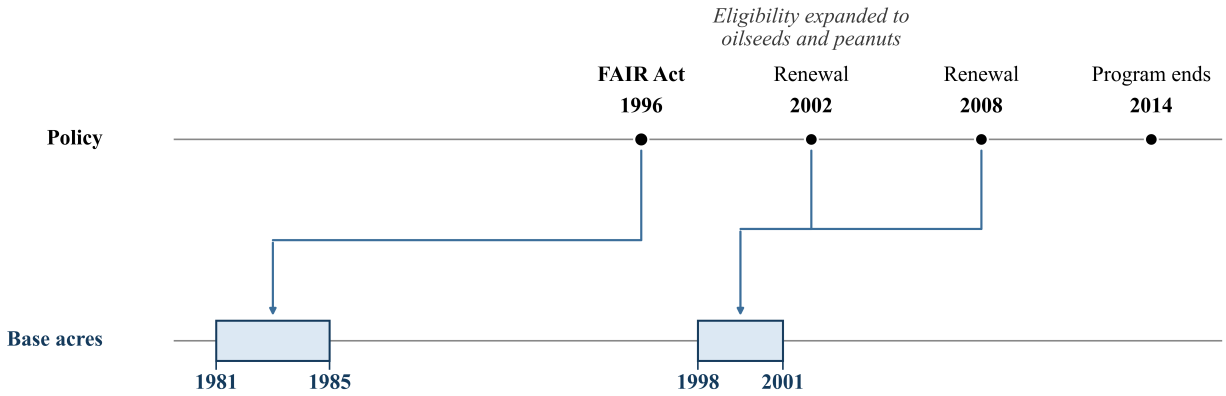
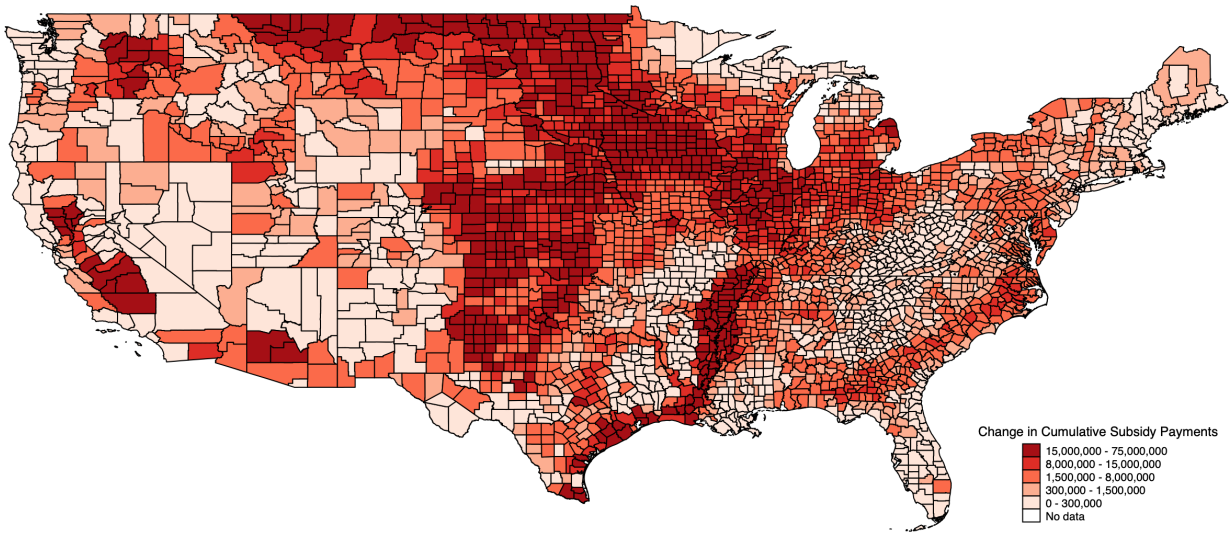


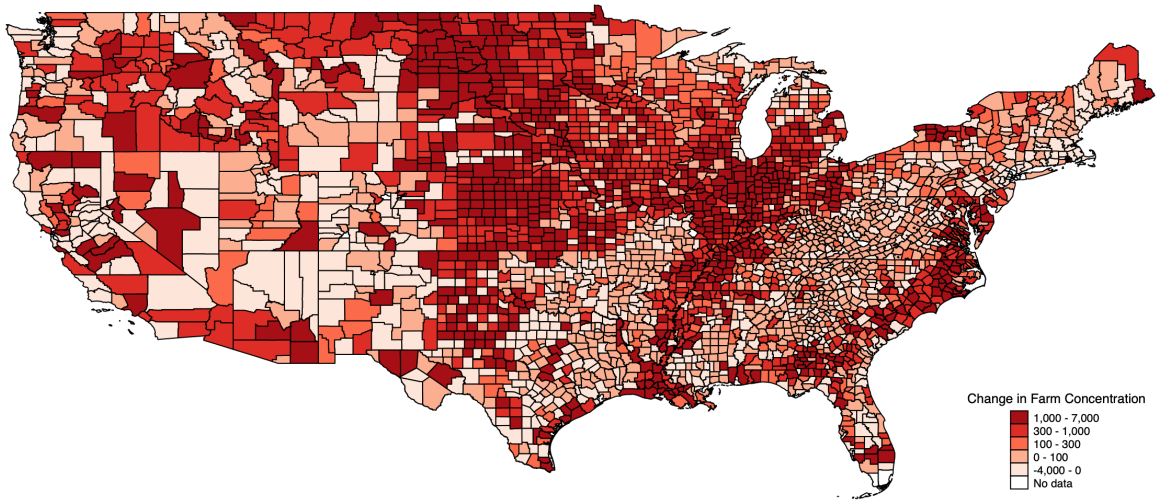
**Figure 2.** Changes in County-Level Subsidy Payments, Farm Concentration, Land Values, and Deposits

This figure maps the change in subsidy payments and in the three candidate channels across counties. Panel A shows the change in cumulative subsidy payments per county between 1992 and 2012. Panel B shows the change in farm concentration between 1992 and 2012. Panel C shows the change in land values per acre between 1992 and 2012. Panel D shows the change in total deposits per capita between 1994 and 2014.

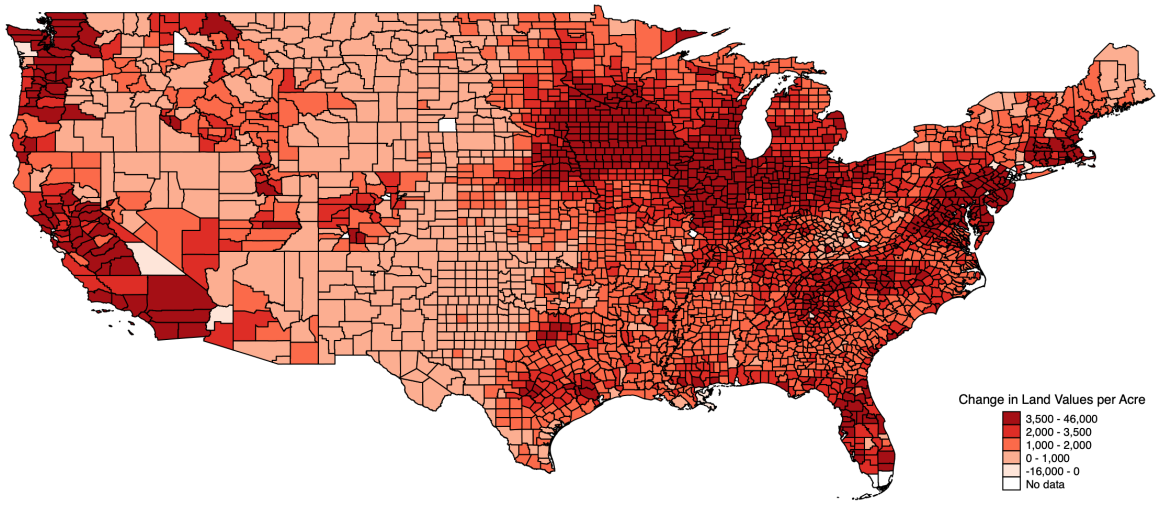
**Panel A.** Change in Cumulative Subsidy Payments between 1992 and 2012




**Panel B.** Change in Farm Concentration between 1992 and 2012



**Panel C.** Change in Land Values per Acre between 1992 and 2012




**Panel D.** Change in Total Deposits per Capita between 1994 and 2014

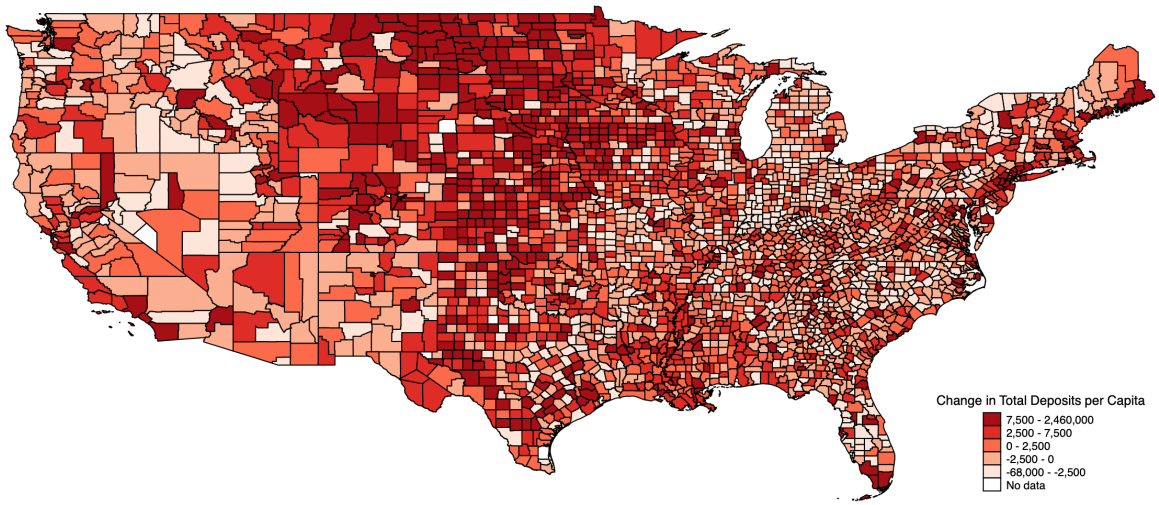
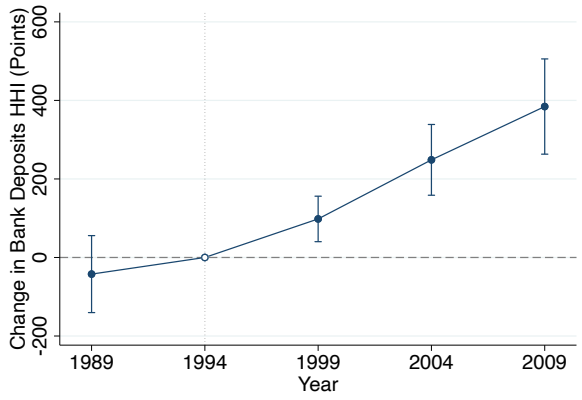


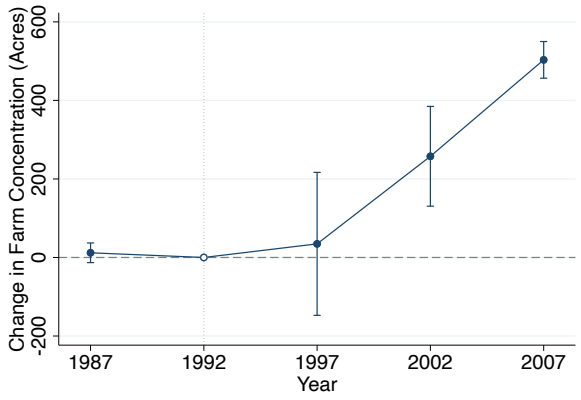
**Figure 3.** Effect of Subsidy Amount on Banking Concentration and Candidate Channels

This figure plots the dynamic effect of Subsidy Amount on the bank deposits HHI, farm concentration, land values, and deposits per capita, with 95% confidence intervals. Estimates are obtained using the difference-in-differences estimator developed by de Chaisemartin and D'Haultfœuille (2026), which accommodates continuous treatments. The specification supplied to the estimator is: $Y_{i,t} = \alpha\, \textit{Subsidy Amount}_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$, and each panel plots the dynamic coefficients for the periods relative to the change in $\textit{Subsidy Amount}_{i,t}$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. Panels A–D report the bank deposits HHI, farm concentration, the log market value of agricultural land and buildings per acre, and total county deposits per capita, respectively. The bank deposits HHI is a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). Panels A and D use banking years 1994–2009, and Panels B and C use Census of Agriculture years 1992–2007. Subsidy Amount is a within-county standardized measure of cumulative total fixed subsidy payments. In Panels A and D, the controls are log income per capita, the employment rate, and indicators for state-level banking deregulation, and Panel D additionally includes the log number of bank branches in the county. In Panels B and C, the controls are log income per capita, the employment rate, and lagged measures of idle cropland, pastureland, and cropland left fallow, lost to crop failure, or otherwise unharvested from the prior Census of Agriculture. All panels allow nonparametric trends across state-level banking deregulation regimes.

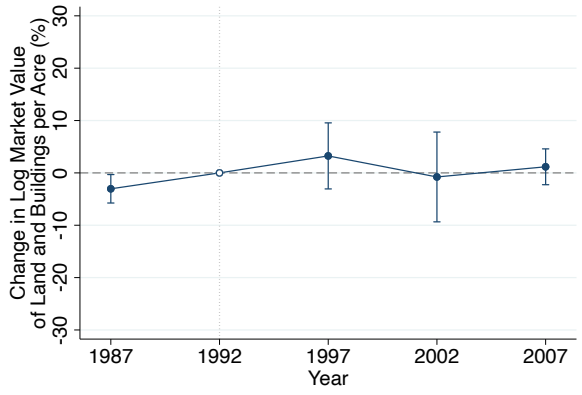
**Panel A.** Bank Deposits HHI




**Panel B.** Farm Concentration




**Panel C.** Log Land Value Per Acre




**Panel D.** Total Deposits per Capita

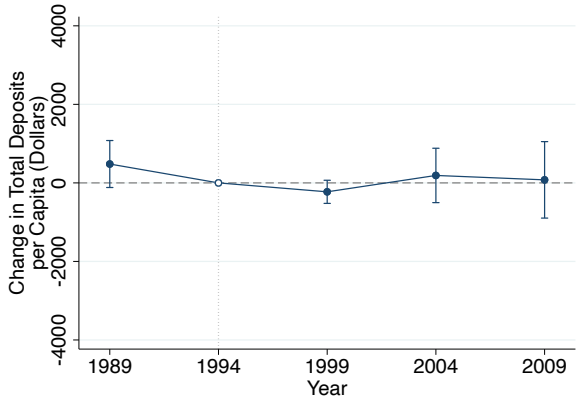


**Figure 4.** Effect of Farm Concentration on Banking Concentration

This figure illustrates the effect of fitted Farm Concentration$_{i,t-2}$ on the Bank Deposits Herfindahl-Hirschman Index$_{i,t}$ (HHI), following Saidi and Streitz (2021). The plotted coefficients are second-stage coefficients on deciles of fitted Farm Concentration$_{i,t-2}$, with 95% confidence intervals. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \sum_{k=2}^{10} \beta_k\, \mathbf{1}\left(Farm\ Concentration_{i,t-2} \in D_k\right) + \beta_{11}\hat{\varepsilon}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$, where $D_k$ denotes the $k$-th decile of fitted farm concentration and the first decile is the omitted category. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). The outcome variable, Bank Deposits HHI$_{i,t}$, is a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. In both stages, the controls are log income per capita, the employment rate, indicators for state-level banking deregulation, and lagged measures of idle cropland, pastureland, and cropland left fallow, lost to crop failure, or otherwise unharvested from the prior Census of Agriculture.

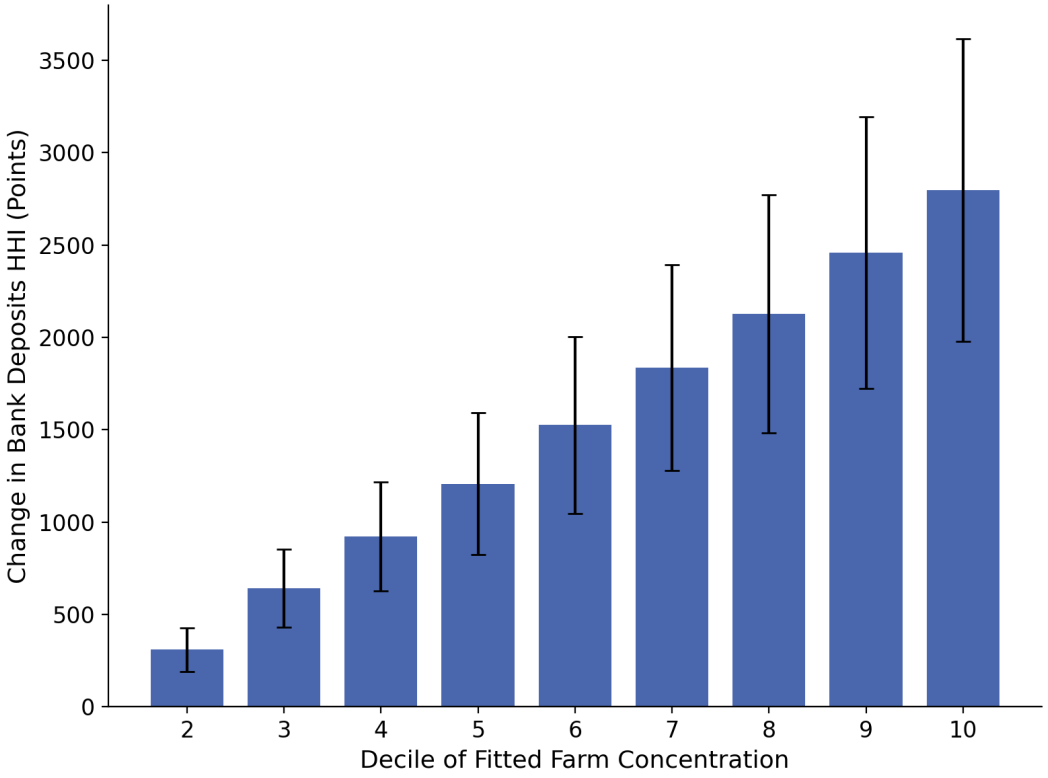


**Table 1.** Descriptive Statistics

The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). Subscript $i$ represents the county and $t$ represents the year. $Farm\ Concentration_{i,t-2}$ is the midpoint farm size at the county level. $Bank\ Deposits\ HHI_{i,t}$ is the Herfindahl-Hirschman Index, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. $Agricultural\ Loan\ HHI_{i,t}$ is calculated using the same method based on agricultural loan shares held by each bank within a county. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. $Probability\ of\ Large\ Bank\ Entry_{i,t}$ and $Probability\ of\ Small\ Bank\ Entry_{i,t}$ are indicator variables for whether a large or small bank, respectively, is present in a county in a given year. $Number\ of\ Large\ Bank\ Branches_{i,t}$ and $Number\ of\ Small\ Bank\ Branches_{i,t}$ refer to the number of branches operated by large and small banks at the county level, respectively. $Large\ Banks'\ Deposits\ Per\ Capita_{i,t}$ and $Small\ Banks'\ Deposits\ Per\ Capita_{i,t}$ measure total deposits per capita held at large and small banks, respectively. $Large\ Acquires\ Small_{i,t}$ measures the total number of mergers or acquisitions in which a large bank acquires a small bank, aggregated over five-year intervals. $Small\ Acquires\ Small_{i,t}$ and $Large\ Acquires\ Large_{i,t}$ measure the number of mergers between two small or two large banks, respectively, also aggregated over five-year intervals. $Total\ Agricultural\ Loans_{i,t}$ is the total imputed agricultural loan volume at the county level, disaggregated by bank size. $Real\ Estate\ Farm\ Loans_{i,t}$ and $Non\text{-}Real\ Estate\ Farm\ Loans_{i,t}$ represent county-level imputed loan volumes secured by farmland and used to finance agricultural production, respectively. $Total\ Agricultural\ Loan\ Growth_{i,t}$, $Real\ Estate\ Loan\ Growth_{i,t}$, and $Non\text{-}Real\ Estate\ Loan\ Growth_{i,t}$ are large or small banks' lending in each loan type relative to the prior period's total volume of that loan type. $Direct\ Loan\ Amount_{i,t}$ and $Number\ of\ Direct\ Loans_{i,t}$ are the dollar volume, in thousands, and the number of direct farm ownership and operating loans the Farm Service Agency issued in the county-year. $Probability\ of\ Large\ Agricultural\ Bank\ Entry_{i,t}$ and $Probability\ of\ Small\ Agricultural\ Bank\ Entry_{i,t}$ are indicator variables for whether a large or small agricultural bank, respectively, is present in a county in a given year, where agricultural banks are those whose farm loans exceed 5% of total loans in any of the preceding six years. $Number\ of\ Large\ Agricultural\ Bank\ Branches_{i,t}$ and $Number\ of\ Small\ Agricultural\ Bank\ Branches_{i,t}$ are the number of branches operated by large and small agricultural banks, respectively, and $Large\ Agricultural\ Banks'\ Deposits\ Per\ Capita_{i,t}$ and $Small\ Agricultural\ Banks'\ Deposits\ Per\ Capita_{i,t}$ are total deposits per capita held at large and small agricultural banks, respectively. $Savings\ Deposit\ Rate_{i,t}$ is the real interest rate offered on $10,000 money market accounts, and $Time\ Deposit\ Rate_{i,t}$ is the real interest rate offered on 12-month $10,000 certificates of deposit. $Checking\ Deposit\ Rate_{i,t}$ is the real interest rate offered on interest-bearing checking accounts. $Auto\ Loan\ Interest\ Rate_{i,t}$ is the real interest rate offered on auto loans, and $Mortgage\ Interest\ Rate_{i,t}$ is the real interest rate offered on residential mortgages. The instrument, $Subsidy\ Amount_{i,t}$, is a continuous measure of the predicted value of total fixed subsidy payments in a given county-year. The control variables for county-level land availability are idle cropland, pastureland, and fallow or unharvested cropland from the prior Census of Agriculture. The control variables for local economic conditions and pre-existing banking infrastructure are the county-level employment rate, log income per capita, and number of branches, all measured two years before the outcome. $Population_{i,t-2}$ is the county population from the Bureau of Economic Analysis, and $State\text{-}Level\ Deregulation\ Index_{i,t}$ is the interstate branching deregulation index of Célérier and Matray (2019).

| | Mean | SD | P10 | P50 | P90 | N |
|---|---|---|---|---|---|---|
| *Main Dependent Variables* | | | | | | |
| Farm Concentration$_{i,t-2}$ (Acres) | 752 | 833 | 65 | 327 | 2,072 | 13,808 |
| Bank Deposits HHI$_{i,t}$ | 3,123 | 2,011 | 1,308 | 2,540 | 5,474 | 13,808 |
| Agricultural Loan HHI$_{i,t}$ | 4,472 | 2,661 | 1,720 | 3,785 | 10,000 | 13,808 |
| Probability of Large Bank Entry$_{i,t}$ | 0.78 | 0.42 | 0.00 | 1.00 | 1.00 | 13,808 |
| Probability of Small Bank Entry$_{i,t}$ | 0.96 | 0.20 | 1.00 | 1.00 | 1.00 | 13,808 |
| Number of Large Bank Branches$_{i,t}$ | 13 | 29 | 0 | 3 | 33 | 13,808 |
| Large Banks' Deposits Per Capita$_{i,t}$ ($) | 7,498 | 24,887 | 0 | 5,684 | 15,437 | 13,808 |
| Number of Small Bank Branches$_{i,t}$ | 9 | 10 | 1 | 7 | 19 | 13,808 |
| Small Banks' Deposits Per Capita$_{i,t}$ ($) | 11,814 | 9,679 | 1,552 | 9,836 | 24,462 | 13,808 |
| Large Acquires Small$_{i,t}$ | 0.35 | 0.78 | 0.00 | 0.00 | 1.00 | 13,808 |
| Small Acquires Small$_{i,t}$ | 0.43 | 0.84 | 0.00 | 0.00 | 1.00 | 13,808 |
| Large Acquires Large$_{i,t}$ | 0.67 | 1.21 | 0.00 | 0.00 | 2.00 | 13,808 |
| Large Banks' Total Agricultural Loans$_{i,t}$ ($1,000s) | 12,900 | 32,656 | 0 | 1,534 | 35,074 | 13,808 |
| Small Banks' Total Agricultural Loans$_{i,t}$ ($1,000s) | 28,027 | 40,403 | 1 | 11,944 | 77,869 | 13,808 |
| Large Banks' Real Estate Farm Loans$_{i,t}$ ($1,000s) | 6,089 | 15,127 | 0 | 730 | 16,667 | 13,808 |
| Small Banks' Real Estate Farm Loans$_{i,t}$ ($1,000s) | 13,491 | 19,336 | 0 | 5,973 | 36,340 | 13,808 |
| Large Banks' Non-Real Estate Farm Loans$_{i,t}$ ($1,000s) | 6,632 | 18,965 | 0 | 400 | 17,198 | 13,808 |
| Small Banks' Non-Real Estate Farm Loans$_{i,t}$ ($1,000s) | 14,321 | 23,329 | 0 | 4,265 | 43,186 | 13,808 |
| Large Banks' Total Agricultural Loan Growth$_{i,t}$ | 0.39 | 0.56 | 0.00 | 0.16 | 1.04 | 13,808 |
| Small Banks' Total Agricultural Loan Growth$_{i,t}$ | 0.70 | 0.57 | 0.00 | 0.66 | 1.33 | 13,808 |
| Large Banks' Real Estate Loan Growth$_{i,t}$ | 0.41 | 0.62 | 0.00 | 0.14 | 1.12 | 13,808 |
| Small Banks' Real Estate Loan Growth$_{i,t}$ | 0.84 | 0.80 | 0.00 | 0.73 | 1.62 | 13,808 |
| Large Banks' Non-Real Estate Loan Growth$_{i,t}$ | 0.39 | 0.62 | 0.00 | 0.14 | 1.04 | 13,808 |
| Small Banks' Non-Real Estate Loan Growth$_{i,t}$ | 0.62 | 0.60 | 0.00 | 0.55 | 1.25 | 13,808 |
| Direct Loan Amount$_{i,t}$ ($1,000s) | 431 | 890 | 0 | 89 | 1,258 | 13,808 |
| Number of Direct Loans$_{i,t}$ | 12 | 24 | 0 | 3 | 35 | 13,808 |
| *Agricultural Banks' Dependent Variables* | | | | | | |
| Probability of Large Agricultural Bank Entry$_{i,t}$ | 0.32 | 0.47 | 0.00 | 0.00 | 1.00 | 13,808 |
| Probability of Small Agricultural Bank Entry$_{i,t}$ | 0.77 | 0.42 | 0.00 | 1.00 | 1.00 | 13,808 |
| Number of Large Agricultural Bank Branches$_{i,t}$ | 1.32 | 3.94 | 0.00 | 0.00 | 4.00 | 13,808 |
| Large Agricultural Banks' Deposits Per Capita$_{i,t}$ ($) | 1,431 | 3,519 | 0 | 0 | 4,951 | 13,808 |
| Number of Small Agricultural Bank Branches$_{i,t}$ | 4.56 | 4.96 | 0.00 | 3.00 | 11.00 | 13,808 |
| Small Agricultural Banks' Deposits Per Capita$_{i,t}$ ($) | 8,803 | 10,271 | 0 | 5,258 | 22,882 | 13,808 |
| *Interest Rate Dependent Variables* | | | | | | |
| Savings Deposit Rate$_{i,t}$ (%) | -1.21 | 0.45 | -1.76 | -1.19 | -0.65 | 36,539 |
| Time Deposit Rate$_{i,t}$ (%) | -0.21 | 1.07 | -1.58 | -0.18 | 1.40 | 36,544 |
| Checking Deposit Rate$_{i,t}$ (%) | -1.63 | 0.41 | -2.09 | -1.66 | -0.93 | 36,489 |
| Auto Loan Interest Rate$_{i,t}$ (%) | 4.64 | 1.56 | 2.37 | 4.91 | 6.42 | 27,141 |
| Mortgage Interest Rate$_{i,t}$ (%) | 3.19 | 1.10 | 1.72 | 3.43 | 4.41 | 24,336 |
| *Instrument* | | | | | | |
| Subsidy Amount$_{i,t}$ ($1,000s) | 1,369 | 2,193 | 0 | 277 | 4,265 | 13,808 |
| *Control Variables* | | | | | | |
| Income Per Capita$_{i,t-2}$ ($) | 36,523 | 9,217 | 27,227 | 34,930 | 47,124 | 13,808 |
| Employment Rate$_{i,t-2}$ | 0.50 | 0.13 | 0.34 | 0.49 | 0.66 | 13,808 |
| Population$_{i,t-2}$ | 63,234 | 115,890 | 5,909 | 25,012 | 149,639 | 13,808 |
| State-Level Deregulation Index$_{i,t}$ | 1 | 2 | 0 | 1 | 4 | 13,808 |
| Number of Branches$_{i,t-2}$ | 22 | 35 | 3 | 11 | 48 | 13,808 |
| Idle Cropland$_{i,t-7}$ (Acres) | 9,959 | 16,403 | 751 | 4,851 | 23,274 | 13,808 |
| Unharvested Cropland$_{i,t-7}$ (Acres) | 21,009 | 42,398 | 1,047 | 7,868 | 48,167 | 13,808 |
| Pastureland$_{i,t-7}$ (Acres) | 148,400 | 328,613 | 6,864 | 40,478 | 387,947 | 13,808 |

**Table 2.** Flat vs. Rugged Counties: Effect of Subsidy Amount on Farm Concentration and the Presence, Branches, and Deposits of Large and Small Banks

This table reports reduced-form estimates of the effect of subsidy payments on farm concentration and on the presence, branches, and deposits of large and small banks in flat counties (Panel A) and rugged counties (Panel B). The specifications are: $\textit{Farm Concentration}_{i,t} = \beta_1 \textit{Subsidy Amount}_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ in Column 1, $Y_{i,t} = \alpha\, \textit{Subsidy Amount}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ in Columns 2 and 5, and $E\big[Y_{i,t} \,\big|\, \textit{Subsidy Amount}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\alpha\, \textit{Subsidy Amount}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). Counties are classified as rugged if their average Terrain Ruggedness Index (TRI), calculated by the Economic Research Service (ERS), is in the top quartile of the national distribution, and flat otherwise. In Column 1, the outcome variable is $\textit{Farm Concentration}_{i,t}$, the midpoint farm size at the county level. In Columns 2 and 5, the outcome variable is an indicator that a large or small bank, respectively, is present in the county in a given year. In Columns 3 and 6, the outcome variable is $\textit{Branches}_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 4 and 7, the outcome variable is $\textit{Deposits Per Capita}_{i,t}$, which is the deposits per capita in large and small banks, respectively. $\textit{Subsidy Amount}_{i,t}$ is a continuous measure of the predicted value of total fixed subsidy payments in a given county-year, standardized by its within-county standard deviation within each subsample. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The control variables for local economic conditions are the county-level employment rate and log income per capita, and the control variables for county-level land availability are lagged measures of idle cropland, pastureland, and cropland left fallow, lost to crop failure, or otherwise unharvested from the prior Census of Agriculture. Column 1 includes both sets of control variables. Columns 2 through 7 include the control variables for local economic conditions and additionally control for pre-existing banking infrastructure by including the log number of bank $\text{branches}_{i,t-2}$ in the county. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages in Columns 2 through 7. Means of the dependent variable are calculated within the relevant subsample. Standard errors clustered at the county level are reported in parentheses. $^{***}$, $^{**}$, and $^{*}$ represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| | | Large Banks | | | Small Banks | | |
| | Farm Concentration | Probability of Entry | Branches | Deposits Per Capita | Probability of Entry | Branches | Deposits Per Capita |
| *Panel A: Flat Counties* | | | | | | | |
| Subsidy Amount$_{i,t-2}$ | 85.11*** | 0.014*** | 0.031*** | 0.065* | 0.003** | 0.020*** | 0.021*** |
| | (5.68) | (0.005) | (0.010) | (0.038) | (0.001) | (0.005) | (0.004) |
| Observations | 10,376 | 10,368 | 9,080 | 9,080 | 10,368 | 10,313 | 10,313 |
| R-squared | 0.84 | 0.72 | 0.89 | 0.80 | 0.58 | 0.62 | 0.87 |
| $F$-statistic | 224.52 | 224.52 | 224.52 | 224.52 | 224.52 | 224.52 | 224.52 |
| Mean of Dep. Variable | 892 | 0.77 | 15 | 8,279 | 0.97 | 10 | 12,897 |
| *Panel B: Rugged Counties* | | | | | | | |
| Subsidy Amount$_{i,t-2}$ | 16.23*** | -0.002 | 0.008 | -0.012 | -0.002 | 0.008 | 0.016* |
| | (4.31) | (0.006) | (0.007) | (0.010) | (0.002) | (0.005) | (0.008) |
| Observations | 3,442 | 3,440 | 3,151 | 3,146 | 3,440 | 3,354 | 3,354 |
| R-squared | 0.86 | 0.71 | 0.84 | 0.76 | 0.65 | 0.57 | 0.80 |
| $F$-statistic | 14.18 | 14.18 | 14.18 | 14.18 | 14.18 | 14.18 | 14.18 |
| County Controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 328 | 0.81 | 12 | 9,015 | 0.93 | 7 | 8,981 |

**Table 3.** Effect of Subsidy Amount on Farm Concentration and Banking Concentration and of Farm Concentration on Banking Concentration

This table reports reduced-form estimates of the effect of subsidy payments on farm concentration and banking concentration (Panel A) and instrumental variable (IV) and ordinary least squares (OLS) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on banking concentration (Panel B). The specifications are: $Y_{i,t} = \alpha\, Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ in Panel A and $Y_{i,t} = \beta_2\, \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Panel B. In Panel B, Columns 1 and 2 report the IV estimates, and Columns 3 and 4 report the OLS estimates. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Columns 1 and 2 of Panel A, the outcome variable is $Farm\ Concentration_{i,t}$, the midpoint farm size at the county level. In the remaining columns of Panel A and in Panel B, the outcome variable is $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. $Subsidy\ Amount_{i,t}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. The control variables for local economic conditions are the county-level employment rate and log income per capita, and the control variables for county-level land availability are lagged measures of idle cropland, pastureland, and cropland left fallow, lost to crop failure, or otherwise unharvested from the prior Census of Agriculture. $F$-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Farm Concentration | | Bank Deposits HHI | |
| *Panel A: Effect of Subsidy Amount* | | | | |
| Subsidy Amount$_{i,t-2}$ | 105.22*** | 107.42*** | 57.39*** | 63.55*** |
| | (5.49) | (5.88) | (8.45) | (8.84) |
| Log Income Per Capita$_{i,t-2}$ | | 183.02*** | | 20.89 |
| | | (69.75) | | (115.56) |
| Employment Rate$_{i,t-2}$ | | -123.53 | | 244.67 |
| | | (156.75) | | (313.49) |
| Idle Cropland$_{i,t-7}$ | | 20.13*** | | |
| | | (4.99) | | |
| Unharvested Cropland$_{i,t-7}$ | | -11.11*** | | |
| | | (3.58) | | |
| Pastureland$_{i,t-7}$ | | 4.21 | | |
| | | (4.30) | | |
| Observations | 13,808 | 13,808 | 13,808 | 13,808 |
| $F$-statistic | 367.03 | 333.82 | 367.03 | 356.99 |
| Deregulation Index FE | No | Yes | No | Yes |
| County & Year FE | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 752 | 752 | 3,123 | 3,123 |

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Bank Deposits HHI | | | |
| *Panel B: Effect of Farm Concentration* | | | | |
| $\widehat{\text{Farm Concentration}}_{i,t-2}$ | 252.23*** | 275.48*** | | |
| | (38.59) | (41.21) | | |
| Farm Concentration$_{i,t-2}$ | | | 20.81** | 17.71** |
| | | | (9.06) | (9.00) |
| Log Income Per Capita$_{i,t-2}$ | | -88.77 | | 23.99 |
| | | (125.05) | | (117.02) |
| Employment Rate$_{i,t-2}$ | | 319.50 | | 321.75 |
| | | (328.18) | | (314.65) |
| Idle Cropland$_{i,t-7}$ | | -12.32 | | -4.34 |
| | | (7.87) | | (7.34) |
| Unharvested Cropland$_{i,t-7}$ | | 7.77 | | -8.90 |
| | | (7.10) | | (7.19) |
| Pastureland$_{i,t-7}$ | | -1.88 | | 0.43 |
| | | (5.89) | | (4.92) |
| Observations | 13,808 | 13,808 | 13,808 | 13,808 |
| $F$-statistic | 367.03 | 333.82 | 5.28 | 4.96 |
| Deregulation Index FE | No | Yes | No | Yes |
| County & Year FE | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 3,123 | 3,123 | 3,123 | 3,123 |

**Table 4.** Effect of Farm Concentration on the Presence, Branches, and Deposits of Large and Small Banks

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on the presence, branches, and deposits of large and small banks. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ Farm\ \widehat{Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Columns 1 and 4 and $E\big[Y_{i,t} \,\big|\, Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Columns 1 and 4, the outcome variable is an indicator that a large or small bank, respectively, is present in the county in a given year. In Columns 2 and 5, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 3 and 6, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small banks, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than \$1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 3. I additionally control for pre-existing banking infrastructure by including the log number of bank branches$_{i,t-2}$ in the county. Log population$_{i,t-2}$ is included in the first-stage regression for Columns 3 and 6 to adjust for the per capita scaling of the outcome variable. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. $^{***}$, $^{**}$, and $^{*}$ represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Large Banks | | | Small Banks | | |
| | Probability of Entry | Branches | Deposits Per Capita | Probability of Entry | Branches | Deposits Per Capita |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | 0.046** | 0.160*** | 0.327** | 0.021*** | 0.099*** | 0.225*** |
| | (0.022) | (0.050) | (0.128) | (0.006) | (0.023) | (0.031) |
| Observations | 13,808 | 12,231 | 12,226 | 13,808 | 13,667 | 13,667 |
| $F$-statistic | 322.68 | 322.68 | 174.91 | 322.68 | 322.68 | 174.91 |
| County Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 0.78 | 14 | 8,468 | 0.96 | 9 | 11,936 |

**Table 5.** Effect of Farm Concentration on Bank Acquisitions, by Bank Size

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on bank acquisitions by bank size. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t} \,\big|\, Farm\ \ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Column 1, the outcome variable is $Large\ Acquires\ Small_{i,t}$, which is the number of mergers or acquisitions in which a large bank acquires a small bank, aggregated over five-year intervals. In Columns 2 and 3, the outcome variables are $Small\ Acquires\ Small_{i,t}$ and $Large\ Acquires\ Large_{i,t}$, which are the number of mergers or acquisitions between two small banks and between two large banks, respectively, also aggregated over five-year intervals. Each merger or acquisition is counted in every county where the target bank operated branches, with bank sizes and branch counties measured at the last Summary of Deposits before the transaction date. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 4. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) Large Acquires Small | (2) Small Acquires Small | (3) Large Acquires Large |
|---|---|---|---|
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | 0.153 | 0.234** | -0.325*** |
| | (0.138) | (0.103) | (0.076) |
| Observations | 8,902 | 10,236 | 9,848 |
| $F$-statistic | 322.68 | 322.68 | 322.68 |
| County Controls | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes |
| Mean of Dep. Variable | 0.55 | 0.58 | 0.94 |

**Table 6.** Effect of Farm Concentration on Large Banks' Farm Lending by Farm Revenue and Loan Size

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on large banks' farm lending by farm revenue and loan size. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t}\,\big|\,Farm\ \ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ \ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data at the annual level (see Section 5 for variable and sample construction details). The sample is restricted to 1996 to 2014, the years with CRA reporting. The outcome variable is the total volume of farm loans of $500,000 or less that large banks originate in a given county-year, for loans to farms with gross annual revenue above $1 million in Column 1 and $1 million or less in Column 2. In Columns 3 through 5, the outcome variable is the total volume of farm loans at or below $100,000, between $100,000 and $250,000, and between $250,000 and $500,000, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size, carried forward annually until the next census observation. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 4. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| | Farm Revenue | | Loan Size | | |
| | >$1M | ≤$1M | ≤$100K | $100K–$250K | $250K–$500K |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | 0.299*** | 0.017 | -0.046 | 0.031 | 0.177** |
| | (0.082) | (0.059) | (0.051) | (0.076) | (0.077) |
| Observations | 52,533 | 52,665 | 52,675 | 51,212 | 48,919 |
| $F$-statistic | 326.06 | 326.38 | 326.37 | 318.33 | 301.52 |
| County Controls | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 1,039 | 3,393 | 1,512 | 1,516 | 1,552 |

**Table 7.** Effect of Farm Concentration on Agricultural Loan Growth

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on agricultural loan growth by bank size. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t}\,\big|\,Farm\ \ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Columns 1 and 2, the outcome variable is $Total\ Agricultural\ Loan\ Growth_{i,t}$, in Columns 3 and 4 it is $Real\ Estate\ Loan\ Growth_{i,t}$, and in Columns 5 and 6 it is $Non\text{-}Real\ Estate\ Loan\ Growth_{i,t}$, each measuring large or small banks' growth in the loan type relative to the prior period's total volume of that loan type. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 4. The first stage also includes the prior period's log total loan volume by loan type. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. $^{***}$, $^{**}$, and $^{*}$ represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Total Agricultural Loan Growth | | Real Estate Loan Growth | | Non-Real Estate Loan Growth | |
| | Large Banks | Small Banks | Large Banks | Small Banks | Large Banks | Small Banks |
| $\widehat{\text{Farm Concentration}}_{i,t-2}$ | 0.455*** | 0.028 | 0.398** | 0.015 | 0.460*** | 0.060* |
| | (0.124) | (0.032) | (0.164) | (0.037) | (0.120) | (0.036) |
| Observations | 11,719 | 13,465 | 11,709 | 13,418 | 11,683 | 13,306 |
| $F$-statistic | 323.27 | 323.27 | 322.45 | 322.45 | 321.62 | 321.62 |
| County Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 0.46 | 0.73 | 0.49 | 0.87 | 0.47 | 0.65 |

**Table 8.** Effect of Farm Concentration on Small Banks' Agricultural Lending by Loan Size and Core Funding

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on small banks' agricultural lending by loan size and core deposit share. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t} \,\big|\, Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Panels A and B, the outcome variable is high- and low-core small banks' growth in real estate farm lending, respectively, and in Panels C and D it is high- and low-core small banks' growth in non-real estate farm lending, each relative to the prior period's total volume of that loan type. In Columns 1 through 4, the outcome variable is the growth in loans at or below $100,000, between $100,000 and $250,000, between $250,000 and $500,000, and above $500,000, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. High-core small banks are those with a pre-program (1990–1995) core deposit share above the within-small-bank median, and low-core small banks otherwise. The controls are the same set of first- and second-stage controls as in Table 4. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. $^{***}$, $^{**}$, and $^{*}$ represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Agricultural Loan Growth | | | |
| | ≤$100K | $100K–$250K | $250K–$500K | >$500K |
| *Panel A: Real Estate Loans, High-Core Funding* | | | | |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | -0.199*** | -0.090 | 0.133 | 0.941*** |
| | (0.049) | (0.080) | (0.118) | (0.273) |
| Observations | 10,848 | 10,069 | 9,592 | 10,406 |
| $F$-statistic | 322.65 | 322.65 | 322.65 | 322.65 |
| Mean of Dep. Variable | 0.28 | 0.13 | 0.12 | 0.14 |
| *Panel B: Real Estate Loans, Low-Core Funding* | | | | |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | -0.176*** | -0.158** | -0.079 | -0.002 |
| | (0.055) | (0.075) | (0.095) | (0.151) |
| Observations | 11,617 | 11,187 | 10,931 | 11,416 |
| $F$-statistic | 322.65 | 322.65 | 322.65 | 322.65 |
| Mean of Dep. Variable | 0.27 | 0.12 | 0.11 | 0.17 |
| *Panel C: Non-Real Estate Loans, High-Core Funding* | | | | |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | -0.063 | 0.193* | 0.446** | 0.805** |
| | (0.051) | (0.113) | (0.175) | (0.362) |
| Observations | 10,724 | 9,315 | 8,684 | 10,205 |
| $F$-statistic | 322.65 | 322.65 | 322.65 | 322.65 |
| Mean of Dep. Variable | 0.28 | 0.07 | 0.07 | 0.10 |
| *Panel D: Non-Real Estate Loans, Low-Core Funding* | | | | |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | -0.170*** | -0.353*** | -0.249*** | -0.033 |
| | (0.052) | (0.069) | (0.085) | (0.145) |
| Observations | 11,536 | 10,380 | 9,876 | 11,024 |
| $F$-statistic | 322.65 | 322.65 | 322.65 | 322.65 |
| Mean of Dep. Variable | 0.29 | 0.07 | 0.07 | 0.09 |
| County Controls | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes |

**Table 9.** Effect of Farm Concentration on Agricultural Loan Concentration and on Large and Small Agricultural Banks

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on the concentration of agricultural lending and on the presence, branches, and deposits of large and small agricultural (ag) banks. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Columns 1, 2, and 5 and $E\big[Y_{i,t}\,\big|\,Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Column 1, the sample includes all banks, and the outcome variable is $Agricultural\ Loan\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared agricultural loan shares held by each bank within a county. In the remaining columns, the sample is restricted to ag banks, those with farm loans exceeding 5% of total loans in any of the preceding six years. In Columns 2 and 5, the outcome variable is an indicator that a large or small ag bank, respectively, is present in the county in a given year. In Columns 3 and 6, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small ag banks, respectively. In Columns 4 and 7, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small ag banks, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. In Column 1, the controls are the same set of first- and second-stage controls as in Table 3, and in the remaining columns as in Table 4. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages in Columns 2 through 7. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| | | Large Ag Banks | | | Small Ag Banks | | |
| | Agricultural Loan HHI | Probability of Entry | Branches | Deposits Per Capita | Probability of Entry | Branches | Deposits Per Capita |
| $\widehat{\text{Farm Concentration}}_{i,t-2}$ | 628.51*** | 0.154*** | 0.546*** | 0.752*** | 0.016 | -0.003 | 0.237*** |
| | (92.74) | (0.024) | (0.170) | (0.228) | (0.010) | (0.021) | (0.034) |
| Observations | 13,808 | 13,808 | 8,559 | 8,554 | 13,808 | 12,218 | 12,218 |
| $F$-statistic | 333.82 | 333.74 | 333.74 | 187.52 | 333.74 | 333.74 | 187.52 |
| County Controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 4,472 | 0.32 | 2 | 2,310 | 0.77 | 5 | 9,948 |

**Table 10.** Effect of Farm Concentration on Offered Interest Rates, 2002 to 2014

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on the real interest rates offered by banks, for all banks (Panel A) and separately for small banks (Panel B) and large banks (Panel C). The specifications for the first and second stages, respectively, are: $\textit{Farm Concentration}_{i,t} = \beta_1 \textit{Subsidy Amount}_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2 \widehat{\textit{Farm Concentration}}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The county-year data are from merged bank and agriculture data at the annual level (see Section 5 for variable and sample construction details). In Panel A, each observation is a county-level average of the June interest rates offered by bank branches within the county. In Panels B and C, the second stage is estimated at the bank-county-year level within each size subsample, with bank-by-county and year fixed effects in place of the county fixed effects $\gamma_i$, and each observation is a bank's average offered rate across its branches within a county in June of a given year. In Columns 1 through 5, the outcome variables are the offered interest rates on savings deposit accounts ($10,000 money market accounts), time deposit accounts (12-month $10,000 certificates of deposit), checking deposit accounts (interest-bearing checking accounts), auto loans, and residential mortgage loans, all expressed in real, inflation-adjusted terms. Coefficients are reported in percentage points. $\textit{Farm Concentration}_{i,t-2}$ is the lagged within-county standardized midpoint farm size, carried forward annually until the next census observation. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 3. In both stages, the control variables for local credit demand are county-level annual log total deposits and log total farm interest expenses. $F$-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| | Deposit Rates | | | Loan Rates | |
| | Savings | Time | Checking | Auto | Mortgage |
| *Panel A: All Banks* | | | | | |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | -0.316*** | 0.125*** | -0.223*** | 1.028*** | 0.009 |
| | (0.055) | (0.041) | (0.035) | (0.208) | (0.065) |
| Observations | 34,828 | 34,834 | 34,786 | 25,770 | 23,178 |
| $F$-statistic | 323.16 | 323.16 | 323.16 | 323.16 | 323.16 |
| County Controls | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | -1.20 | -0.17 | -1.64 | 4.69 | 3.23 |
| *Panel B: Small Banks* | | | | | |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | -0.118** | 0.302*** | -0.132*** | -0.369*** | -0.104 |
| | (0.054) | (0.051) | (0.029) | (0.130) | (0.125) |
| Observations | 110,968 | 109,764 | 107,648 | 30,366 | 18,209 |
| $F$-statistic | 323.16 | 323.16 | 323.16 | 323.16 | 323.16 |
| County Controls | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes |
| Bank × County & Year FE | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | -1.07 | -0.06 | -1.55 | 4.83 | 3.58 |
| *Panel C: Large Banks* | | | | | |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | -0.221*** | -0.181*** | -0.041* | 0.971*** | 0.107** |
| | (0.070) | (0.063) | (0.023) | (0.284) | (0.053) |
| Observations | 106,470 | 103,140 | 101,628 | 70,313 | 59,102 |
| $F$-statistic | 323.16 | 323.16 | 323.16 | 323.16 | 323.16 |
| County Controls | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes |
| Bank × County & Year FE | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | -1.36 | -0.49 | -1.74 | 4.35 | 2.96 |

# Appendix A. Additional Analyses

**Figure A.1.** Distribution of Counties by Average Terrain Ruggedness

This figure shows the distribution of average terrain ruggedness across U.S. counties, measured using the 2010 Terrain Ruggedness Index (TRI) from the USDA Economic Research Service (ERS). Counties are divided into quartiles based on the national TRI distribution, with the top quartile corresponding to an average ruggedness index greater than or equal to 28.

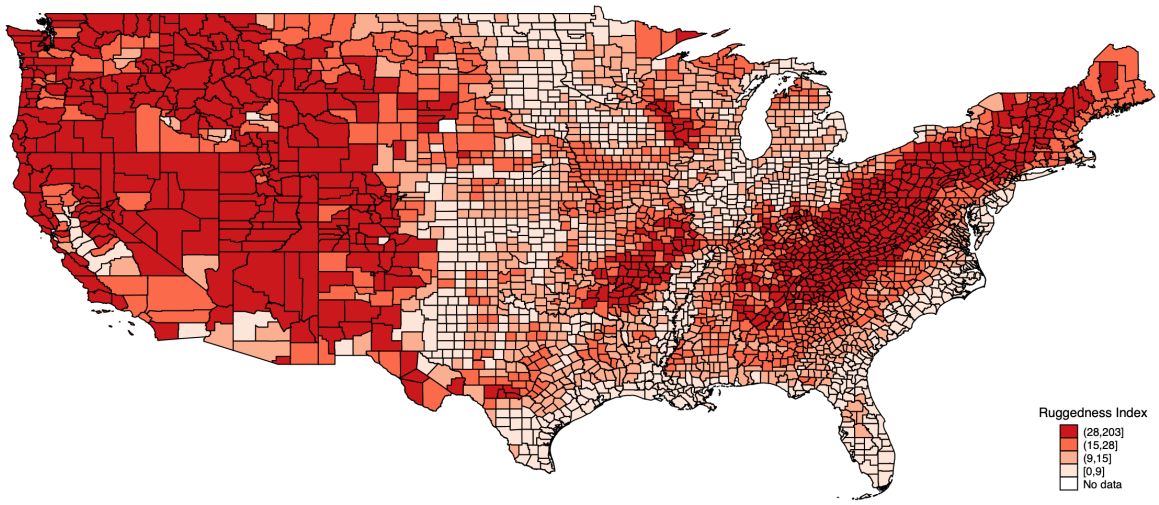


**Figure A.2.** Distribution of High- and Low-Cost Counties

This figure shows the distribution of U.S. counties classified as high- or low-cost based on per acre economic costs. The measure is constructed as a time-invariant, county-level weighted average of per acre economic costs, where weights are each crop's share of harvested acreage from the 1992 Census of Agriculture. Economic costs per acre for each major commodity crop are calculated as the six-year average (1990–1995) from the Farm Costs and Returns Survey (FCRS) compiled by the U.S. Department of Agriculture's Economic Research Service (ERS). Counties with weighted average costs above the national median are classified as high-cost, and those below the median as low-cost.

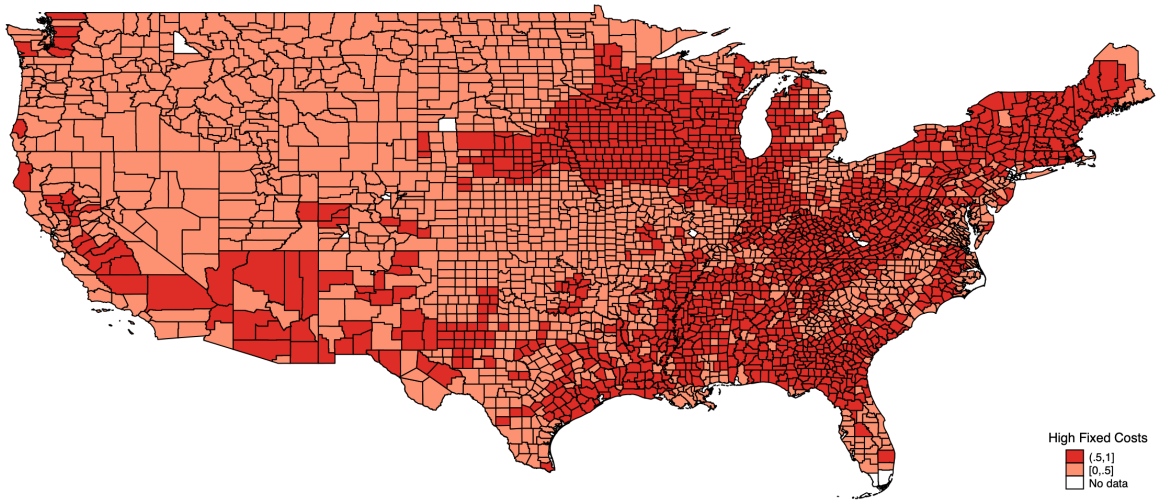


**Figure A.3.** Distribution of Changes in Land Values by Farming Cost

This figure shows the distribution of percent changes in market value of agricultural land and buildings per acre between 1992 and 2012 for high- and low-cost counties. Counties are classified as high-cost if their weighted average per acre economic cost exceeds the nationwide median, and low-cost otherwise. Economic costs per acre for each major commodity crop are calculated as the six-year average (1990–1995) from the Farm Costs and Returns Survey (FCRS) compiled by the U.S. Department of Agriculture's Economic Research Service (ERS). The measure is constructed as a time-invariant, county-level weighted average of per acre economic costs, where weights are each crop's share of harvested acreage from the 1992 Census of Agriculture.

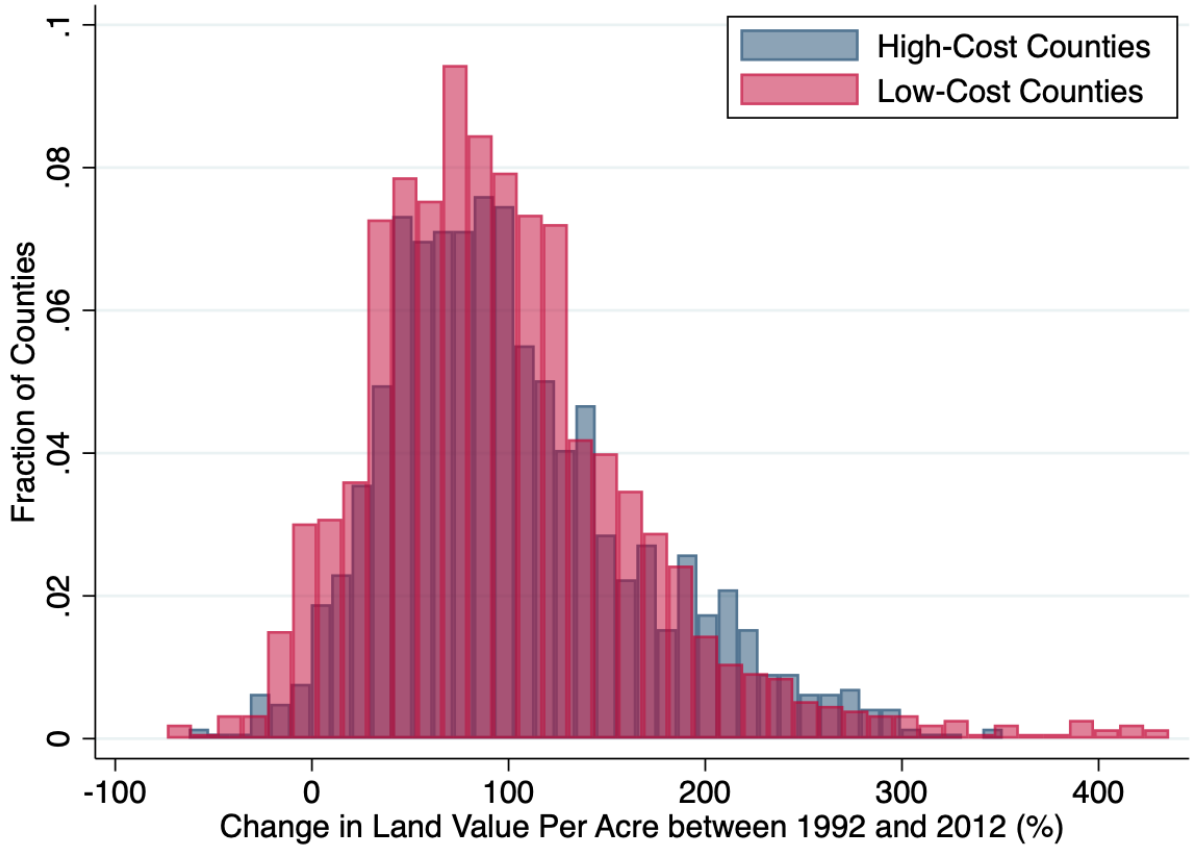


**Table A.1.** Pre-Program Shares: Effect of Subsidy Amount and Farm Concentration on Banking Concentration and the Presence, Branches, and Deposits of Large and Small Banks

This table reports reduced-form estimates of the effect of subsidy payments (Panel A) and second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration (Panel B) on banking concentration and on the presence, branches, and deposits of large and small banks. In Panel A, the specification is $Y_{i,t} = \alpha\, Subsidy\ Amount_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$. In Panel B, the specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Columns 1, 2, and 5 and $E\big[Y_{i,t} \,\big|\, Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Column 1, the outcome variable is $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. In Columns 2 and 5, the outcome variable is an indicator that a large or small bank, respectively, is present in the county in a given year. In Columns 3 and 6, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 4 and 7, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small banks, respectively. $Subsidy\ Amount_{i,t-2}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year, constructed from pre-program shares, using the 1996 base acre shares for the seven original program crops in all years and county shares of national 1992 Census of Agriculture harvested acres for the crops added in 2002. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than \$1 billion in inflation-adjusted assets, and large otherwise. In Column 1, the controls are the same set of controls as in Table 3, and in Columns 2 through 7, the controls are the same set of controls as in Table 4. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages in Columns 2 through 7. Standard errors clustered at the county level are reported in parentheses. $^{***}$, $^{**}$, and $^{*}$ represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| | | Large Banks | | | Small Banks | | |
| | Bank Deposits HHI | Probability of Entry | Branches | Deposits Per Capita | Probability of Entry | Branches | Deposits Per Capita |
| *Panel A: Effect of Subsidy Amount* | | | | | | | |
| Subsidy Amount$_{i,t-2}$ | 49.63*** | 0.007* | 0.025*** | 0.066* | 0.003*** | 0.017*** | 0.020*** |
| | (7.18) | (0.004) | (0.008) | (0.037) | (0.001) | (0.004) | (0.003) |
| Observations | 13,808 | 13,808 | 12,231 | 12,226 | 13,808 | 13,667 | 13,667 |
| R-squared | 0.92 | 0.72 | 0.88 | 0.79 | 0.61 | 0.61 | 0.86 |
| Mean of Dep. Variable | 3,123 | 0.78 | 14 | 8,468 | 0.96 | 9 | 11,936 |
| *Panel B: Effect of Farm Concentration* | | | | | | | |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | 268.77*** | 0.047** | 0.155*** | 0.369** | 0.021*** | 0.095*** | 0.219*** |
| | (41.89) | (0.022) | (0.052) | (0.149) | (0.006) | (0.023) | (0.032) |
| Observations | 13,808 | 13,808 | 12,231 | 12,226 | 13,808 | 13,667 | 13,667 |
| $F$-statistic | 337.03 | 325.95 | 325.95 | 177.39 | 325.95 | 325.95 | 177.39 |
| County Controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 3,123 | 0.78 | 14 | 8,468 | 0.96 | 9 | 11,936 |

**Table A.2.** Alternative Lag Structures: Effect of Farm Concentration on Banking Concentration and Large and Small Banks

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on banking concentration and on the branches and deposits of large and small banks with alternative time lags for the second stage. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ \widehat{Farm\ Concentration}_{i,t-k} + \mathbf{X}'_{i,t-k}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Column 1 and $E\big[Y_{i,t} \,\big|\, Farm\ Concentration_{i,t-k}, \hat{\varepsilon}_{i,t-k}, \mathbf{X}'_{i,t-k}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-k} + \mathbf{X}'_{i,t-k}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-k} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-k}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Panels A and B, the first stage is lagged by three and four years, respectively. In Column 1, the outcome variable is $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. In Columns 2 and 4, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 3 and 5, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small banks, respectively. $Farm\ Concentration_{i,t-k}$ is the within-county standardized midpoint farm size, lagged by either three or four years. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. In Column 1, the controls are the same set of first- and second-stage controls as in Table 3. In Columns 2 through 5, the controls are the same set of first- and second-stage controls as in Table 4. All controls carry the same lag as $Farm\ Concentration_{i,t-k}$. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages in Columns 2 through 5. Standard errors clustered at the county level are reported in parentheses. $^{***}$, $^{**}$, and $^{*}$ represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| | | Large Banks | | Small Banks | |
| | Bank Deposits HHI | Branches | Deposits Per Capita | Branches | Deposits Per Capita |
| *Panel A: Three-Year Lag* | | | | | |
| Farm $\widehat{\text{Concentration}}_{i,t-3}$ | 224.66*** | 0.169*** | 0.414*** | 0.102*** | 0.236*** |
| | (38.32) | (0.065) | (0.154) | (0.022) | (0.031) |
| Observations | 13,808 | 12,276 | 12,271 | 13,653 | 13,653 |
| $F$-statistic | 94.49 | 92.15 | 79.15 | 92.15 | 79.15 |
| Mean of Dep. Variable | 3,120 | 15 | 8,787 | 9 | 11,759 |
| *Panel B: Four-Year Lag* | | | | | |
| Farm $\widehat{\text{Concentration}}_{i,t-4}$ | 227.53*** | 0.117** | 0.469*** | 0.108*** | 0.244*** |
| | (38.02) | (0.050) | (0.144) | (0.019) | (0.032) |
| Observations | 13,808 | 12,357 | 12,347 | 13,654 | 13,651 |
| $F$-statistic | 93.58 | 91.21 | 77.71 | 91.21 | 77.71 |
| County Controls | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 3,114 | 15 | 9,110 | 9 | 11,629 |

**Table A.3.** Effect of Subsidy Amount on Farm Size and Land Tenure

This table reports reduced-form estimates of the effect of subsidy payments on farm size and land tenure. The specification is $Y_{i,t} = \alpha\, \textit{Subsidy Amount}_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Column 1, the outcome variable is the log of the number of farms with harvested cropland. In Column 2, it is the log of total harvested acres. In Columns 3 and 4, it is rented acres and owned acres per part owner farm, where part owners operate both owned and rented land. In Columns 5 through 8, it is the share of the county's harvested acres on farms harvesting 1 to 49, 50 to 499, 500 to 999, and 1,000 or more acres. Rented and owned acres are from the Census of Agriculture tenure tables. $\textit{Subsidy Amount}_{i,t}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year. The controls are the same set of first-stage controls as in Table 3. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| | | | | | Share of Harvested Acres | | | |
| | Log Number of Farms | Log Harvested Acres | Rented Acres | Owned Acres | 1–49 | 50–499 | 500–999 | 1,000+ |
| Subsidy Amount$_{i,t}$ | -0.033*** | 0.003 | 32.66*** | 20.15*** | 0.004*** | -0.018*** | -0.016*** | 0.030*** |
| | (0.002) | (0.002) | (2.90) | (2.51) | (0.001) | (0.001) | (0.001) | (0.002) |
| Observations | 13,817 | 13,772 | 13,595 | 13,646 | 13,772 | 13,772 | 13,772 | 13,772 |
| R-squared | 0.98 | 0.98 | 0.93 | 0.91 | 0.92 | 0.88 | 0.76 | 0.93 |
| County Controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 467 | 103,622 | 553 | 488 | 0.14 | 0.43 | 0.16 | 0.28 |

**Table A.4.** Effect of Subsidy Amount on Farm Concentration, Land Values, and Deposits per Capita

This table reports reduced-form estimates of the effect of subsidy payments on farm concentration, land values, and deposits per capita, estimated jointly in a stacked system. The specification is $Y_{i,t}^{j} = \alpha_j \textit{ Subsidy Amount}_{i,t} + \mathbf{X}_{i,t}'\mathbf{\Gamma}_j + \gamma_i^j + \delta_t^j + \phi_{d(i,t)}^j + \varepsilon_{i,t}^j$. Column $j$ reports equation $j$, where $j = 1, 2, 3$ denotes the farm concentration, land value, and deposits per capita equations, respectively. Subscript $i$ represents the county, $t$ represents the year, $d$ represents state-level banking deregulation, and $j$ represents the outcome equation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). The common sample includes county-years with nonmissing outcomes and controls in all three equations and uses banking years 1994–2009. The outcomes are county-level midpoint farm size in Column 1, log market value of agricultural land and buildings per acre in Column 2, and total county deposits per capita in Column 3. In the deposits per capita equation, the subsidy payments and the controls are measured two years before the outcome, as in Table 2. Each outcome and $\textit{Subsidy Amount}_{i,t}$, a continuous measure of predicted total fixed subsidy payments, is divided by its pooled within-county standard deviation on this sample. The controls are the same set of first-stage controls as in Table 3. Control coefficients are equation-specific, and the system covariance matrix allows residuals to be correlated across outcomes within counties. Farm Concentration Exceeds Alternatives is the $p$-value from a Wald test that the coefficient on Subsidy Amount in the farm concentration equation is larger in absolute value than the coefficients in the land value and deposits per capita equations. Coefficients report the change in outcome standard deviations associated with a one-standard-deviation increase in Subsidy Amount. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) Farm Concentration | (2) Land Values | (3) Deposits Per Capita |
|---|---|---|---|
| Subsidy Amount$_{i,t}$ | 0.243*** | -0.031*** | 0.015 |
| | ( 0.014) | ( 0.007) | ( 0.010) |
| Farm Concentration Exceeds Alternatives | | $< 0.001$ | |
| Observations | 11,146 | 11,146 | 11,146 |
| R-squared | 0.86 | 0.97 | 0.71 |
| County Controls | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes |
| Mean of Dep. Variable | 697 | 7.67 | 18,868 |

**Table A.5.** Effect of Subsidy Amount on Investment, Crop Mix, and Production Outcomes

This table reports reduced-form estimates of the effect of subsidy payments on county-level agricultural production outcomes from the Census of Agriculture. The specification is $Y_{i,t} = \alpha\, \textit{Subsidy Amount}_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Column 1, the outcome variable is the Törnqvist crop-yield index, which aggregates per acre yield growth across the nine census crops with quantity data using revenue-share weights. In Column 2, the outcome variable is the inverse hyperbolic sine of the real value of total county machinery in thousands of dollars, and in Column 3 the inverse hyperbolic sine of the county count of recently manufactured tractors. In Column 4, the outcome variable is the program crop share, the share of county crop acres planted in the seven program crops of the 1996 FAIR Act (wheat, corn, sorghum, barley, oats, upland cotton, and rice). $\textit{Subsidy Amount}_{i,t}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year. The controls are the same set of first-stage controls as in Table 3. Standard errors clustered at the county level are reported in parentheses. $^{***}$, $^{**}$, and $^{*}$ represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) Törnqvist Yield Index | (2) Machinery Stock | (3) Tractor Investment | (4) Program Crop Share |
|---|---|---|---|---|
| Subsidy Amount$_{i,t}$ | -0.0031 | -0.0144$^{***}$ | -0.0326$^{***}$ | 0.0007 |
| | (0.0052) | (0.0023) | (0.0051) | (0.0009) |
| Observations | 12,134 | 13,808 | 13,808 | 13,808 |
| R-squared | 0.97 | 0.96 | 0.86 | 0.96 |
| County Controls | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | -10 | 11 | 5 | 0.35 |

**Table A.6.** Flat vs. Rugged Counties: Effect of Subsidy Amount on Farm Concentration and the Presence, Branches, and Deposits of Large and Small Banks, Controlling for Population, Urbanization, and Land Values

This table reports reduced-form estimates of the effect of subsidy payments on farm concentration and on the presence, branches, and deposits of large and small banks in flat counties (Panel A) and rugged counties (Panel B), allowing the effect of payments to vary with pre-program population, urbanization, and land values. The specifications are: $\textit{Farm Concentration}_{i,t} = \beta_1 \textit{Subsidy Amount}_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ in Column 1, $Y_{i,t} = \alpha\, \textit{Subsidy Amount}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ in Columns 2 and 5, and $E\big[Y_{i,t} \,\big|\, \textit{Subsidy Amount}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\alpha\, \textit{Subsidy Amount}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). Counties are classified as rugged if their average Terrain Ruggedness Index (TRI), calculated by the Economic Research Service (ERS), is in the top quartile of the national distribution, and flat otherwise. In Column 1, the outcome variable is $\textit{Farm Concentration}_{i,t}$, the midpoint farm size at the county level. In Columns 2 and 5, the outcome variable is an indicator that a large or small bank, respectively, is present in the county in a given year. In Columns 3 and 6, the outcome variable is $\textit{Branches}_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 4 and 7, the outcome variable is $\textit{Deposits Per Capita}_{i,t}$, which is the deposits per capita in large and small banks, respectively. $\textit{Subsidy Amount}_{i,t}$ is a continuous measure of the predicted value of total fixed subsidy payments in a given county-year, standardized by its within-county standard deviation within each subsample. Each specification additionally includes the products of *Subsidy Amount* with $Z_{m,i}$, which collects pre-program population, urbanization, and land values, each standardized across counties. Pre-program characteristics are county means over banking years 1994 to 1997 of log population, the 1990 National Center for Health Statistics (NCHS) urbanization code, which runs from 2 for large fringe metropolitan counties to 6 for noncore counties, and the market value of agricultural land and buildings per acre. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of controls as in Table 2. Coefficients are reported as decimal percentages in Columns 2 through 7. Means of the dependent variable are calculated within the relevant subsample. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| | | Large Banks | | | Small Banks | | |
| | Farm Concentration | Probability of Entry | Branches | Deposits Per Capita | Probability of Entry | Branches | Deposits Per Capita |
| *Panel A: Flat Counties* | | | | | | | |
| Subsidy Amount$_{i,t-2}$ | 86.00*** | 0.017*** | 0.098*** | 0.053*** | 0.003** | 0.030*** | 0.012*** |
| | (6.60) | (0.005) | (0.013) | (0.020) | (0.001) | (0.004) | (0.005) |
| Subsidy Amount$_{i,t-2}$ × Population$_i$ | -30.59*** | -0.013** | -0.038*** | 0.049 | 0.001 | -0.019*** | -0.037*** |
| | (7.78) | (0.006) | (0.009) | (0.052) | (0.001) | (0.006) | (0.005) |
| Subsidy Amount$_{i,t-2}$ × Urbanization$_i$ | -23.74*** | 0.018** | 0.008 | 0.062*** | -0.001 | -0.001 | 0.007 |
| | (8.87) | (0.008) | (0.011) | (0.022) | (0.002) | (0.005) | (0.005) |
| Subsidy Amount$_{i,t-2}$ × Land Value$_i$ | -12.49 | 0.034*** | 0.003 | -0.027 | -0.001 | -0.003 | 0.005 |
| | (11.67) | (0.009) | (0.010) | (0.061) | (0.002) | (0.007) | (0.008) |
| Observations | 9,625 | 9,625 | 8,586 | 8,586 | 9,625 | 9,605 | 9,605 |
| R-squared | 0.83 | 0.71 | 0.89 | 0.80 | 0.52 | 0.61 | 0.87 |
| $F$-statistic | 207.38 | 207.38 | 207.38 | 207.38 | 207.38 | 207.38 | 207.38 |
| Mean of Dep. Variable | 865 | 0.78 | 16 | 8,268 | 0.98 | 10 | 12,738 |
| *Panel B: Rugged Counties* | | | | | | | |
| Subsidy Amount$_{i,t-2}$ | 17.82*** | -0.008 | 0.024* | -0.018 | -0.000 | 0.005 | 0.010 |
| | (4.37) | (0.008) | (0.012) | (0.013) | (0.003) | (0.007) | (0.009) |
| Subsidy Amount$_{i,t-2}$ × Population$_i$ | -11.23 | -0.005 | -0.016 | 0.026 | 0.008** | -0.006 | -0.009 |
| | (7.08) | (0.009) | (0.015) | (0.023) | (0.004) | (0.012) | (0.012) |
| Subsidy Amount$_{i,t-2}$ × Urbanization$_i$ | -8.78 | 0.006 | 0.017 | 0.031 | 0.009** | 0.009 | 0.033*** |
| | (12.51) | (0.017) | (0.018) | (0.029) | (0.004) | (0.012) | (0.013) |
| Subsidy Amount$_{i,t-2}$ × Land Value$_i$ | -4.11 | -0.008 | 0.010 | -0.006 | 0.007** | -0.012 | 0.022 |
| | (13.63) | (0.010) | (0.012) | (0.018) | (0.003) | (0.009) | (0.018) |
| Observations | 3,068 | 3,068 | 2,844 | 2,839 | 3,068 | 3,020 | 3,020 |
| R-squared | 0.84 | 0.69 | 0.84 | 0.74 | 0.61 | 0.57 | 0.79 |
| $F$-statistic | 15.20 | 15.20 | 15.20 | 15.20 | 15.20 | 15.20 | 15.20 |
| County Controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 302 | 0.82 | 13 | 8,720 | 0.94 | 8 | 9,023 |

**Table A.7.** Effect of Alternative Measures of Farm Concentration on Banking Concentration

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on banking concentration using two alternative measures of farm concentration. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). The outcome variable is $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. In Column 1, farm concentration is measured by $Share\ of\ Acres\ on$ 1,000+ $Acre\ Farms_{i,t-2}$, the share of harvested acres on farms of 1,000 acres or more. In Column 2, it is measured by $Log\ Number\ of\ Farms_{i,t-2}$, the log number of farms, for which a negative coefficient corresponds to greater consolidation. Both measures are lagged, constructed from the Census of Agriculture acreage bins, and standardized within county. The controls are the same set of first- and second-stage controls as in Table 3. $F$-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) |
|---|---|---|
| | Bank Deposits HHI | |
| $\widehat{\text{Share of Acres on 1,000+ Acre Farms}}_{i,t-2}$ | 189.89*** | |
| | (27.63) | |
| $\widehat{\text{Log Number of Farms}}_{i,t-2}$ | | -335.81*** |
| | | (51.98) |
| Observations | 13,762 | 13,808 |
| $F$-statistic | 369.86 | 167.14 |
| County Controls | Yes | Yes |
| Deregulation Index FE | Yes | Yes |
| County & Year FE | Yes | Yes |
| Mean of Dep. Variable | 3,114 | 3,123 |

**Table A.8.** First Differences: Effect of Farm Concentration on Banking Concentration

This table reports first- and second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation change in farm concentration on the change in banking concentration. This specification uses first differences, where the first and second stages, respectively, are: $\Delta Farm\ Concentration_{i,t} = \beta_1\ \Delta Subsidy\ Amount_{i,t} + \Delta \mathbf{X}'_{i,t}\mathbf{\Gamma} + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $\Delta Y_{i,t} = \beta_2\ \Delta Farm\ \widehat{Concentration}_{i,t-2} + \Delta \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Columns 1 and 2, the outcome variable is $\Delta Farm\ Concentration_{i,t}$, the change in the midpoint farm size at the county level. In Columns 3 and 4, the outcome variable is $\Delta Bank\ Deposits\ HHI_{i,t}$, the change in $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. $\Delta Subsidy\ Amount_{i,t}$ is a continuous within-county standardized measure of the predicted change in total fixed subsidy payments in a given county-year. $\Delta Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized change in the midpoint farm size. The controls are the same set of first- and second-stage controls as in Table 3, but in first differences. $F$-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | First Stage | | Second Stage | |
| | Δ Farm Concentration | | Δ Bank Deposits HHI | |
| Δ $\widehat{\text{Farm Concentration}}_{i,t-2}$ | | | 123.30* | 158.00** |
| | | | (74.12) | (65.66) |
| Δ Subsidy Amount$_{i,t-2}$ | 36.55*** | 44.78*** | | |
| | (3.92) | (4.22) | | |
| Δ Log Income Per Capita$_{i,t-2}$ | | 80.89 | | -53.02 |
| | | (65.59) | | (90.13) |
| Δ Employment Rate$_{i,t-2}$ | | -83.82 | | 60.81 |
| | | (127.95) | | (275.95) |
| Δ Idle Cropland$_{i,t-7}$ | | 20.43*** | | 4.54 |
| | | (4.34) | | (6.35) |
| Δ Unharvested Cropland$_{i,t-7}$ | | 15.78*** | | -7.20 |
| | | (4.18) | | (7.00) |
| Δ Pastureland$_{i,t-7}$ | | 5.89* | | -6.39 |
| | | (3.34) | | (5.39) |
| Observations | 13,166 | 13,166 | 13,166 | 13,166 |
| $F$-statistic | 86.94 | 112.83 | 86.94 | 112.83 |
| Deregulation Index FE | No | Yes | No | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 119 | 119 | 13 | 13 |

**Table A.9.** Effect of Subsidy Amount and Farm Concentration on Banks' Interest Expense Betas

This table reports reduced-form estimates of the effect of subsidy payments and second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on county-level bank interest expense betas. Columns 1 and 3 report the reduced form: $Y_{i,t} = \alpha\, Subsidy\ Amount_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$. Columns 2 and 4 report second-stage IV estimates, where the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\, Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\, \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Columns 1 and 2, the outcome variable is the total interest expense beta of Drechsler, Savov, and Schnabl (2021), the sensitivity of banks' total interest expense per dollar of assets to the federal funds rate. In Columns 3 and 4, the outcome variable is the deposit interest expense beta, the sensitivity of banks' interest expense on deposits per dollar of deposits to the federal funds rate. A lower beta indicates greater deposit market power. The outcome variables are standardized by their within-county standard deviations. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. The controls are the same set of first- and second-stage controls as in Columns 1, 2, 4, and 5 of Table 4. $F$-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. $^{***}$, $^{**}$, and $^{*}$ represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Total Interest Expense Beta | | Deposit Interest Expense Beta | |
| Subsidy Amount$_{i,t-2}$ | -0.058*** | | -0.041*** | |
| | (0.016) | | (0.014) | |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | | -0.234*** | | -0.183*** |
| | | (0.068) | | (0.063) |
| Observations | 13,621 | 13,621 | 13,621 | 13,621 |
| $F$-statistic | 4.63 | 89.57 | 3.56 | 89.57 |
| County Controls | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 17 | 17 | 16 | 16 |

**Table A.10.** Effect of Farm Concentration on Banking Concentration, Correcting for Attenuation Bias

This table reports ordinary least squares (OLS) and instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on banking concentration in twenty-year differences, where $\Delta$ denotes the change between 1994 and 2014. Column 1 reports the OLS specification: $\Delta Y_{i,t} = \beta\, \Delta \textit{Farm Concentration}_{i,t-2} + \Delta \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \delta_t + \varepsilon_{i,t}$. Columns 2 and 3 report second-stage IV estimates, where the second stage is $\Delta Y_{i,t} = \beta_2\, \Delta \widehat{\textit{Farm Concentration}}_{i,t-2} + \Delta \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \delta_t + \epsilon_{i,t}$. In Column 2, the first stage is $\Delta \textit{Farm Concentration}_{i,t} = \pi_1\, \Delta^{1999,2009} \textit{Farm Share}_i + \pi_2\, \Delta^{1999,2009} \textit{Land Share}_i + \Delta \mathbf{X}'_{i,t}\mathbf{\Gamma} + \delta_t + \nu_{i,t}$, where $\Delta^{1999,2009}$ denotes the change between 1999 and 2009. In Column 3, the first stage is $\Delta \textit{Farm Concentration}_{i,t} = \beta_1\, \Delta \textit{Subsidy Amount}_{i,t} + \Delta \mathbf{X}'_{i,t}\mathbf{\Gamma} + \delta_t + \nu_{i,t}$. Subscript $i$ represents the county and $t$ represents the year. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). The sample is restricted to counties observed in both 1994 and 2014 whose farm share and land share are also observed in 1999 and 2009. The outcome variable, $\Delta Y_{i,t}$, is the change in *Bank Deposits HHI*$_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. *Farm Concentration*$_{i,t-2}$ is the lagged within-county standardized midpoint farm size. *Farm Share*$_{i,t}$ is the within-county standardized share of farms with harvested cropland that harvest at least 1,000 acres, and *Land Share*$_{i,t}$ is the within-county standardized share of land in farms on farms of at least 1,000 acres. *Subsidy Amount*$_{i,t}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year. The controls are the same set of first- and second-stage controls as in Column 2 of Panel B of Table 3, in twenty-year differences and without the deregulation index. Columns 1 and 2 include only the second-stage controls. $F$-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. $^{***}$, $^{**}$, and $^{*}$ represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) |
|---|---|---|---|
| | $\Delta$ Bank Deposits HHI | | |
| $\Delta$ Farm Concentration$_{i,t-2}$ | 35.16** | | |
| | (17.09) | | |
| $\Delta$ Farm $\widehat{\text{Concentration}}_{i,t-2}$ | | 221.59*** | 317.29*** |
| | | (48.92) | (71.44) |
| Observations | 2,340 | 2,340 | 2,340 |
| $F$-statistic | 2.63 | 142.53 | 76.47 |
| County Controls | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes |
| County FE | No | No | No |
| Mean of Dep. Variable | -33 | -33 | -33 |

**Table A.11.** Decomposition of the IV–OLS Gap in the Effect of Farm Concentration on Banking Concentration

This table decomposes the gap between the instrumental variable (IV) and ordinary least squares (OLS) estimates of the effect of farm concentration on banking concentration following Ishimaru (2024). The baseline specification is: $Y_{i,t} = \beta \, \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). The outcome variable is $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. The decomposition separates the IV–OLS gap, $\hat{\beta}_{IV} - \hat{\beta}_{OLS}$, into three components: differences in covariate weights, treatment-level weights, and marginal effects. Columns 1 and 3 use a cubic polynomial in farm concentration as the basis function. Columns 2 and 4 use a fifth-order polynomial. The baseline controls are the same set of first- and second-stage controls as in Column 2 of Panel B of Table 3. Columns 3 and 4 augment the baseline controls with the log market value of agricultural land and buildings per acre from the Census of Agriculture and the log of total county deposits. $F$-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. $^{***}$, $^{**}$, and $^{*}$ represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Bank Deposits HHI | | | |
| Δ Covariate Weight | -1.97 | -1.97 | 4.67 | 4.67 |
| | ( 3.92) | ( 3.92) | ( 4.62) | ( 4.62) |
| Δ Treatment-Level Weight | 4.49 | 10.81* | 4.86 | 11.27* |
| | ( 3.86) | ( 6.10) | ( 3.52) | ( 6.03) |
| Δ Marginal Effect | 255.25*** | 248.93*** | 306.34*** | 299.93*** |
| | (39.46) | (39.14) | (40.61) | (40.22) |
| Observations | 13,808 | 13,808 | 13,790 | 13,790 |
| $F$-statistic | 333.80 | 333.80 | 334.14 | 334.14 |
| Marginal Effect / Gap (%) | 99.0 | 96.6 | 97.0 | 95.0 |
| Cubic Polynomial Basis | Yes | No | Yes | No |
| Quintic Polynomial Basis | No | Yes | No | Yes |
| Collateral & Deposit Controls | No | No | Yes | Yes |
| County Controls | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 3,123 | 3,123 | 3,123 | 3,123 |

**Table A.12.** Excluding 2009: Effect of Farm Concentration on Large and Small Banks

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on the presence, branches, and deposits of large and small banks, excluding the 2008 Financial Crisis. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Columns 1 and 4 and $E\big[Y_{i,t}\,\big|\,Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). The sample is further restricted to exclude 2009. In Columns 1 and 4, the outcome variable is an indicator that a large or small bank, respectively, is present in the county in a given year. In Columns 2 and 5, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 3 and 6, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small banks, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 4. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Large Banks | | | Small Banks | | |
| | Probability of Entry | Branches | Deposits Per Capita | Probability of Entry | Branches | Deposits Per Capita |
| $\widehat{\text{Farm Concentration}}_{i,t-2}$ | 0.049* | 0.178*** | 0.361** | 0.024*** | 0.100*** | 0.273*** |
| | (0.027) | (0.058) | (0.170) | (0.008) | (0.029) | (0.043) |
| Observations | 11,154 | 9,759 | 9,755 | 11,154 | 11,021 | 11,021 |
| $F$-statistic | 246.65 | 246.65 | 131.41 | 246.65 | 246.65 | 131.41 |
| County Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 0.76 | 14 | 8,128 | 0.96 | 9 | 12,111 |

**Table A.13.** Controlling for Land Values, Deposits, and Production Outcomes: Effect of Subsidy Amount and Farm Concentration on Large and Small Banks

This table reports reduced-form estimates of the effect of subsidy payments (Panel A) and second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration (Panel B) on the presence, branches, and deposits of large and small banks, controlling directly, in both stages, for land values, total county deposits, and the production outcomes reported in Table A.5. In Panel A, the specifications are $Y_{i,t} = \alpha\, \textit{Subsidy Amount}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ in Columns 1 and 4, and $E\big[Y_{i,t} \,\big|\, \textit{Subsidy Amount}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\alpha\, \textit{Subsidy Amount}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. In Panel B, the specification for the first stage is the specification of Columns 1 and 4 in Panel A with $\textit{Farm Concentration}_{i,t-2}$ as the outcome variable. The specifications for the second stages are $Y_{i,t} = \beta_2\, \widehat{\textit{Farm Concentration}}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Columns 1 and 4 and $E\big[Y_{i,t} \,\big|\, \textit{Farm Concentration}_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\, \textit{Farm Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\, \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns using a control function approach. In the table, I include a hat on $\textit{Farm Concentration}_{i,t-2}$ to indicate this is an IV regression. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). The sample uses banking years 1994–2009. In Columns 1 and 4, the outcome variable is an indicator that a large or small bank, respectively, is present in the county in a given year. In Columns 2 and 5, the outcome variable is $\textit{Branches}_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 3 and 6, the outcome variable is $\textit{Deposits Per Capita}_{i,t}$, which is the deposits per capita in large and small banks, respectively. $\textit{Subsidy Amount}_{i,t-2}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year. $\textit{Farm Concentration}_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than \$1 billion in inflation-adjusted assets, and large otherwise. The baseline controls are the same set of controls as in Columns 2 through 7 of Table 2 in Panel A and Table 4 in Panel B, augmented by the following set of controls. The first is $\textit{Log Land Value Per Acre}_{i,t-2}$, the log market value per acre of agricultural land and buildings from the Census of Agriculture. The second is $\textit{Log Total County Deposits}_{i,t-2}$. The third is the production controls, the Törnqvist crop-yield index, machinery stock, tractor investment, and program crop share. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. $^{***}$, $^{**}$, and $^{*}$ represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | | Large Banks | | | Small Banks | |
| | Probability of Entry | Branches | Deposits Per Capita | Probability of Entry | Branches | Deposits Per Capita |
| *Panel A: Effect of Subsidy Amount* | | | | | | |
| Subsidy Amount$_{i,t-2}$ | 0.008* | 0.026*** | 0.029*** | 0.002 | 0.015*** | 0.026*** |
| | (0.005) | (0.009) | (0.011) | (0.001) | (0.004) | (0.004) |
| Observations | 11,146 | 9,723 | 9,723 | 11,146 | 11,027 | 11,027 |
| R-squared | 0.73 | 0.88 | 0.78 | 0.64 | 0.62 | 0.88 |
| Mean of Dep. Variable | 0.76 | 14 | 7,883 | 0.97 | 9 | 12,127 |
| *Panel B: Effect of Farm Concentration* | | | | | | |
| Farm Concentration$_{i,t-2}$ | 0.059** | 0.157** | 0.453*** | 0.014** | 0.079*** | 0.283*** |
| | (0.029) | (0.067) | (0.140) | (0.007) | (0.026) | (0.045) |
| Observations | 11,146 | 9,723 | 9,723 | 11,146 | 11,027 | 11,027 |
| $F$-statistic | 262.68 | 262.68 | 146.20 | 262.68 | 262.68 | 146.20 |
| County Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 0.76 | 14 | 7,883 | 0.97 | 9 | 12,127 |

**Table A.14.** Small Banks: Effect of Farm Concentration on Farm Lending by Farm Revenue and Loan Size

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on small banks' farm lending by farm revenue and loan size. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t}\,\big|\,Farm\ \ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ \ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data at the annual level (see Section 5 for variable and sample construction details). The sample is restricted to 1996 to 2004, the years in which small banks report under the CRA. The outcome variable is the total volume of farm loans of $500,000 or less that small banks originate in a given county-year, for loans to farms with gross annual revenue above $1 million in Column 1 and $1 million or less in Column 2. In Columns 3 through 5, the outcome variable is the total volume of farm loans at or below $100,000, between $100,000 and $250,000, and between $250,000 and $500,000, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size, carried forward annually until the next census observation. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 6. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| | Farm Revenue | | Loan Size | | |
| | >$1M | ≤$1M | ≤$100K | $100K–$250K | $250K–$500K |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | 0.332 | -0.258 | -0.242 | -0.289 | -0.209 |
| | (0.515) | (0.201) | (0.200) | (0.208) | (0.234) |
| Observations | 20,301 | 24,351 | 24,351 | 20,898 | 17,934 |
| $F$-statistic | 38.34 | 48.34 | 48.11 | 46.88 | 46.72 |
| County Controls | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 412 | 2,563 | 1,250 | 1,111 | 951 |

**Table A.15.** Alternative County Allocation: Effect of Farm Concentration on Agricultural Loan Growth

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on agricultural loan growth by bank size. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t}\,\big|\,Farm\ \ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). Each bank's agricultural lending is allocated across counties under the alternative allocation described in Appendix B.1. In Columns 1 and 2, the outcome variable is $Total\ Agricultural\ Loan\ Growth_{i,t}$, in Columns 3 and 4 it is $Real\ Estate\ Loan\ Growth_{i,t}$, and in Columns 5 and 6 it is $Non\text{-}Real\ Estate\ Loan\ Growth_{i,t}$, each measuring large or small banks' growth in the loan type relative to the prior period's total volume of that loan type. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 7. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. $^{***}$, $^{**}$, and $^{*}$ represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Total Agricultural Loan Growth | | Real Estate Loan Growth | | Non-Real Estate Loan Growth | |
| | Large Banks | Small Banks | Large Banks | Small Banks | Large Banks | Small Banks |
| $\widehat{\text{Farm Concentration}}_{i,t-2}$ | 0.529*** | 0.025 | 0.438** | -0.004 | 0.403*** | 0.051 |
| | (0.144) | (0.040) | (0.179) | (0.053) | (0.122) | (0.039) |
| Observations | 13,780 | 13,735 | 13,769 | 13,711 | 13,766 | 13,702 |
| $F$-statistic | 323.26 | 323.26 | 322.48 | 322.48 | 321.98 | 321.98 |
| County Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 0.49 | 0.73 | 0.54 | 0.89 | 0.56 | 0.66 |

**Table A.16.** Effect of Farm Concentration on Small Banks' Agricultural Lending by Loan Size, Core Funding, and Funding Utilization

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on small banks' agricultural lending, splitting small banks on two predetermined funding measures at once. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t} \,\big|\, Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. High-core small banks are those with a pre-program (1990–1995) core deposit share above the within-small-bank median, and low-core small banks otherwise. Slack small banks are those with a pre-program (1990–1995) ratio of loans to deposits below the within-small-bank median, and at limit otherwise. In Columns 1 and 3, the outcome variable is small banks' growth in real estate farm lending, and in Columns 2 and 4 it is small banks' growth in non-real estate farm lending, each relative to the prior period's total volume of that loan type. In Columns 1 and 2, the outcome variable is the growth in loans above $500,000, and in Columns 3 and 4 it is the growth in loans at or below $100,000. The controls are the same set of first- and second-stage controls as in Table 8. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Agricultural Loan Growth | | | |
| | >$500K | | ≤$100K | |
| | Real Estate | Non-Real Estate | Real Estate | Non-Real Estate |
| High-Core, Slack | 1.391** | 3.008*** | -0.184*** | -0.017 |
| | (0.560) | (0.799) | (0.065) | (0.059) |
| High-Core, At Limit | 0.823** | 0.236 | -0.288*** | -0.162** |
| | (0.349) | (0.374) | (0.071) | (0.071) |
| Low-Core, Slack | 0.098 | 0.388 | -0.303*** | -0.183** |
| | (0.230) | (0.245) | (0.065) | (0.077) |
| Low-Core, At Limit | -0.012 | -0.207 | -0.070 | -0.168** |
| | (0.217) | (0.184) | (0.085) | (0.071) |
| High-Core, Slack − Low-Core, At Limit | 1.404** | 3.215*** | -0.114 | 0.151* |
| | (0.703) | (0.885) | (0.099) | (0.083) |
| High-Core, Slack − High-Core, At Limit | 0.568 | 2.772** | 0.104 | 0.145* |
| | (0.967) | (1.188) | (0.083) | (0.083) |
| High-Core, Slack − Low-Core, Slack | 1.293* | 2.621** | 0.119 | 0.166* |
| | (0.663) | (1.030) | (0.076) | (0.093) |
| Observations | 13,821 | 13,821 | 13,821 | 13,821 |
| $F$-statistic | 322.65 | 322.65 | 322.65 | 322.65 |
| County Controls | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 0.11 | 0.06 | 0.18 | 0.19 |

**Table A.17.** High- vs. Low-Cost Counties: Effect of Farm Concentration on the Presence, Branches, and Deposits of Large and Small Banks

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on the presence, branches, and deposits of large and small banks, in high-cost counties (Panel A) and low-cost counties (Panel B). The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ Farm\ \widehat{Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Columns 1 and 4 and $E\big[Y_{i,t}\,\big|\,Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). Counties are classified as high-cost if their average per acre economic costs are above the nationwide median, and low-cost otherwise. Economic costs per acre for each major commodity crop are calculated as the six-year average (1990–1995) from the Farm Costs and Returns Survey (FCRS) compiled by the U.S. Department of Agriculture's Economic Research Service (ERS). The measure is constructed as a time-invariant, county-level weighted average of per acre economic costs, where weights are each crop's share of harvested acreage from the 1992 Census of Agriculture. In Columns 1 and 4, the outcome variable is an indicator that a large or small bank, respectively, is present in the county in a given year. In Columns 2 and 5, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 3 and 6, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small banks, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size, standardized within the relevant subsample. Small banks are those with less than \$1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 4. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages. Means of the dependent variable are calculated within the relevant subsample. Standard errors clustered at the county level are reported in parentheses. $^{***}$, $^{**}$, and $^{*}$ represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | | Large Banks | | | Small Banks | |
| | Probability of Entry | Branches | Deposits Per Capita | Probability of Entry | Branches | Deposits Per Capita |
| *Panel A: High-Cost Counties* | | | | | | |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | 0.041* | 0.120** | 0.411** | 0.010* | 0.062*** | 0.060** |
| | (0.022) | (0.048) | (0.191) | (0.006) | (0.021) | (0.025) |
| Observations | 6,910 | 6,274 | 6,274 | 6,910 | 6,882 | 6,882 |
| $F$-statistic | 56.84 | 56.84 | 45.47 | 56.84 | 56.84 | 45.47 |
| Mean of Dep. Variable | 0.80 | 14 | 8,181 | 0.97 | 10 | 11,250 |
| *Panel B: Low-Cost Counties* | | | | | | |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | 0.005 | 0.077** | 0.070 | 0.013** | 0.122*** | 0.197*** |
| | (0.022) | (0.037) | (0.057) | (0.006) | (0.026) | (0.028) |
| Observations | 6,898 | 5,957 | 5,952 | 6,898 | 6,785 | 6,785 |
| $F$-statistic | 53.76 | 53.76 | 50.83 | 53.76 | 53.76 | 50.83 |
| County Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 0.76 | 15 | 8,772 | 0.95 | 8 | 12,632 |

**Table A.18.** Alternative Definition of Agricultural Banks: Effect of Farm Concentration on Large and Small Banks

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on the presence, branches, and deposits per capita of large and small agricultural (ag) banks. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Columns 1 and 4 and $E\big[Y_{i,t} \,\big|\, Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). The sample is restricted to ag banks, those with farm loans exceeding 1% of total loans in any of the preceding six years. In Columns 1 and 4, the outcome variable is an indicator that a large or small ag bank, respectively, is present in the county in a given year. In Columns 2 and 5, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small ag banks, respectively. In Columns 3 and 6, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small ag banks, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Columns 2 through 7 of Table 9. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. $^{***}$, $^{**}$, and $^{*}$ represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Large Ag Banks | | | Small Ag Banks | | |
| | Probability of Entry | Branches | Deposits Per Capita | Probability of Entry | Branches | Deposits Per Capita |
| $\widehat{\text{Farm Concentration}}_{i,t-2}$ | 0.069*** | 0.231*** | 0.607*** | 0.023*** | 0.037* | 0.228*** |
| | (0.023) | (0.064) | (0.120) | (0.008) | (0.022) | (0.032) |
| Observations | 13,808 | 11,912 | 11,897 | 13,808 | 13,432 | 13,432 |
| $F$-statistic | 331.31 | 331.31 | 185.11 | 331.31 | 331.31 | 185.11 |
| County Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 0.67 | 7 | 5,032 | 0.90 | 7 | 10,502 |

**Table A.19.** Effect of Farm Concentration on the Best and Worst Offered Interest Rates, 2002 to 2014

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on the best and worst real interest rates offered by banks in the county. The specifications for the first and second stages, respectively, are: $\textit{Farm Concentration}_{i,t} = \beta_1 \textit{Subsidy Amount}_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2 \widehat{\textit{Farm Concentration}}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The county-year data are from merged bank and agriculture data at the annual level (see Section 5 for variable and sample construction details). In Panel A, the outcome variable is the best interest rate offered by any bank branch in the county, which is the highest rate for deposit accounts and the lowest rate for auto loans and residential mortgage loans. In Panel B, the outcome variable is the worst interest rate offered by any bank branch in the county, which is the lowest rate for deposit accounts and the highest rate for auto loans and residential mortgage loans. Each branch's rate is its June rate. In Columns 1 through 5, the outcome variables are the offered interest rates on savings deposit accounts ($10,000 money market accounts), time deposit accounts (12-month $10,000 certificates of deposit), checking deposit accounts (interest-bearing checking accounts), auto loans, and residential mortgage loans, all expressed in real, inflation-adjusted terms. Coefficients are reported in percentage points. $\textit{Farm Concentration}_{i,t-2}$ is the lagged within-county standardized midpoint farm size, carried forward annually until the next census observation. The controls are the same set of first- and second-stage controls as in Table 10. $F$-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| | Deposit Rates | | | Loan Rates | |
| | Savings | Time | Checking | Auto | Mortgage |
| *Panel A: Best Offered Rate in the County* | | | | | |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | -0.298*** | 0.106** | -0.273*** | 1.084*** | 0.050 |
| | (0.079) | (0.044) | (0.051) | (0.228) | (0.065) |
| Observations | 34,828 | 34,834 | 34,786 | 25,770 | 23,178 |
| $F$-statistic | 323.16 | 323.16 | 323.16 | 323.16 | 323.16 |
| Mean of Dep. Variable | -0.76 | 0.17 | -1.40 | 3.85 | 2.94 |
| *Panel B: Worst Offered Rate in the County* | | | | | |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | -0.181*** | 0.048 | -0.080*** | 0.577*** | -0.093 |
| | (0.048) | (0.059) | (0.028) | (0.186) | (0.104) |
| Observations | 34,828 | 34,834 | 34,786 | 25,770 | 23,178 |
| $F$-statistic | 323.16 | 323.16 | 323.16 | 323.16 | 323.16 |
| County Controls | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | -1.54 | -0.53 | -1.78 | 5.62 | 3.59 |

**Table A.20.** First Differences: Effect of Farm Concentration on Offered Interest Rates, 2003 to 2014

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation change in farm concentration on the change in the real interest rates offered by banks. This specification uses first differences, where the first and second stages, respectively, are: $\Delta Farm\ Concentration_{i,t} = \beta_1\ \Delta Subsidy\ Amount_{i,t} + \Delta \mathbf{X}'_{i,t}\mathbf{\Gamma} + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $\Delta Y_{i,t} = \beta_2\ \Delta Farm\ \widehat{Concentration}_{i,t-2} + \Delta \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The county-year data are from merged bank and agriculture data at the annual level (see Section 5 for variable and sample construction details). In Columns 1 through 5, the outcome variables are the changes in the county-level average offered interest rates on savings deposit accounts ($10,000 money market accounts), time deposit accounts (12-month $10,000 certificates of deposit), checking deposit accounts (interest-bearing checking accounts), auto loans, and residential mortgage loans, all expressed in real, inflation-adjusted terms. Coefficients are reported in percentage points. $\Delta Farm\ Concentration_{i,t-2}$ is the change in the lagged within-county standardized midpoint farm size, interpolated annually until the next census observation. The controls are the same set of first- and second-stage controls as in Table 10, but in first differences. $F$-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. $^{***}$, $^{**}$, and $^{*}$ represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| | Δ Deposit Rates | | | Δ Loan Rates | |
| | Savings | Time | Checking | Auto | Mortgage |
| Δ Farm $\widehat{\text{Concentration}}_{i,t-2}$ | -0.191*** | 0.093** | -0.123*** | 0.391*** | 0.047 |
| | (0.048) | (0.041) | (0.032) | (0.134) | (0.074) |
| Observations | 29,718 | 29,720 | 29,687 | 20,985 | 18,805 |
| $F$-statistic | 86.68 | 86.68 | 86.68 | 86.68 | 86.68 |
| County Controls | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | -0.07 | -0.13 | -0.00 | -0.31 | -0.17 |

**Table A.21.** Falsification Test: Effect of Farm Concentration on Federal Government Loans

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on direct loans from the Farm Service Agency. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t}\,\big|\,Farm\ \ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Columns 1 and 2, the outcome variables are $Direct\ Loan\ Amount_{i,t}$, in \$1,000s, and $Number\ of\ Direct\ Loans_{i,t}$, respectively, for direct loans issued at the county level by the federal government. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. The controls are the same set of first- and second-stage controls as in Table 4. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) Direct Loan Amount | (2) Number of Direct Loans |
|---|---|---|
| $\widehat{\text{Farm Concentration}}_{i,t-2}$ | 0.170 | -0.142 |
| | (0.211) | (0.135) |
| Observations | 12,556 | 12,556 |
| $F$-statistic | 322.68 | 322.68 |
| County Controls | Yes | Yes |
| Deregulation Index FE | Yes | Yes |
| County & Year FE | Yes | Yes |
| Mean of Dep. Variable | 474 | 13 |

**Table A.22.** Effect of Farm Concentration on Banking Concentration, Controlling for Crop Insurance Coverage

This table reports first- and second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on banking concentration, controlling for crop insurance coverage. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \beta_3 Log\ Crop\ Insurance\ Policies_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ \widehat{Farm\ Concentration}_{i,t-2} + \beta_4 Log\ Crop\ Insurance\ Policies_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Columns 1 and 2, the outcome variable is $Farm\ Concentration_{i,t}$, the midpoint farm size at the county level. In Columns 3 and 4, the outcome variable is $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. $Subsidy\ Amount_{i,t}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. $Log\ Crop\ Insurance\ Policies_{i,t}$ is the log number of commodity crop insurance policies in the county-year. The controls are the same set of first- and second-stage controls as in Table 3. $F$-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | First Stage | | Second Stage | |
| | Farm Concentration | | Bank Deposits HHI | |
| Farm $\widehat{\text{Concentration}}_{i,t-2}$ | | | 251.84*** | 274.57*** |
| | | | (38.67) | (41.53) |
| Log Crop Insurance Policies$_{i,t-2}$ | 23.41*** | 22.87*** | 10.63 | 5.81 |
| | (4.29) | (4.34) | (9.16) | (9.47) |
| Subsidy Amount$_{i,t-2}$ | 105.03*** | 106.58*** | | |
| | (5.52) | (5.90) | | |
| Log Income Per Capita$_{i,t-2}$ | | 192.10*** | | -86.11 |
| | | (69.56) | | (125.27) |
| Employment Rate$_{i,t-2}$ | | -109.82 | | 322.74 |
| | | (156.08) | | (328.18) |
| Idle Cropland$_{i,t-7}$ | | 20.72*** | | -12.13 |
| | | (4.98) | | (7.90) |
| Unharvested Cropland$_{i,t-7}$ | | -11.91*** | | 7.55 |
| | | (3.61) | | (7.14) |
| Pastureland$_{i,t-7}$ | | 3.99 | | -1.93 |
| | | (4.30) | | (5.87) |
| Observations | 13,808 | 13,808 | 13,808 | 13,808 |
| $F$-statistic | 362.21 | 326.25 | 362.21 | 326.25 |
| Deregulation Index FE | No | Yes | No | Yes |
| County & Year FE | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 752 | 752 | 3,123 | 3,123 |

**Table A.23.** Effect of Subsidy Amount and Farm Concentration on Banking Concentration and Large and Small Banks, Controlling for Countercyclical Payments (CCP)

This table reports reduced-form estimates of the effect of subsidy payments on farm concentration and banking concentration (Columns 1 and 2) and second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on the branches and deposits of large and small banks (Columns 3 through 6), controlling for countercyclical payments (CCP). In Columns 1 and 2, the specification is $Y_{i,t} = \alpha\, Subsidy\ Amount_{i,t-2} + \theta\, CCP_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$. In Columns 3 through 6, the specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \theta_1\, CCP_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t}\,\big|\, Farm\ \ Concentration_{i,t-2}, CCP_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \theta_2\, CCP_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$, estimated using a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Column 1, the outcome variable is $Farm\ Concentration_{i,t}$, the midpoint farm size at the county level. In Column 2, the outcome variable is $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. In Columns 3 and 5, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 4 and 6, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small banks, respectively. $Subsidy\ Amount_{i,t-2}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year. $CCP_{i,t-2}$ is a continuous within-county standardized measure of the predicted value of countercyclical payments in a given county-year. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first-stage controls as in Table 3 in Column 1, the same set of controls as in Column 4 of Panel A of Table 3 in Column 2, and the same set of first- and second-stage controls as in Table 4 in Columns 3 through 6. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages in Columns 3 through 6. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | | | Large Banks | | Small Banks | |
| | Farm Concentration | Bank Deposits HHI | Branches | Deposits Per Capita | Branches | Deposits Per Capita |
| Subsidy Amount$_{i,t-2}$ | 107.89*** | 63.57*** | | | | |
| | (5.91) | (8.85) | | | | |
| $\widehat{\text{Farm Concentration}}_{i,t-2}$ | | | 0.164*** | 0.333** | 0.097*** | 0.222*** |
| | | | (0.051) | (0.132) | (0.023) | (0.031) |
| CCP$_{i,t-2}$ | 11.19*** | 3.45 | 0.001 | -0.000 | -0.009*** | -0.009*** |
| | (2.55) | (3.57) | (0.002) | (0.005) | (0.002) | (0.002) |
| Observations | 13,808 | 13,808 | 12,231 | 12,226 | 13,667 | 13,667 |
| $F$-statistic | 333.61 | 333.61 | 322.44 | 174.79 | 322.44 | 174.79 |
| County Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 752 | 3,123 | 14 | 8,468 | 9 | 11,936 |

**Table A.24.** Effect of Farm Concentration on Banking Concentration and Large and Small Banks, Controlling for Farm Credit System (FCS) Branches

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on banking concentration and on the branches and deposits of large and small banks, controlling for Farm Credit System (FCS) branches. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \beta_3 Log\ FCS\ Branches_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ \widehat{Farm\ Concentration}_{i,t-2} + \beta_4 Log\ FCS\ Branches_{i,t} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Column 1 and $E\big[Y_{i,t} \,\big|\, Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, Log\ FCS\ Branches_{i,t}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \beta_4 Log\ FCS\ Branches_{i,t} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_5\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript $i$ represents the county, $t$ represents the year, and $d$ represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Column 1, the outcome variable is $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. In Columns 2 and 4, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 3 and 5, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small banks, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. $Log\ FCS\ Branches_{i,t}$ is the log number of Farm Credit System (FCS) branches in the county in year $t$. Small banks are those with less than \$1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 4. $F$-statistics are from the first stage. Coefficients are reported as decimal percentages in Columns 2 through 5. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| | | Large Banks | | Small Banks | |
| | Bank Deposits HHI | Branches | Deposits Per Capita | Branches | Deposits Per Capita |
| $\widehat{\text{Farm Concentration}}_{i,t-2}$ | 273.03*** | 0.159*** | 0.313*** | 0.101*** | 0.235*** |
| | (40.74) | (0.050) | (0.115) | (0.023) | (0.032) |
| Log FCS Branches$_{i,t}$ | | 0.007 | 0.082 | -0.015 | -0.069*** |
| | | (0.024) | (0.138) | (0.023) | (0.023) |
| Observations | 13,808 | 12,231 | 12,226 | 13,667 | 13,667 |
| $F$-statistic | 329.23 | 316.79 | 172.33 | 316.79 | 172.33 |
| County Controls | Yes | Yes | Yes | Yes | Yes |
| Deregulation Index FE | Yes | Yes | Yes | Yes | Yes |
| County & Year FE | Yes | Yes | Yes | Yes | Yes |
| Mean of Dep. Variable | 3,123 | 14 | 8,468 | 9 | 11,936 |

# Appendix B. Data Methods and Technical Details

This appendix provides additional detail on the procedures used to construct key variables and allocate bank- and loan-level data to the county level for empirical analysis.

## B.1 Estimating County-Level Agricultural Lending

Because agricultural loan volumes are reported at the bank level in the FFIEC Call Reports, I allocate each bank's lending across counties based on its branches at the start of each five-year period. For the period ending in year $t$, I use the counties in which the bank operated a branch in year $t-5$, identified from the FDIC's Summary of Deposits. The five-year period matches the frequency with which the Census of Agriculture updates the interest expense weights.

I allocate each bank's agricultural lending in year $t$ across these counties using farm interest expenses from the most recent Census of Agriculture as of year $t-5$. I use interest expenses on real estate loans for real estate farm lending, interest expenses on non-real estate loans for non-real estate farm lending, and their sum for total agricultural lending. For example, if a bank operated branches in three counties with equal farm interest expenses in year $t-5$, each county receives one third of the bank's agricultural lending in year $t$. I allocate the prior period's lending across the same counties using the same weights. I repeat this procedure for every bank and period, winsorize the allocated lending of each bank-county pair at the 1st and 99th percentiles, and aggregate it by county separately for small and large banks.

Table A.15 reports results under an alternative allocation that also assigns lending to counties in which a bank lends without operating a branch. The alternative footprint follows Key, Burns, and Lyons (2019) and Ifft et al. (2024), which use the CRA data to identify counties in which a bank lends outside its branch network. The alternative footprint adds to the bank's branch counties in year $t$ the counties in which the bank originated farm loans during the five years ending in year $t$, identified using the CRA data. I use originations with original amounts of \$500,000 or less and match CRA respondents to the Call Reports. Because CRA reporting begins in 1996 and applies only to institutions above an asset threshold, this footprint contains branch counties alone for all years before 1996 and whenever CRA data are unavailable. The alternative allocation uses farm interest expenses from the most recent Census of Agriculture as of year $t$ and decreases the weight assigned to counties in which the bank does not operate a branch. I multiply the county's interest expenses by $(1+r_{bi})^{\theta}$, where $r_{bi}$ is the distance in kilometers between county $i$'s banking centroid and the nearest branch of bank $b$. I define the banking centroid as the average coordinates of all bank

branches in the county. Counties in which the bank operates a branch receive no distance adjustment. I estimate $\theta$ from a Poisson regression of the bank's farm loan originations in each county on log distance, log farm interest expenses, and an indicator for whether the bank operates a branch in the county, with fixed effects for each bank and year combination. The estimate of $\theta$ is $-0.23$, consistent with the distance deterrence in commercial lending documented by Brevoort and Hannan (2006). I apply the same distance adjustment to real estate and non-real estate farm lending.

To assess the distance adjustment, I estimate the decay parameter using a random half of the CRA reporting banks and evaluate the resulting county shares using the remaining banks. Among the remaining banks, the correlation between the allocated county shares and observed origination shares is 0.63, compared with 0.56 when I use the same interest expense weights without the distance adjustment. For example, suppose a bank's lending set contains three counties with equal farm interest expenses. The bank operates branches in two counties and recently originated farm loans in a third county 100 kilometers from its nearest branch. Each branch county receives an unadjusted weight of one, and the third county receives a weight of $(1 + 100)^{-0.23} = 0.35$, after rounding. Normalizing these weights assigns 42.6% of the bank's agricultural lending to each branch county and 14.7% to the county outside its branch network. I repeat this procedure for every bank and year and aggregate the allocated lending by county, separately for small and large banks.

Two limitations apply to both allocations. First, the main allocation assumes that a bank lends only in its branch counties, and the CRA data that relax this assumption cover only institutions above the reporting threshold and only farm loans of $500,000 or less. Second, both allocations assume that past interest expenses measure current county loan demand, and that lending remains proportional to the resulting weights.

## B.2 Loan Size Categories and Funding Measures

Call Reports list each bank's outstanding farm loans of $500,000 or less by original loan amount in three categories, $100,000 or less, between $100,000 and $250,000, and between $250,000 and $500,000. The Call Reports provide these categories separately for real estate farm loans and non-real estate farm loans. I construct the category above $500,000 by subtracting the three reported categories from the bank's total farm lending in the Call Reports. The reporting thresholds are fixed in nominal terms. I express loan volumes in constant dollars using the CPI but retain the reported origination categories. I allocate each bank's loans in each category across the set of counties described above using the same interest expense weights, then aggregate the allocated lending by county, separately for small and large banks.

The funding measures are fixed before the subsidy program begins. The core deposit share equals total deposits less time deposits of $100,000 or more and brokered deposits, divided by total deposits, averaged over 1990–1995. Core deposits are the stable component of bank funding, and institutions that rely more heavily on them sustain lending when funding conditions tighten (Berlin and Mester, 1999; Cornett et al., 2011). The loans-to-deposits ratio, averaged over the same period, measures how much of the deposit base is already deployed in lending. A lower ratio indicates that more deposit funding remains available for additional loans (Acharya and Mora, 2015). I calculate each median among small banks, classify small banks as above or below it, and hold these classifications fixed throughout the analysis.

## B.3 Community Reinvestment Act Loans

Table 6 uses farm loan originations that banks report under the CRA. Each reporting bank lists the dollar amount of its farm loans with original amounts of $500,000 or less in each year by the county of the borrowing farm. I use originations from 1996 to 2014 and match each CRA lender to the Call Reports.

Only banks above an asset threshold report these originations. Through 2004, a bank reports if it has $250 million or more in total assets or belongs to a holding company with $1 billion or more in total assets. From 2005, the threshold is about $1 billion in total assets, adjusted each year for inflation. Most small banks stop reporting after 2004, so the small bank sample ends in 2004.

The CRA data report loans to farms with revenue of $1 million or less. I measure loans to farms with revenue above $1 million as the remaining loans, which also include loans to farms whose revenue the bank does not report. In 113 of the 353,207 bank-county-year observations, all in 1996, the reported amount to farms with revenue of $1 million or less exceeds the bank's total originations. I set the remaining loans in these observations to zero. Each bank also divides its originations into three categories of original amount, $100,000 or less, above $100,000 and up to $250,000, and above $250,000 and up to $500,000. These thresholds are fixed in nominal terms. I express amounts in thousands of 2020 dollars using the CPI and aggregate them by county and year, separately for small and large banks, with bank size measured in the year of the origination.

## B.4 Instrument Construction and Interpolation

I construct predicted subsidy payments from county base acres and national crop-level subsidy allocations. Base acres initially reflect planting between 1981 and 1985. Beginning in 2002, I use updated base acres based on planting between 1998 and 2001 and include

oilseeds and peanuts, which became eligible under the Farm Security and Rural Investment Act. Actual payments also depended on historical program yields, but I exclude yields because they may be correlated with local productivity. Including historical yields produces similar estimates.

For each county and year, I multiply its share of national base acres in each crop by the corresponding national subsidy allocation and sum across crops. Base acres remain fixed within each policy period, and national subsidy allocations vary across crops and over time. Counties therefore experience different changes in predicted payments based on their historical crop shares. County fixed effects absorb permanent differences across counties, and year fixed effects absorb national changes common to all counties. The resulting measure is the continuous treatment used in the reduced-form and first-stage specifications.

In the five-year panel, I calculate predicted payments in each Census of Agriculture year and use them as the instrument for farm concentration. The annual RateWatch panel requires annual values of farm concentration, but the Census measures farm structure every five years. I estimate fitted farm concentration in each census year and carry the most recent fitted value forward until the next census. Annual local economic controls continue to vary in the second-stage regressions. Observed farm concentration changes slowly, increasing by 2.5% per year on average over the sample. To estimate the specification in first differences (Table A.20), I instead interpolate fitted farm concentration linearly between census years.

## B.5 Lag Structure and Timing Alignment

In the first stage, I relate farm concentration to predicted payments in the same Census of Agriculture year. In the banking specifications, predicted payments, fitted farm concentration, and the agricultural controls enter with a two-year lag. The lag allows changes in farm structure to precede the banking outcomes. The SOD records the location and deposits of each branch as of June 30 each year. I use each branch's June RateWatch rate, the average of its weekly June quotes, so the interest rate and SOD outcomes refer to the same month.

## B.6 USDA Direct Loan Program

To study whether borrower consolidation affects access to federal agricultural credit, I use administrative data on direct loans from the USDA's Farm Service Agency (FSA). These loans primarily serve family farms and are intended for small, new, or socially disadvantaged farmers who cannot obtain commercial credit at reasonable rates. The FSA defines a family farm as an operation in which members of one family own and operate most of the business. Direct loans support farm ownership, operation, or expansion.

Eligibility does not depend on a minimum credit score. Although applicants must demonstrate reasonable ability to repay and cannot be delinquent on other federal debt, individuals with no formal credit history may still qualify. The program includes both ownership and operating credit. Farm ownership loans can be used to purchase farmland or expand physical infrastructure and carry repayment terms of up to 40 years. Operating loans are designed for inputs such as seed, livestock, or equipment and carry repayment terms of up to seven years. Over the sample period, the loan limit for both programs was $200,000 through 2007 and $300,000 from 2008 onward, following the loan limit increase in the 2008 Farm Bill.

I aggregate annual FSA direct loan records to the county level from 1994 to 2014. The two outcomes are the number of loans issued and their total dollar amount, combining farm ownership and operating loans. These outcomes measure whether borrower consolidation changes the use of federally supported agricultural credit.

## B.7 Total and Deposit Interest Expense Betas

I estimate the interest expense betas in Table A.9 using the quarterly Call Report panel distributed by Drechsler, Savov, and Schnabl (2021). The sample includes commercial banks from 1984 to 2017 with at least 60 quarterly observations and excludes merger quarters and quarters with extreme asset growth. For each bank, I regress the quarterly change in total interest expense per dollar of assets on the contemporaneous change in the federal funds rate and three lags. The total interest expense beta equals the sum of these four coefficients. I construct the deposit interest expense beta in the same manner, using interest expense on deposits divided by total deposits (Narayanan, Ratnadiwakara, and Strahan, 2025). Following Drechsler, Savov, and Schnabl (2021), I winsorize both betas at the 5th and 95th percentiles. County-level betas are deposit-weighted averages across banks operating in the county, using each bank's branch deposits from the SOD in the corresponding census year.

# Internet Appendix

## IA.1 A Model of Farm Consolidation and Bank Lending

This section presents a one-period model of farm borrowing and bank lending. The purpose of the model is to formalize a simple loan pricing framework in which fixed cash payments tied to land, such as agricultural subsidies, lower interest rates and expand access to credit by increasing the borrower's pledgeable assets. In the model, cash is pledgeable in default and acts as collateral, both increasing the lender's recovery value and reducing the set of states in which default occurs. As a result, higher cash holdings allow lenders to extend credit at lower interest rates, and farms that were credit rationed obtain credit. The model also shows that larger farms borrow at lower interest rates, and that merging farms borrow at lower interest rates when consolidation increases returns to scale.

The model then adds two bank types, small and large, that differ in their information about borrowers, their underwriting costs, their recovery rates in default, and how they fund their loans. When farms expand or merge, credit demand becomes concentrated in fewer, larger loans. Small banks' informational advantage declines as farms become larger, and fixed underwriting costs decrease per dollar loaned as loans become larger. Larger loans are also more costly for small banks to fund, because their cost of funds increases with lending relative to their core deposits. As a result, borrower consolidation increases large banks' market share, while only small banks with more funding capacity remain better positioned to lend to larger farms.

I use the model to study four channels through which subsidy payments may affect which bank lends to the farm. First, I show that as farms consolidate, loans become larger, and larger loans are more costly for small banks to fund than for large banks. Second, I show that higher collateral values lower both banks' interest rates at every loan size. Third, I show that deposit inflows benefit small banks. Both the collateral and the deposit effects are larger at small banks with less funding capacity. Fourth, I show that higher returns to production affect which bank lends to the farm only through differences in recovery rates. These predictions are the basis for the empirical tests in the main paper.

Empirically, I find that subsidy payments increase farm concentration and that large banks gain market share. A within-county one-standard-deviation increase in subsidy payments increases farm concentration by 105–107 acres. A within-county one-standard-deviation increase in farm concentration increases lending above $500,000 only at small banks with high-core funding, and mergers among small banks increase by about 23%. Subsidy payments do not increase land values, deposits per capita, or farm production.

### IA.1.1 Setup

Assume there is a farm with size $S$, measured in acres, and let $q$ be the market price of farmland per acre. The farm seeks bank financing for a project requiring upfront funding $L$. The project has a payoff $X \sim N(\mu L, \sigma_j^2 L^2)$, so expected returns and risk scale with project size, where $\mu$ is the expected return per dollar invested and $\sigma_j$ measures payoff uncertainty. The subscript $j$ indexes the lender and is defined below. Assume the farm also receives fixed cash payments $A$, which represent the subsidies tied to its land. I assume cash and investment needs scale with size, $A = \pi_1 S$ and $L = \pi_2 S$, which implies that $A = \pi L$ for some constant $\pi$. If the farm receives a loan at interest rate $r$, it owes $L(1 + r)$ after one period. The farm repays when the project payoff $X$, its cash $A$, and the value of its land $qS$ together cover this obligation, and defaults when $X + A + qS < L(1 + r)$. If the farm defaults, it liquidates and the lender recovers a fraction of its liquidation value, $A + qS + X$, when it is positive. The lending market is competitive, and the farm borrows from the bank that offers the lowest interest rate.

Two bank types, small ($s$) and large ($\ell$), can lend to the farm. They differ in three ways. First, the small bank has a comparative advantage in relationship lending that relies on soft information (DeYoung, Hunter, and Udell, 2004; Berger et al., 2005), so payoff uncertainty when the small bank lends is $\sigma_s(S) = \sigma - \delta(S)$, with $\delta(S) > 0$, $\delta'(S) < 0$, and $\delta(S) \to 0$ as $S$ increases, while payoff uncertainty when the large bank lends is $\sigma_\ell(S) = \sigma$. The advantage $\delta(S)$ is larger for small, opaque farms and declines as borrowers become larger and more transparent (Petersen and Rajan, 1994; Berger et al., 2005). The functional form for $\sigma_j(S)$ should not be interpreted as farm size affecting the farm's risk in absolute terms. It only means that, for a given loan, the small bank's willingness to acquire soft information decreases its uncertainty relative to the large bank. In principle, $\sigma$ could also be a function of size, but for simplicity I assume that it is constant.

Second, the banks differ in underwriting costs and recovery rates. Underwriting a loan costs bank $j$ a fixed amount $F_j$, with $F_\ell > F_s$, because large banks specialize in standardized and scale-intensive lending, which has a high fixed cost (DeYoung, Hunter, and Udell, 2004). The large bank's cost per dollar decreases quickly with loan size, while the small bank's fixed cost is low. If the farm defaults, bank $j$ recovers a fraction $\ell_j \in [0, 1)$ of its liquidation value. I refer to $\ell_j$ as the recovery rate. Whether higher collateral values or higher returns favor the small or the large bank depends on the difference between the two recovery rates, $\ell_\ell - \ell_s$. I assume that the large bank recovers at least as large a fraction as the small bank, $\ell_\ell \geq \ell_s$, because large banks have a comparative advantage in lending against standardized assets such as farmland and equipment (Berger and Udell, 2002; Berger et al., 2005).

Third, the banks differ in how they fund loans. Bank $j$ pays a cost of funds $w_j$ per dollar

loaned. The small bank raises its funds from a local core deposit base $D$, the core deposits it has available for new lending after its existing commitments, and its cost of funds is

$$w_s = w\left(\frac{L}{D}\right),$$

where $w$ is increasing without bound. As long as its lending is small relative to its deposit base, the small bank funds loans with low-cost core deposits. Once lending exceeds the deposit base, the bank must turn to wholesale and brokered funds. Small banks' lending increases with their local deposits, and non-core funds cost more than core deposits (Stein, 1998; Jayaratne and Morgan, 2000; Kashyap and Stein, 2000). The large bank can draw on internal capital markets across its branch network. I therefore assume that $w_\ell$ is constant in $L$ (Stein, 1997; Houston, James, and Marcus, 1997; Campello, 2002). Consistent with this assumption, large banks move funding across markets in the data (Gilje, Loutskina, and Strahan, 2016; Cortés and Strahan, 2017). If large banks simply had larger deposits $D$, the results below would be similar. A bank whose existing loans use most of its deposits has a small $D$, and a merger of two banks pools their unused deposits. In the empirical tests, funding capacity is measured by the pre-program core deposit share and the loans-to-deposits ratio, which correspond to the bank's lending relative to its unused deposit base $D$.

The definitions of $w_s$ and $F_s$ imply that small and large banks face different cost tradeoffs as a function of the size of the loan. Small banks have relatively small fixed costs of lending, but their funding costs imply that large loans are prohibitively expensive, particularly for small banks without larger deposit bases $D$. Large banks have larger fixed costs, but as the loan size increases, this fixed cost disadvantage becomes irrelevant and their cheaper cost of funds $w_\ell$ dominates their lending costs.

### IA.1.2 Equilibrium

For an interest rate $r$, bank $j$'s expected profit from lending $L$ to a farm of type $(S, q)$ is

$$\begin{aligned}\Pi_j(r; S, q) \;=\; E\Big[&\mathbb{1}_{\{X+A+qS<L(1+r)\}}\,\ell_j \max\big(0,\; A + qS + X\big) \\ &+\; \mathbb{1}_{\{X+A+qS\geq L(1+r)\}}\,L(1+r)\Big] \;-\; L\,(1+w_j) \;-\; F_j,\end{aligned}$$

where $w_s = w(L/D)$ and $w_\ell$ is constant. The deposit base $D$ affects the small bank's profits through its cost of funds. Dividing by the loan size $L$, the return per dollar $X/L$ is normal with mean $\mu$ and standard deviation $\sigma_j$. The farm's cash and land are worth $a \equiv \pi + qS/L$ per dollar loaned. I refer to $a$ as the farm's pledgeable resources per dollar. The farm defaults when the return per dollar is below $(1 + r) - a$, and its liquidation value is positive when the return exceeds $-a$. Using the formula for the expected value of a truncated normal distribution, I can rewrite expected profits per dollar loaned as

$$\frac{\Pi_j(r;S,q)}{L} \;=\; (1+r)\Big(1-\Phi(\bar{z})\Big) \;+\; \ell_j\Big[(\mu+a)\big(\Phi(\bar{z})-\Phi(z_0)\big)-\sigma_j\big(\phi(\bar{z})-\phi(z_0)\big)\Big] \\ -\;(1+w_j)\;-\;\frac{F_j}{L}, \qquad \bar{z}=\frac{(1+r)-a-\mu}{\sigma_j}, \quad z_0=\frac{-a-\mu}{\sigma_j},$$

where $\phi$ and $\Phi$ denote the standard normal p.d.f. and c.d.f., respectively, and $\Phi(\bar{z})$ is the probability of default. The first term is the repayment when the farm stays solvent. The second term is the expected recovery when the farm defaults, over the states in which its liquidation value is positive. The last two terms are the funds advanced, including the cost of funds, and the fixed underwriting cost. I denote the first two terms, the expected payoff per dollar, by $g(r,a,\sigma_j)$. Note that the mean return and the farm's pledgeable resources enter only through the sum $\mu+a$. As a result, a dollar of expected project return, a dollar of subsidy cash, and a dollar of collateral value increase the lender's expected payoff by the same amount. Every proposition below uses this property.

The lending market is competitive, so each bank's break-even interest rate $r_j^*(S,q)$ is the minimum rate such that its expected profits are nonnegative, which solves

$$g\big(r_j^*,\,a,\,\sigma_j\big) \;=\; 1+w_j+\frac{F_j}{L},$$

if a solution exists. The right-hand side, $1+w_j+F_j/L$, is bank $j$'s cost per dollar of lending, and I refer to it as the cost per dollar throughout. If no solution exists, the bank does not lend. The farm borrows from the bank with the lower break-even rate, so the equilibrium lender is $j^*(S,q)=\arg\min_j\, r_j^*(S,q)$.

Throughout, I assume two regularity conditions. First, the probability of default at the break-even rate is below one half, which matches the single-digit delinquency rates on agricultural loans in the data. Second, $\partial g/\partial r>0$ at $r_j^*$, which implies that expected profit crosses zero strictly from below at the break-even rate. Because $r_j^*$ is the lowest solution and profit is negative at lower rates, the weak inequality holds automatically. The assumption that the crossing is strict rules out the case in which profit touches zero without crossing it.[27] The next lemma determines the sign of the derivatives of the break-even rate with respect to pledgeable resources, payoff uncertainty, and the cost of lending.

LEMMA 1: *Under the regularity conditions, at the break-even rate $r_j^*$: (i) $\dfrac{\partial g}{\partial a}=(1-\ell_j)(1+r)\dfrac{\phi(\bar{z})}{\sigma_j}+\ell_j\big(\Phi(\bar{z})-\Phi(z_0)\big)>0$, so $r_j^*$ is strictly decreasing in the farm's pledgeable resources per dollar $a$ and $r_j^*$ is strictly decreasing in the mean return $\mu$ with the same derivative. (ii) $\dfrac{\partial g}{\partial \sigma_j}=(1-\ell_j)(1+r)\,\bar{z}\,\dfrac{\phi(\bar{z})}{\sigma_j}+\ell_j\big(\phi(z_0)-\phi(\bar{z})\big)<0$, so $r_j^*$ is strictly increasing in payoff*

[27] The probability of default is below one half at every break-even rate shown in the figures. The largest default probability at any plotted break-even rate is about 9%, which is at the low end of the mean return range in Figure IA.9.

*uncertainty $\sigma_j$. (iii) $r_j^*$ is strictly increasing in the cost per dollar $1 + w_j + F_j/L$. For the small bank, $r_s^*$ is strictly decreasing in the deposit base $D$. (iv) The bank is willing to lend at some rate if and only if $\max_r g(r, a, \sigma_j) \geq 1 + w_j + F_j/L$, and the left side is strictly increasing in $a$, so farms whose pledgeable resources are below a threshold cannot borrow at any rate, and the threshold is increasing in the cost per dollar.*

*Proof.* (i) Increasing $a$ by $da$ has two effects. It decreases the default boundary, transferring probability mass $\phi(\bar{z})\, da/\sigma_j$ from default to repayment. At the default boundary the farm's liquidation value equals the repayment $(1+r)$ per dollar, so each transferred state increases profit by $(1+r) - \ell_j(1+r) = (1-\ell_j)(1+r)$. And it increases the recovery by $\ell_j\, da$ in every default state with positive liquidation value. This event has probability $\Phi(\bar{z}) - \Phi(z_0)$. At the boundary $z_0$ the farm's liquidation value equals zero, so states crossing $z_0$ do not affect $g$. Summing these effects yields the derivative in the lemma, which is strictly positive. Because $g$ is a function of $\mu$ and $a$ only through $\mu + a$, the derivative in $\mu$ is identical. By the second regularity condition and the implicit function theorem, $dr_j^*/da = -(\partial g/\partial a)/(\partial g/\partial r) < 0$, and similarly with $\mu$. (ii) Differentiating $g$ in $\sigma_j$ and simplifying using the identities $\mu + a + \sigma_j \bar{z} = 1 + r$ and $\mu + a + \sigma_j z_0 = 0$ yields the result. Under the first regularity condition $\bar{z} < 0$, so the first term is strictly negative, and $z_0 < \bar{z} < 0$ implies $\phi(z_0) < \phi(\bar{z})$, so the second term is strictly negative as well. (iii) and (iv) By the second regularity condition $\partial g/\partial r > 0$ at $r_j^*$, so the implicit function theorem implies $dr_j^*/d\theta = \big(\partial k/\partial\theta - \partial g/\partial\theta\big)\big/\big(\partial g/\partial r\big)$ for any parameter $\theta$, where $k = 1 + w_j + F_j/L$ is the cost per dollar. $k$ is increasing in $w_j$ and $F_j/L$ and, for the small bank, it is decreasing in $D$ because $w' > 0$. Because $g$ is strictly increasing in $a$, so is $\max_r g$ by the envelope theorem, which yields the participation result. Moreover, $\max_r g$ exists because $g$ is continuous, $g$ tends to $\ell_j\, E[\max(0, a + X/L)]$ as $r \to \infty$, and, because $\ell_j < 1$, $g$ exceeds this limiting value at a finite interest rate, so the maximum is attained. □

### IA.1.3 How Subsidies and Scale Increase Borrowing

I first study how fixed payments make it easier for farms to borrow. With subsidy payments, farms that were credit rationed obtain credit, farms that were not credit rationed borrow at lower interest rates, and a merged farm borrows at a lower interest rate than the two separate farms would. To focus on the borrower side of the model, I eliminate differences between banks and consider a single competitive lender with recovery rate $\ell$, payoff uncertainty $\sigma$, fixed underwriting cost $F$, a constant cost of funds normalized to zero, and no collateral value ($q = 0$). The farm's pledgeable resources per dollar are then its cash, $a = \pi$. Proposition 1 summarizes the main results.

PROPOSITION 1 (Payments, size, and the terms of credit): *Assume the regularity conditions stated above hold. (i) A farm with more base acres per dollar loaned (higher $\pi$) borrows at a strictly lower equilibrium rate, and the highest lending cost $F$ at which it can borrow at all is strictly larger, so farms with few base acres are more likely to be rationed. (ii) For fixed $\pi$, a larger farm (higher $S$) borrows at a strictly lower rate, and among farms facing the same lending cost, small farms are more likely to be rationed. (iii) A merged farm that requires a loan of twice the size to finance its project, incurs twice the lending cost, and has mean return $\mu' > \mu$ borrows at a strictly lower rate than the two separate farms, and it can borrow at lending costs at which they would be rationed.*

*Proof.* (i) When $q = 0$, $a = \pi$, so Lemma 1(i) implies the result regarding interest rates and Lemma 1(iv) implies the result regarding rationing. (ii) Increasing $S$ for fixed $\pi$ does not change $a$ and $\sigma$ and reduces the cost per dollar $F/L$. Lemma 1(iii) implies the rate result, and rationing decreases because the right-hand side of the condition in Lemma 1(iv) decreases while the left-hand side does not change. (iii) Doubling $L$ and $F$ does not change the cost per dollar $F/L$, and increasing the mean return from $\mu$ to $\mu'$ increases $g$ pointwise by Lemma 1(i), so the break-even rate decreases and rationing decreases. □

I describe the bank's decision problem first, then give the intuition for each of the three parts of the proposition. Figure IA.1 illustrates the bank's decision problem. It decomposes the lender's expected payoff per dollar loaned, before the cost of lending, into the gains from the solvent firm and the gains from the defaulting firm. As the interest rate increases, the farm's probability of default increases. Conditional on repayment, the lender's gains increase mechanically with the interest rate, but the lender is repaid in fewer states. Conditional on default, the lender recovers only a fraction of the farm's liquidation value, which is less than what it is owed, so the gains from the defaulting firm are negative. As rates increase, the set of payoff realizations that trigger default expands, so the lender bears the loss of default in more states. However, the gains upon default are lower than upon solvency for a given rate, so increasing rates past a point leads to lower expected profits. If the lending cost exceeds the maximum expected payoff, there is no interest rate at which the lender would lend.

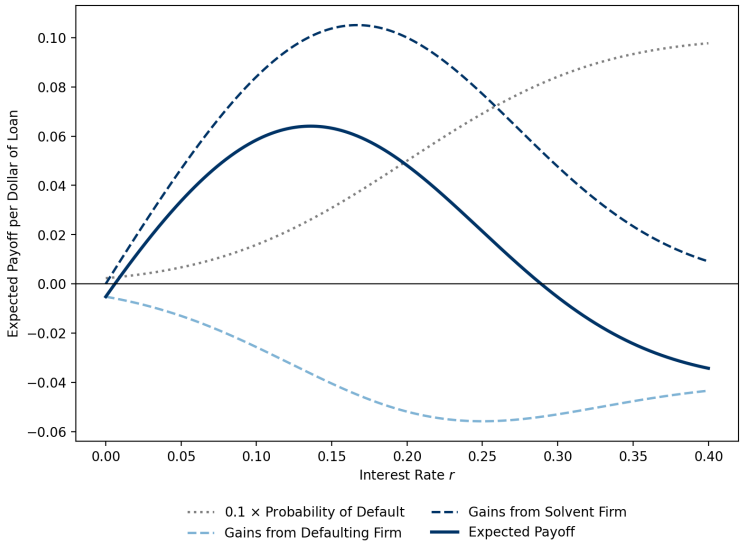



**Figure IA.1.** Lender's Expected Payoff by Interest Rate

This figure illustrates the components of the lender's decision problem. The gains from the solvent firm are the interest earned when the farm repays, weighted by the probability of repayment. The gains from the defaulting firm are the lender's recovery less the dollar advanced, weighted by the probability of default. The expected payoff is their sum. The dotted line plots one tenth of the probability of default. The parameter values are $\pi = 0.15$, $\mu = 1.05$, $\sigma = 0.1$, and $\ell = 0.8$.

Figure IA.2 illustrates part (i) of Proposition 1. It shows that the bank's profits before lending costs are increasing in the farm's cash holdings $A$, which for a fixed loan size increase with the farm's base acres, because fixed payments are tied to base acres. If a farm has more base acres, there are more states in which it does not default, and the lender recovers more in the states in which it does, leading to higher payoffs for the lender. These are the two effects in Lemma 1(i). The dotted horizontal lines indicate the level of lending costs, and the $x$ values of the solid dots denote equilibrium interest rates. When the cost of lending is high, there is no interest rate at which the lender would lend to the farm with low base acreage, because profits would be negative at any interest rate. In this case, such farms are credit rationed. When the cost is sufficiently low, the bank is willing to lend to farms with low base acreage, but only at a higher interest rate than for farms with greater base acreage.

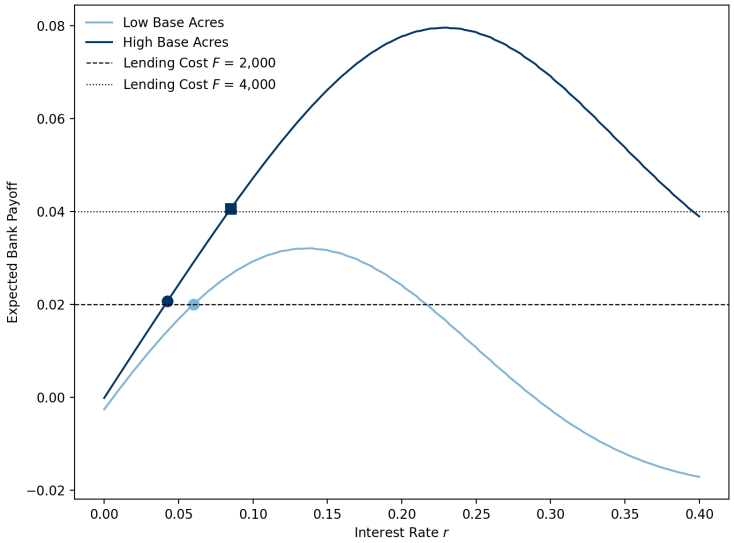



**Figure IA.2.** Bank Payoffs by Base Acreage

This figure shows bank payoffs as a function of rates for farms with a large and a small number of base acres. $\pi$ is set to 0.15 (light blue) and 0.25 (dark blue). The other parameter values are $L$ = 50,000, $\mu$ = 1.05, $\sigma$ = 0.1, and $\ell$ = 0.8. Two lending cost levels, $F$ = 2,000 and 4,000, are considered. The solid dots mark equilibrium rates.

Figure IA.3 presents the analogous comparison for farms of different sizes and illustrates part (ii). Expected lender profits per dollar increase with farm size because larger farms undertake larger projects, over which the fixed cost of lending is spread. The payoff per dollar does not depend on farm size. Because subsidies are tied to land, larger farms mechanically have greater base acreage and higher cash holdings, so pledgeable resources per dollar loaned stay the same as the loan grows. As a result, sufficiently small farms are credit rationed when the cost of lending is high, holding lending costs fixed across farms. When the cost of lending is sufficiently low, the bank is willing to lend to both small and large farms, but charges the large farm a lower equilibrium interest rate. This result suggests that larger farms are better positioned to borrow at lower rates following the introduction of fixed subsidies, allowing them to fund larger upfront investments such as farmland acquisitions.

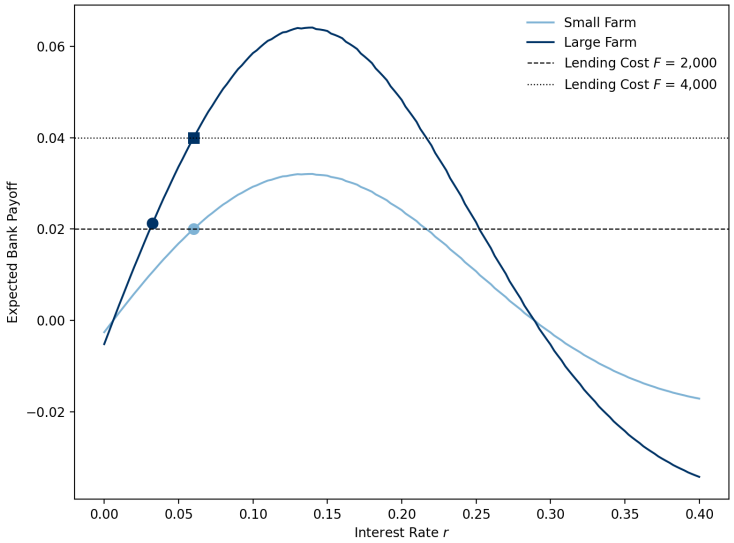



**Figure IA.3.** Bank Payoffs by Farm Size

This figure shows bank payoffs as a function of rates for a large and a small farm. $L$ is varied from 50,000 to 100,000. The other parameter values are $\mu = 1.05$, $\sigma = 0.1$, $\pi = 0.15$, and $\ell = 0.8$. Two lending cost levels, $F = 2{,}000$ and 4,000, are considered. The solid dots mark equilibrium rates.

Last, I consider a setting in which farms may merge and, upon merging, their projects exhibit increasing returns to scale. I compare the bank's lending decision for two identical separate farms with its decision for the merged farm. I model increasing returns to scale by assuming that the merged farm has a higher mean project return $\mu' > \mu$. The merged farm's project requires double the upfront investment and incurs double the lending cost, $2F$, relative to a single farm. Holding the total lending cost at $2F$ is conservative, because reducing the cost of lending to the merged farm below $2F$ would only strengthen the results. Figure IA.4 illustrates part (iii). It shows that when two farms merge and exhibit increasing returns to scale, the bank's expected profit from lending to the merged farm, at a given interest rate, exceeds the combined expected profit from lending to the two farms separately. Moreover, the merged farm is less likely to be credit rationed and, conditional on the bank being willing to lend to both the two separate farms and the merged farm, the merged farm can borrow at a lower equilibrium interest rate than the two separate farms.

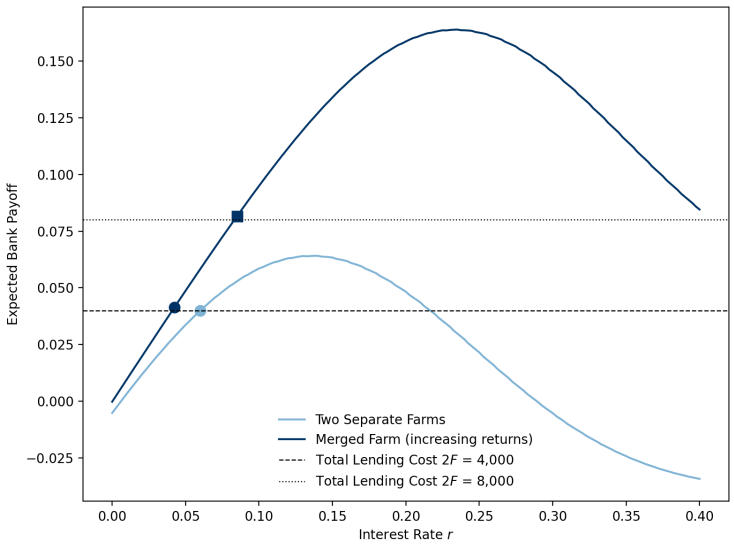



**Figure IA.4.** Bank Payoffs under a Merger

This figure compares expected bank payoffs under a merger with increasing returns to scale to payoffs when the bank lends to two farms separately. $\mu = 1.05$ when the farms are separate. Under a merger, $\mu = 1.1 \cdot 1.05$. Each farm has $L = 50{,}000$. The other parameter values are $\sigma = 0.1$, $\pi = 0.15$, and $\ell = 0.8$. Two total lending cost levels, $2F = 4{,}000$ and $8{,}000$, are considered, and the total cost of lending to either the two separate farms or the merged farm is the same.

### IA.1.4 Empirical Implications for the Demand Side

Proposition 1 generates the demand-side predictions that two results in the main paper test. The first-stage estimates show that farm concentration increases more in counties receiving larger fixed payments, consistent with part (i), under which payments increase pledgeable resources per dollar, $a = \pi$, and ease the terms of credit. A within-county one-standard-deviation increase in subsidy payments increases farm concentration by 105–107 acres (Table 3). The event-study estimates show that farm concentration increases gradually after payments begin, accumulating to about 500 acres during the first decade of the program (Figure 3). Part (iii) predicts that merging farms borrow at lower rates when consolidation increases mean returns. Increasing returns in farming make this assumption plausible (MacDonald, Korb, and Hoppe, 2013; Key, 2019), but the main prediction does not require it. Evidence that farms with a greater share of base acres receive slightly lower interest rates on operating loans also supports the rate prediction in part (i) (Kropp and Whitaker, 2011).

### IA.1.5 Examining Four Channels

In the analysis that follows, I study which bank lends to the farm as farms consolidate. Subsidy payments may affect local banking markets through four parameters of the model.

Larger farms have a higher $S$. Capitalization of payments into farmland increases the land price $q$. Payments deposited in local banks increase the deposit base $D$. Increasing returns increase the mean return $\mu$. I consider each in turn. Scale benefits large banks, because larger farms demand loans that small banks cannot finance from their core deposits at low cost. Deposits benefit small banks, because deposit inflows increase the loans they can finance at low cost. Collateral and production increase what both banks recover in default, so their effect on which bank lends to the farm depends on the difference in recovery rates.

Proposition 2 describes the effect of farm scale $S$.

PROPOSITION 2 (Farm scale and bank funding): *Assume that $q$, $D$, and $\mu$ are fixed and let farm scale $S$ vary, with $L = \pi_2 S$. Suppose that if the two banks offer the same rate, the small bank's cost of funds increases with the loan at least as fast as its average fixed cost decreases, $w'(L/D)/D \geq F_s/L^2$, and that there is some rate such that lending by the large bank is profitable, that is, $\max_r g_\ell > 1 + w_\ell$. Then (i) the large bank's break-even rate is strictly decreasing in $S$, and if the small bank offers the lower rate for some farm scale, there is a unique cutoff $\bar{S}(q, D)$ such that, among farms that some bank is willing to lend to, the small bank lends to those with $S < \bar{S}$ and the large bank lends to those with $S > \bar{S}$. (ii) $\bar{S}(q, D)$ is strictly increasing in $D$, so small banks with more funding capacity continue to lend to larger farms. (iii) Across markets with the same pledgeable resources per dollar loaned, $a = \pi + q/\pi_2$, $\bar{S}$ is strictly decreasing in the loan each acre requires, $\pi_2$. When farming is more capital intensive, the large bank's break-even rate is below the small bank's at smaller farm scales $S$.*

*Proof.* (i) Fix $q$. Then $a = \pi + qS/L = \pi + q/\pi_2$ does not vary with $S$, so scale affects the break-even rates through the costs per dollar $F_j/L$ and $w_j$ and through $\sigma_s = \sigma - \delta(S)$. For the large bank only $F_\ell/L$ is affected, and it decreases, so $r^*_\ell$ is strictly decreasing in $S$ by Lemma 1(iii). Let $H(S) = r^*_s(S) - r^*_\ell(S)$. By the implicit function theorem, at any scale for which the two banks offer the same interest rate,

$$H'(S) \;=\; \frac{\pi_2\left(\dfrac{w'(L/D)}{D} - \dfrac{F_s}{L^2}\right) + \left(-\dfrac{\partial g}{\partial \sigma}\right)\left(-\delta'(S)\right)}{\partial g_s/\partial r} \;+\; \frac{\pi_2 F_\ell/L^2}{\partial g_\ell/\partial r}.$$

The first numerator is nonnegative by the funding-slope condition and because $\delta' < 0$ and $\partial g/\partial \sigma < 0$ by Lemma 1(ii), and the second numerator is strictly positive. Both denominators are positive by the second regularity condition. So $H$ is strictly increasing at any zero, which means $H$ has at most one zero and crosses from below. This monotonicity implies a crossing is unique, and it remains to show that one exists. By part (i), $r^*_\ell$ is strictly decreasing in $S$, hence bounded above by the interest rate at the smallest scale. For the small bank, $w(\pi_2 S/D)$ increases without bound while the payoff per dollar is bounded, $\max_r g \leq E[\max(0, a + X/L)]$.

This bound is finite and does not depend on $S$, so for scales larger than a given finite value the small bank cannot break even at any rate. A bank that declines to lend is assigned $r = \infty$. By the participation condition in the statement of the proposition, $\max_r g_\ell > 1 + w_\ell$, so the large bank participates at all sufficiently large $S$. Its participation decision is monotone in $S$ because $F_\ell/L$ decreases and $\max_r g_\ell$ is constant in $S$. Then $H > 0$ for farm scales larger than the value at which the small bank exits. If $H$ is still negative for scales smaller than the value at which the small bank exits, the cutoff is the exit scale. If instead $H$ changes sign before the exit scale, continuity implies there is one crossing, which defines $\bar{S}(q, D)$. (ii) Increasing $D$ strictly decreases $r_s^*$ for every scale $S$ by Lemma 1(iii) and does not affect $r_\ell^*$, so $H$ decreases pointwise. Because $H$ crosses zero from below, the crossing shifts right. (iii) With $a$ fixed, the gap in costs per dollar is $w(\pi_2 S/D) - w_\ell + (F_s - F_\ell)/(\pi_2 S)$, whose derivative in $\pi_2$ is $w'S/D + (F_\ell - F_s)/(\pi_2^2 S) > 0$ because $F_\ell > F_s$. The funding-slope condition $w'/D \geq F_s/L^2$ is equivalent to $w'S/D \geq F_s/(\pi_2^2 S)$, so at any zero of $H$, $dH/d\pi_2 = \big(w'S/D - F_s/(\pi_2^2 S)\big)\big/(\partial g_s/\partial r) + \big(F_\ell/(\pi_2^2 S)\big)\big/(\partial g_\ell/\partial r)$ has a nonnegative first term and a strictly positive second term. $H$ increasing in $\pi_2$ at its zero shifts the crossing left. □

Figure IA.5 illustrates parts (i) and (ii) of Proposition 2 by plotting both banks' expected profits against the interest rate. The farm borrows from the bank with the lower break-even interest rate. I consider three cases, changing one parameter at a time. Panel A considers a farm with 550 acres, which I refer to as the small farm. Its loan is small relative to the small bank's deposit base, so the small bank's funding costs are low and its informational advantage is large. It breaks even about 0.8 percentage points below the large bank. Panel B considers a farm of 850 acres, which I refer to as the large farm, holding the other parameters fixed. The larger loan increases the small bank's funding costs, while the large bank finances the loan through its internal capital market and spreads its fixed underwriting cost over more dollars. The ranking reverses, and the large bank breaks even about 1.2 percentage points below the small bank. The dotted line holds the small bank's cost of funds fixed at its core deposit rate while keeping the differences in information and underwriting costs. In this case the small bank breaks even at about 3%, compared with 5.9% for the large bank. Hence the increase in funding costs reverses the ranking. Panel C considers a small bank with double the deposit base, reflecting a merger between two equally sized small banks. Lower funding costs restore the small bank's advantage. This result suggests that the small bank loses larger farms when their loans become large relative to its deposit base, and that a deposit inflow or merger allows it to continue to lend to these farms.

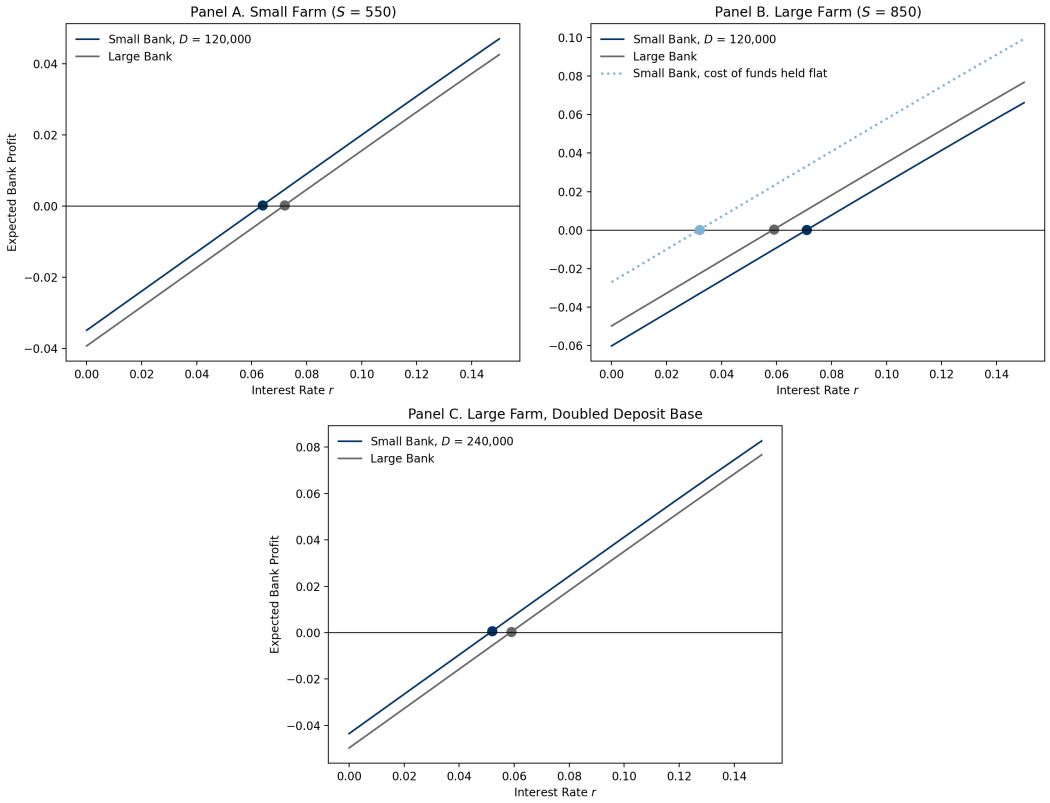



**Figure IA.5.** Break-Even Rates of Small and Large Banks by Farm Size and Deposit Base

This figure shows expected bank profits, net of funding and fixed underwriting costs, as a function of the interest rate, for the small bank and the large bank. The solid dots mark break-even rates, and the farm borrows from the bank with the lower break-even rate. Panel A considers a small farm ($S = 550$ acres) and a small bank with core deposit base $D = 120{,}000$. Panel B considers a large farm ($S = 850$ acres) with the same deposit base. The dotted line displays the small bank's profits with its cost of funds held fixed at the core rate of 2%, so that its funding cost no longer increases with the loan while the informational and fixed cost differences remain. Panel C considers the same large farm when the small bank's deposit base has doubled to $D = 240{,}000$, as after a merger of two equal small banks. Parameter values: $\mu = 1.05$, $\sigma = 0.10$, $\pi = 0.15$, $L = \pi_2 S$ with $\pi_2 = \$100$ per acre, collateral per dollar loaned fixed at 0.2, $\delta(S) = 0.02\, e^{-S/500}$, $\ell_s = 0.75$, $\ell_\ell = 0.85$, $F_s = 1{,}000$, $F_\ell = 2{,}000$, $w_s = 0.02 + 0.055\, L/D$, and $w_\ell = 0.035$.

Figure IA.6 shows how farm size $S$ and the small bank's deposit base $D$ determine which bank lends to the farm. The figure increases $S$ from 500 to 1,000 acres while holding the land price $q$ fixed, so pledgeable resources per dollar do not change. Farm size affects break-even interest rates only through average underwriting costs $F_j/L$, the small bank's funding costs $w(L/D)$, and its informational advantage $\delta(S)$. The small bank's break-even interest rate increases with farm size over most of the range. When $D = 180{,}000$ it first declines slightly, because the fixed underwriting cost spread over a larger loan decreases faster than the funding cost increases, and then increases once the funding cost increases faster. The large bank's interest rate decreases throughout. When $D = 120{,}000$, the two rates cross at

a cutoff $\bar{S}$ of about 660 acres. Increasing $D$ to 180,000 moves $\bar{S}$ to about 870 acres. When the small bank's cost of funds is instead held fixed, its break-even interest rate decreases from 4.0 to 3.0% and never crosses the large bank's. The cutoff $\bar{S}$ is the farm size at which higher funding costs offset the small bank's informational and underwriting advantages, and the larger deposit base implies this cutoff is at a larger farm size.

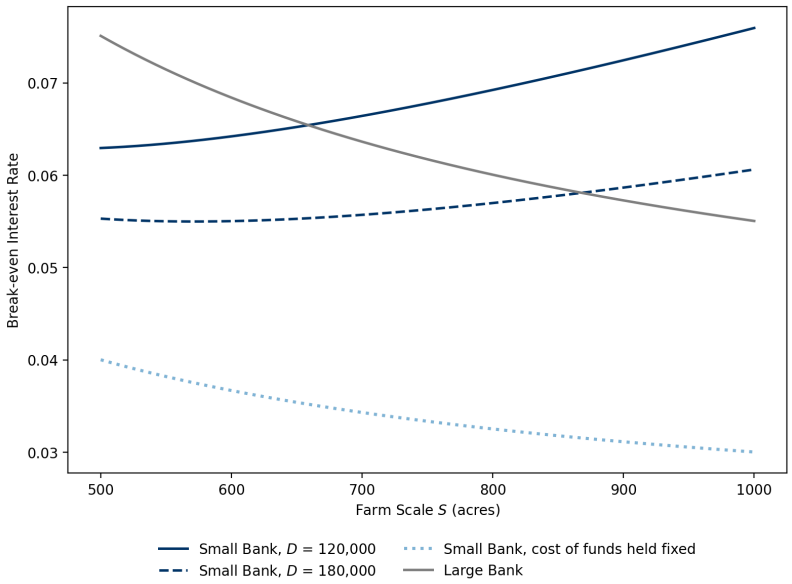



**Figure IA.6.** Break-Even Interest Rates by Farm Scale

This figure plots each bank's break-even interest rate against farm scale $S$ at a fixed collateral value, for two levels of the small bank's core deposit base $D$. The crossings mark the cutoff $\bar{S}(q, D)$. The dotted line displays the small bank with its cost of funds held fixed at the core rate of 2%, so that its funding cost no longer increases with the loan while the informational and fixed cost differences remain. Its rate decreases with scale and never crosses the large bank's. Parameter values are as in Figure IA.5, with collateral per dollar fixed at 0.2.

Next I consider variation in the price of farmland $q$ instead of farm scale $S$. Because both the pledged land $qS$ and the loan $L$ scale with acres, pledgeable resources per dollar loaned, $a = \pi + q/\pi_2$, are the same at every farm size. An increase in the land price $q$ therefore increases pledgeable resources per dollar by the same amount on every loan, large or small, and it does not change the deposit base $D$ or the cost of funds $w$. Its effect on a bank's break-even rate depends on the interest rate the bank charges, which is higher when the bank's cost of funds is higher.

Lemma 1(i) splits the gain from an extra dollar of pledgeable resources into two effects. First, with an extra dollar the farm repays in some states in which it would otherwise have defaulted, and in each of these states the lender's profits increase by the repayment less the recovery it would have received, $(1-\ell_j)(1+r)$. This effect is larger for the bank that recovers

less. Second, in the states in which the farm still defaults, the lender recovers $\ell_j$ more. This effect is larger for the bank that recovers more.

PROPOSITION 3 (Collateral channel): *Assume that $S$, $D$, and $\mu$ are fixed and increase the price of farmland $q$, which increases the farm's pledgeable resources per dollar $a = \pi + q/\pi_2$ for both banks. Then (i) both break-even rates strictly decrease and fewer farms are credit rationed. (ii) Near the cutoff $\bar{S}$, when the small bank's informational advantage is negligible, collateral affects which bank lends to the farm through differences in recovery rates. If the two banks have the same recovery rate, collateral has no effect on which bank lends to the farm. Locally, the difference between the banks' rate changes is proportional to $\ell_\ell - \ell_s$. (iii) The effect is present for every loan size. For the small bank, the decline in the break-even rate from an increase in $q$ is strictly larger when its deposit base $D$ is smaller.*

*Proof.* (i) This follows immediately from Lemma 1(i) and (iv). (ii) From Lemma 1(i), the payoff gain from collateral is the sum of two terms whose weights, $1-\ell_j$ and $\ell_j$, sum to one. The first term, $(1-\ell_j)(1+r)\phi(\bar{z})/\sigma$, the value of pulling marginal borrowers out of default, is weighted by $1-\ell_j$ and is larger for the low-recovery bank. The second term, $\ell_j(\Phi(\bar{z})-\Phi(z_0))$, the value of the additional dollar recovered in each default state that remains, is weighted by $\ell_j$ and is larger for the high-recovery bank. At a cutoff where $r_s^* = r_\ell^*$ and $\sigma_s \approx \sigma_\ell$, the two banks' payoff derivatives $\partial g_j/\partial a$, and likewise the denominators $\partial g_j/\partial r$, differ only through $\ell_j$. Write $\Delta = \ell_\ell - \ell_s$ and view $dr^*/dq$ as a smooth function of the lender's recovery rate, fixing $a$ and $\sigma$ at their common cutoff values. The difference between the two banks' derivatives is then $C\,\Delta + O(\Delta^2)$ by Taylor's theorem, where $C$ is the derivative of $dr^*/dq$ with respect to the recovery rate at the common parameters. The difference is zero when $\Delta = 0$. To determine the sign of $C$, write $m = (1+r)\phi(\bar{z})/\sigma$ and $B = \Phi(\bar{z}) - \Phi(z_0)$. Because $\phi$ is increasing on the negative axis and $z_0 < \bar{z} < 0$, $B < (\bar{z} - z_0)\,\phi(\bar{z}) = m$. The numerator of $|dr^*/da|$, $(1-\ell)m+\ell B$, is therefore strictly decreasing in $\ell$, and its denominator, $1-\Phi(\bar{z})-(1-\ell)m$, which is positive by the second regularity condition, is strictly increasing in $\ell$. Hence $|dr^*/da|$ is strictly decreasing in the recovery rate, and the bank with the lower recovery rate benefits more from collateral. (iii) Because $q$ affects both banks' profits only through $a$, which does not vary with farm size, the rate decreases for every loan size. For the small bank, $D$ appears only through the cost per dollar $k_s = 1 + w(L/D) + F_s/L$, which is strictly decreasing in $D$, so a decrease in $D$ increases $r_s^*$ by Lemma 1(iii). By the implicit function theorem, $dr_s^*/da = -(\partial g_s/\partial a)/(\partial g_s/\partial r)$, where both derivatives are evaluated at $r_s^*$. Differentiating the expression for $\partial g/\partial a$ in Lemma 1(i) with respect to $r$ yields

$$\frac{\partial^2 g}{\partial a\,\partial r} = \Big[(1-\ell_s)\Big(1 - \frac{\bar{z}\,(1+r)}{\sigma_s}\Big) + \ell_s\Big]\frac{\phi(\bar{z})}{\sigma_s} > 0,$$

and differentiating $g$ with respect to $r$ yields

$$\frac{\partial g}{\partial r} = 1 - \Phi(\bar{z}) - (1-\ell_s)(1+r)\frac{\phi(\bar{z})}{\sigma_s}, \qquad \frac{\partial^2 g}{\partial r^2} = -\Big[1 + (1-\ell_s)\Big(1 - \frac{\bar{z}\,(1+r)}{\sigma_s}\Big)\Big]\frac{\phi(\bar{z})}{\sigma_s} < 0.$$

Both of these results use $\bar{z} < 0$, which holds under the first regularity condition. The numerator of $|dr_s^*/da|$ increases in $r$, and its denominator, which is positive by the second regularity condition, decreases in $r$. Hence $|dr_s^*/da|$ is strictly increasing in $r_s^*$ and strictly decreasing in $D$. Because $\sigma_s$, $\ell_s$, and $a$ do not depend on $D$, this comparison holds the farm fixed. □

Figure IA.7 illustrates parts (i) and (ii) of Proposition 3. Panel A plots both banks' break-even interest rates against collateral per dollar. In this numerical example, the decrease in both interest rates is an order of magnitude larger than the change in the gap between them. More collateral decreases both banks' interest rates by similar amounts and the cutoff $\bar{S}$ is largely unchanged. Collateral primarily lowers borrowing costs and expands credit. Part (iii) is quantitatively large. At a farm scale of 600 acres and collateral per dollar loaned of 0.05, with the remaining parameter values as in Figure IA.5, an increase in collateral per dollar to 0.10 decreases the small bank's break-even interest rate by 1.90 percentage points when $D = 60{,}000$, 0.70 when $D = 120{,}000$, and 0.45 when $D = 240{,}000$. A small bank with a smaller deposit base charges a higher interest rate, at which more farms are close to default, so additional collateral reduces default in more states.

Panel B compares banks with different recovery rates for the same cost per dollar of lending, 1.05, and the same payoff uncertainty, $\sigma = 0.10$. The gap in break-even interest rates then measures the effect of the difference in recovery rates. The gap when $\ell_\ell - \ell_s = 0.05$ is about half as large as when $\ell_\ell - \ell_s = 0.10$ over the plotted range, and it is zero when the banks have the same recovery rate. The gap is largest when collateral is low and decreases as collateral increases, because with more collateral there are fewer states in which the farm defaults, so the recovery rate has a smaller effect.

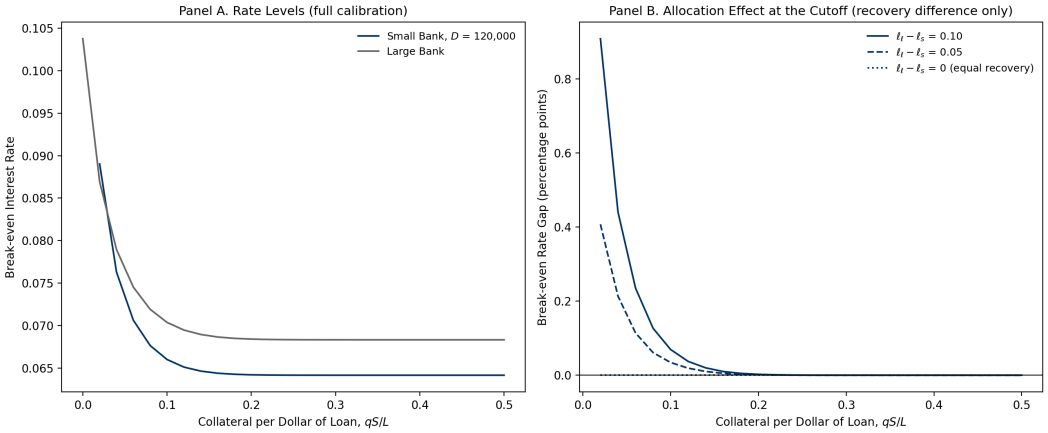



**Figure IA.7.** The Collateral Channel: Break-Even Rates by Collateral per Dollar

This figure illustrates the collateral channel. Panel A plots the break-even interest rates of the small bank ($D = 120{,}000$) and the large bank against collateral per dollar loaned, $qS/L$, at a fixed scale $S = 600$ acres, with the remaining parameter values as in Figure IA.5. Panel B plots the gap between the break-even rates of a low-recovery and a high-recovery lender, in percentage points, holding the two lenders' cost per dollar equal at 1.05 and payoff uncertainty equal at $\sigma = 0.10$. The three lines vary the recovery difference, $\ell_\ell - \ell_s = 0.10$ ($\ell_s = 0.75$), 0.05 ($\ell_s = 0.80$), and 0, with $\ell_\ell = 0.85$ throughout. $\mu = 1.05$ and $\pi = 0.15$.

PROPOSITION 4 (Deposit channel): *Assume that $S$, $q$, and $\mu$ are fixed, let the small bank's deposit base $D$ vary, and let $w$ be weakly convex. Then (i) an increase in the small bank's deposit base $D$ strictly lowers $r_s^*$, does not change $r_\ell^*$, and strictly increases the cutoff $\bar{S}(q, D)$. A deposit inflow into small banks expands their credit, allows them to lend to previously credit rationed farms, and reallocates farms from large banks to small banks. (ii) The shift in the small bank's funding cost from a change in its deposit base, $w'(L/D)\, L/D^2$, is increasing in lending relative to deposits $L/D$, so a given change in the deposit base has the largest effect on the funding cost of small banks whose lending is large relative to their deposits. (iii) A decline in $D$ reverses part (i) for every loan size and increases the funding cost reduction, $w(L/D) - w(L/(2D))$, that two small banks obtain by merging and pooling their deposit bases. This comparative static also holds as $L$ increases for fixed $D$.*

*Proof.* (i) Lemma 1(iii) and Proposition 2(ii) imply the rate and cutoff statements. Participation expands because the cost per dollar $1 + w(L/D) + F_s/L$ decreases while $\max_r g$ does not change, so farms rationed by the small bank at the previous cost can borrow (Lemma 1(iv)). (ii) Write $x = L/D$. The sensitivity of the funding cost to the deposit base is $|\partial w(L/D)/\partial D| = w'(x)\, x/D$, and $w'(x)\, x$ is increasing in $x$ when $w$ is weakly convex, because both factors are nonnegative and nondecreasing. (iii) The reversal results from (i) with the sign of the change in $D$ reversed. Regarding the merger claim, the funding saving from pooling two equal deposit bases $D$ into one bank servicing the loan is $w(L/D) - w(L/(2D))$,

and its derivative in $D$ is $\left[w'(L/D) - \frac{1}{2}w'(L/(2D))\right](-L/D^2) \leq 0$ for weakly convex $w$, because $w'(L/D) \geq w'(L/(2D)) \geq \frac{1}{2}w'(L/(2D))$. The saving decreases in $D$. For the claim in $L$, the derivative of the saving in $L$ is $w'(L/D)/D - w'(L/(2D))/(2D) \geq 0$ for weakly convex $w$, because $w'(L/D) \geq w'(L/(2D)) \geq \frac{1}{2}\,w'(L/(2D))$. □

Figure IA.8 illustrates Proposition 4. Panel A varies the small bank's deposit base $D$ from 60,000 to 300,000. First, the small bank's break-even interest rate decreases from about 11% to below 5% as $D$ increases. Second, it decreases below the large bank's interest rate at about $D = 180{,}000$, at which point the large farm borrows from the small bank. Third, the decrease is largest for low values of $D$, so a given deposit inflow has the largest effect on small banks with the least funding capacity. In contrast, a decrease in $D$ increases the small bank's break-even interest rate and reallocates the large farm to the large bank.

Panel B illustrates the incentive to merge in part (iii). It plots the funding saving, $w(L/D) - w(L/(2D))$, from merging two small banks and pooling their deposit bases. The saving increases from less than 1 percentage point when deposits are high to almost 4 percentage points when deposits are low, so a decrease in $D$ increases the incentive to merge. This saving is consistent with the increase in mergers among small banks in the data.

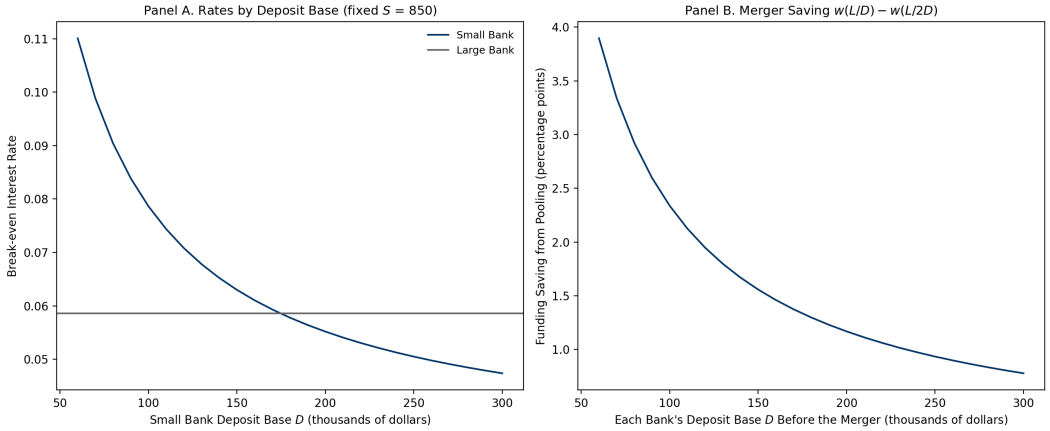



**Figure IA.8.** The Deposit Channel: Break-Even Rates and Merger Savings by Deposit Base

This figure illustrates the deposit channel. Panel A plots the break-even interest rates of the small bank and the large bank against the small bank's core deposit base $D$, for the large farm of Figure IA.5 ($S = 850$ acres) with the remaining parameter values as in that figure. Panel B plots the funding saving from pooling two equal deposit bases of size $D$ into one bank servicing the same loan, $w(L/D) - w(L/(2D))$, in percentage points, against the pre-merger base $D$.

PROPOSITION 5 (Production channel): *Assume that $S$, $q$, and $D$ are fixed and increase the mean return $\mu$. Then (i) both small and large bank break-even rates strictly decrease and fewer farms are credit rationed. (ii) Because expected profit depends on $\mu$ and $a$ only through*

*the sum $\mu + a$, the effect on which bank lends to the farm is the same as that of collateral in Proposition 3. Near the cutoff, when the small bank's informational advantage is negligible, it is proportional to the recovery difference $\ell_\ell - \ell_s$, favors the bank with the lower recovery rate, and is present at every loan size. As with collateral, an increase in $\mu$ decreases the small bank's break-even rate by more when its deposit base $D$ is smaller.*

*Proof.* Part (i) follows from Lemma 1(i) and (iv). For part (ii), $g$ depends on $(\mu, a)$ only through $\mu + a$, so every derivative of the break-even rates in $\mu$ equals the corresponding derivative in $a$, and the result restates Proposition 3(ii) and (iii). □

Figure IA.9 illustrates Proposition 5. It varies only the mean return $\mu$, at a low collateral value so that default risk remains relevant, and shows whether higher returns change which bank lends to the farm. As $\mu$ increases from 1.02 to 1.10, the small bank's break-even interest rate decreases from about 8.9 to 6.6% and the large bank's decreases from about 8.7 to 7.0%. The gap between the two never exceeds about 0.45 percentage points and changes sign at about $\mu = 1.03$. The large bank charges the lower interest rate at low mean returns, and the small bank charges the lower interest rate at high mean returns.

Higher returns decrease both banks' break-even rates. Near the cutoff, which bank gains more depends only on the difference in recovery rates, and the bank with the lower recovery rate, here the small bank, gains more. Because $\mu$ and $a$ affect expected profits through their sum, the production channel and the collateral channel yield the same predictions for lending.

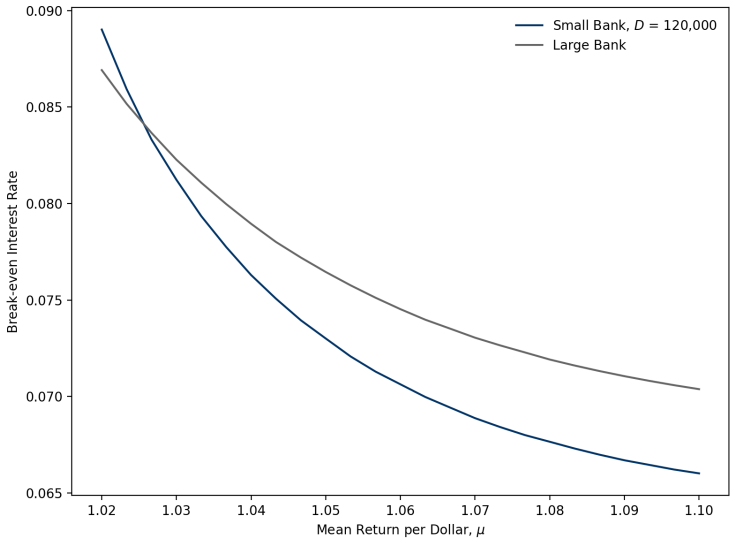



**Figure IA.9.** The Production Channel: Break-Even Rates by Mean Return

This figure illustrates the production channel. It plots the break-even interest rates of the small bank ($D =$ 120,000) and the large bank against the mean return per dollar $\mu$, at a fixed scale $S = 600$ acres and collateral per dollar loaned fixed at 0.05. The remaining parameter values are as in Figure IA.5.

### IA.1.6 Empirical Implications and Evidence for the Channels

**Proposition 2. Farm Scale and Bank Funding.** Proposition 2 has four empirical implications. First, part (i) implies that lending growth should be concentrated in larger loans. The small bank's cost of funds increases steeply once its lending exceeds its core deposit base, so farms whose loans exceed the cutoff $\bar{L}(q, D) = \pi_2 \bar{S}(q, D)$ borrow from the large bank. In the data, these are loans with original amounts above \$500,000, the largest loan size category in the Call Reports. Second, part (ii) implies that within small banks, lending growth should be concentrated at banks with a larger share of core deposits, which have the funding capacity to finance larger loans. Third, part (ii) and Proposition 4(iii) imply that small banks have an incentive to merge and pool their deposit bases, because $\bar{S}(q, D)$ increases with $D$. Fourth, part (iii) implies that large banks should gain the most market share in counties where each acre of farmland requires more credit, that is, where $\pi_2$ is higher. I test the first two implications together in the first test, and the third and fourth in the second and third tests.

The first test compares small banks with a larger and a smaller share of core deposits. The model has one small bank per market, but the same logic applies to small banks with different deposit bases. Farm consolidation increases large banks' market share. Higher land values lower both banks' interest rates and, near the cutoff, favor the small bank, which has the lower recovery rate, but this effect is close to zero in the numerical example. Comparing

large and small banks does not separate farm scale from collateral, so I compare small banks with different shares of core deposits, for which the two make opposite predictions. Because the cutoff $\bar{S}(q, D)$ increases with $D$, consolidation moves the larger loans it creates to small banks with the funding capacity to finance them, regardless of the collateral securing the loan. Under the collateral channel, the opposite holds. Collateral lowers the break-even rate most at small banks with smaller deposit bases (Proposition 3(iii)), so higher land values should increase lending most at small banks with less funding capacity, because borrowers choose the bank with the lower interest rate. In the data, a within-county one-standard-deviation increase in farm concentration increases lending above $500,000 at high-core small banks by about 94% for real estate farm loans and about 81% for non-real estate farm loans (Table 8). High-core small banks are those whose pre-program (1990–1995) core deposit share is above the small bank median. Lending above $500,000 does not increase at low-core small banks, and loans of $100,000 or less do not increase in either group. The same result holds for both real estate farm loans and non-real estate farm loans, and non-real estate farm loans are not secured by farmland, so collateral does not drive this result. Because both groups are small banks, the difference between them is not due to bank size.

The second test concerns bank mergers. Combining two small banks pools their deposit bases, which increases $D$ and with it the cutoff $\bar{S}(q, D)$, while the recovery rate $\ell_s$, the underwriting cost $F_s$, and the informational advantage $\delta(S)$ do not change. Panel C of Figure IA.5 shows that the pooled deposit base lowers the merged bank's break-even rate below the large bank's rate on the large farm. Small banks near the cutoff have an incentive to merge with one another. Higher land values do not change the funding saving from pooling deposits, $w(L/D) - w(L/(2D))$, so they do not change small banks' incentive to merge. In the data, mergers among small banks increase by about 23% as farms consolidate, whereas mergers among large banks decline and acquisitions of small banks by large banks do not change (Table 5).

The third test compares counties where farming requires more or less credit per acre, $\pi_2$. Consolidation does not change $\pi_2$, but for the same increase in $S$, farms require larger loans where $\pi_2$ is higher. Holding pledgeable resources per dollar fixed, the large bank offers the lower interest rate at a smaller farm size, so large banks should gain more market share where $\pi_2$ is higher. I use counties with high and low farming costs to proxy for different values of $\pi_2$. Under the collateral channel, higher land values increase pledgeable resources per dollar loaned, $a = \pi + q/\pi_2$, by less where each acre requires more credit, so similar increases in land values predict, if anything, a smaller effect in high-cost counties, and any effect on which bank lends to the farm favors the bank with the lower recovery rate (Proposition 3(ii)). The collateral channel does not predict that large banks gain market share only in high-

cost counties. In the data, although high- and low-cost counties experience similar increases in land values (Figure A.3), large banks gain market share relative to small banks only in high-cost counties (Table A.17).

**Proposition 3. The Collateral Channel.** Proposition 3 has three empirical implications. First, higher land values affect lending in all counties where farmland values increase, whether or not farms consolidate, because the land price $q$ enters bank profits through pledgeable resources per dollar $a$ at any farm size. Second, the effect is present at every loan size and at both bank types, because $a$ is the same for every farm size. Part (i) shows that higher land values lower both banks' break-even rates, and part (ii) shows that any difference between the banks is proportional to the recovery difference $\ell_\ell - \ell_s$, favors the bank with the lower recovery rate, which is the small bank under $\ell_\ell \geq \ell_s$, and is small in the numerical example. Third, part (iii) shows that among small banks, the decrease in interest rates is largest at banks with smaller deposit bases. I test these three implications, and none of them holds in the data.

First, payments do not increase land values. Part (i) implies that both banks' break-even rates decrease when $q$ increases, so if payments increased land values, credit and banking outcomes would change even without farm consolidation. Payments do not increase land values or deposits per capita (Figure 3, Panels C and D, and Table A.4). In rugged counties, where farms receive payments but expanding acreage is more difficult, payments have a limited effect on farm concentration, and large banks do not enter or expand their branches or deposits (Table 2). Second, the collateral channel predicts an increase at every loan size and at both bank types, but agricultural lending increases most at large banks (Table 7). Third, within small banks, lending above $500,000 increases only at banks with high-core funding (Table 8), whereas the collateral channel predicts the largest increase at small banks with low-core funding. The same result holds for non-real estate farm loans, which are not secured by farmland, so collateral does not drive this result either.

**Proposition 4. The Deposit Channel.** Proposition 4 has three empirical implications. First, part (i) implies that if farms deposit their payments in local banks, deposits per capita should increase when payments begin, and small banks should benefit in all counties that receive payments. Second, part (ii) implies that a given change in deposits has the largest effect on small banks whose lending is large relative to their deposits. Third, part (iii) implies that when loan demand increases faster than deposits, small banks face higher funding costs at every loan size and have a greater incentive to merge and pool their deposits.

Deposits per capita do not change (Panel D of Figure 3). The payments are also small

relative to deposits. A within-county one-standard-deviation increase in annual payments is $1.15 million, about 1% of the median small bank's total deposits. In rugged counties, where expanding acreage is more difficult and payments have a limited effect on farm concentration, large banks do not enter or expand their branches or deposits (Table 2). Large banks gain market share only where payments consolidate farms and farms demand larger loans, which small banks are less able to finance from local deposits.

The second test uses two measures of funding capacity fixed over 1990–1995, the core deposit share and the loans-to-deposits ratio. A lower loans-to-deposits ratio indicates more funding available for additional lending (Acharya and Mora, 2015). Farm concentration increases lending above $500,000 most at small banks with high-core funding and a low loans-to-deposits ratio. Following a within-county one-standard-deviation increase in farm concentration, the difference between these banks and small banks with low-core funding and a high loans-to-deposits ratio is about 140 percentage points for real estate farm lending and about 322 percentage points for non-real estate farm lending. The two groups do not differ in real estate farm lending at or below $100,000 (Table A.16). This result is consistent with part (ii), although the two concern different comparisons. Part (ii) implies that a deposit inflow lowers funding costs most at a bank whose lending is already large relative to its deposits. The test instead holds deposits fixed and increases loan sizes, and a larger loan is least costly to finance at the bank with the most unused funding capacity. The same funding differences are present in deposit rates. A within-county one-standard-deviation increase in farm concentration increases the real rate small banks offer on 12-month certificates of deposit by about 30 basis points, while large banks lower their rates by about 18 basis points (Table 10). Rates on savings and checking accounts decrease at both bank types, consistent with the market power that higher concentration provides banks over core deposits. The opposite signs are consistent with the assumption that small banks fund local lending from local deposits while large banks fund it through internal capital markets. If large banks also funded local lending from local deposits, they would also increase their time deposit rates as their lending increases, even if their cost of funds increased less steeply than small banks'. Instead, large banks lower their time deposit rates, so local deposits are not the source of their local lending.

The third test concerns loan demand relative to deposits. Consolidation increases the size of the loans farms demand while deposits per capita do not change, so $L/D$ increases. By part (iii), this increases the small bank's break-even rate on larger loans, moves those loans to large banks, and increases the funding saving, $w(L/D) - w(L/(2D))$, from merging and pooling deposits. Consistent with this prediction, mergers among small banks increase as farms consolidate, as shown in the second test for Proposition 2 above. These mergers do

not affect the classification of banks in the previous test, because that test classifies banks by core deposit share measured over 1990–1995, before payments began.

**Proposition 5. The Production Channel.** I find no evidence that subsidy payments affect lending through the production channel in Proposition 5. For payments to affect lending through this channel, they must first increase mean returns by changing what farms produce or how they produce it. Subsidy payments do not increase crop yields or the share of acreage planted in program crops, and they decrease machinery stock and tractor investment (Table A.5). Near the cutoff, part (ii) implies that higher mean returns affect which bank lends to the farm only through the recovery difference $\ell_\ell - \ell_s$, as collateral does. Because payments do not increase production or investment, differences in production are unlikely to drive the differences between bank types.

**Summary.** The model predicts that payments lower the cost of financing expansion, that farm consolidation concentrates credit demand in larger loans that small banks cannot fund from their core deposits, and that small banks with a larger share of core deposits keep financing those loans. Empirically, I find that payments consolidate farms, that lending above $500,000 increases only at small banks with a larger share of core deposits, and that small banks merge with one another. Payments increase farms' borrowing capacity and lower the cost of financing consolidation (Proposition 1). As farms consolidate, their larger loans are more costly for small banks to finance from their core deposits. Large banks gain the most market share where each acre requires more credit (Proposition 2). Where farms receive payments but do not substantially consolidate, local banking markets do not change, and payments do not increase land values (Proposition 3). Deposits matter because they limit how much small banks can lend, and payments do not change deposits (Proposition 4). Last, payments do not increase production or investment, which the production channel requires (Proposition 5).